SmartAlarm — Complete screen specifications

Revision 2.0 · 241 identified screens; Vietnamese–English–Japanese. Each screen has a corresponding mockup; state variations do not count as additional screens.

Sample records are not real personal data. Flows with real effects use review/approval/receipt in the product; the prototype only simulates navigation. Role names describe business capabilities, not unrestricted roles.

Shared interaction conventions

Focus follows content order; CTA design size ≥ 44 px; field errors preserve input; Enter does not automatically issue a command in confirmation dialogs; Escape closes a modal without saving. Form system fields are read-only. Mutations use idempotency + version; the UI shows success only after a server receipt. Sensitive details cannot be cached outside scope. Back navigation retains drafts within the same tenant scope and clears them when membership changes.

Individual screen specifications

V2 composition contract

The visual treatment below is normative for layout and information hierarchy. Existing purpose, role, fields, rules, workflow, navigation and acceptance contracts remain unchanged. Legacy Layout labels describe original inventory metadata and do not prescribe a generic renderer. These are design requirements, not a claim that each state has been user-tested. See UX redesign audit.

A01 — Welcome & region

EN: Welcome & region · **JA:** 地域と言語の選択
Requirement: FR-A01 / WF-01 · **Actor:** Citizen · **Surface:** Access · **Layout:** welcome

Purpose: Choose your safety region.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Continue → A02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Region	Europe · EU	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Language	English	Read-only within object scope; display source/time when they affect decisions.
Accessibility	Standard	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A02 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A01 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For welcome & region, what do Region, Language establish, and is “Continue” available to Citizen in this scope?

Primary visual: Region selector with language samples and readable text-size preview

Primary action: Continue → A02

Progressive disclosure: Keep Region, Language in the decision layer. Put the supporting details for Accessibility, full source/version history and FR-A01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a welcome & region fixture for Citizen; change Region and make Accessibility unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Continue” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A02 — Sign in

EN: Sign in · **JA:** サインイン
Requirement: FR-A02 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: Your workspace, securely connected.
Key rule: Verify identity and organization scope before exposing data.
Entry: A01, A05, X08.

Output: Sign in → A03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Email	alex@example.test	Valid email; errors must not reveal whether an account exists.
Workspace	Northstar	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Method	Passkey or enterprise SSO	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A03 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A02 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For sign in, what do Email, Workspace establish, and is “Sign in” available to All roles in this scope?

Primary visual: Workspace identity beside passkey and SSO entry

Primary action: Sign in → A03

Progressive disclosure: Keep Email, Workspace in the decision layer. Put the supporting details for Method, full source/version history and FR-A02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a sign in fixture for All roles; change Email and make Method unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Sign in” reaches A03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A03 — Verify identity

EN: Verify identity · **JA:** 本人確認
Requirement: FR-A03 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: Confirm this sign-in.
Key rule: Verify identity and organization scope before exposing data.
Entry: A02.
Output: Verify → A04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Method	Authenticator	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Verification code	123456	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Trusted device	Off	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A04 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A03 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For verify identity, what do Method, Verification code establish, and is “Verify” available to All roles in this scope?

Primary visual: Verification method and code slots with remaining attempt feedback

Primary action: Verify → A04

Progressive disclosure: Keep Method, Verification code in the decision layer. Put the supporting details for Trusted device, full source/version history and FR-A03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a verify identity fixture for All roles; change Method and make Trusted device unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Verify” reaches A04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A04 — Workspace switcher

EN: Workspace switcher · **JA:** ワークスペースの選択
Requirement: FR-A04 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** cards

Purpose: Choose an organization.
Key rule: Verify identity and organization scope before exposing data.
Entry: A03, X04.
Output: Enter workspace → O01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Northstar	EU · Operator	Read-only within object scope; display source/time when they affect decisions.
Harbor Campus	APAC · Responder	Read-only within object scope; display source/time when they affect decisions.
Personal	EU · Citizen	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O01 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A04 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For workspace switcher, what do Northstar, Harbor Campus establish, and is “Enter workspace” available to All roles in this scope?

Primary visual: Organization tiles showing region and effective role before entry

Primary action: Enter workspace → O01

Progressive disclosure: Keep Northstar, Harbor Campus in the decision layer. Put the supporting details for Personal, full source/version history and FR-A04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a workspace switcher fixture for All roles; change Northstar and make Personal unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and "Enter workspace" reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A05 — Recover access

EN: Recover access · **JA:** アクセスの復旧
Requirement: FR-A05 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: Restore access without exposing account existence.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Send recovery link → A02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Email	alex@example.test	Valid email; errors must not reveal whether an account exists.
Recovery method	Verified recovery contact	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Request	One active request	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A02 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A05 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For recover access, what do Email, Recovery method establish, and is “Send recovery link” available to All roles in this scope?

Primary visual: Recovery method selector and masked destination receipt

Primary action: Send recovery link → A02

Progressive disclosure: Keep Email, Recovery method in the decision layer. Put the supporting details for Request, full source/version history and FR-A05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a recover access fixture for All roles; change Email and make Request unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send recovery link” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A06 — Accept invitation

EN: Accept invitation · **JA:** 招待の承認
Requirement: FR-A06 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: Review your organization invitation.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Accept invitation → A07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Organization	Northstar	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Role	Responder	Select within granted permissions; the server blocks escalation beyond the actor’s rights.
Region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Invitation	Expires in 24 hours	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A07 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A06 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For accept invitation, what do Organization, Role establish, and is “Accept invitation” available to All roles in this scope?

Primary visual: Invitation identity, role scope and expiry comparison

Primary action: Accept invitation → A07

Progressive disclosure: Keep Organization, Role in the decision layer. Put the supporting details for Region, Invitation, full source/version history and FR-A06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a accept invitation fixture for All roles; change Organization and make Invitation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Accept invitation” reaches A07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A07 — Permissions & consent

EN: Permissions & consent · **JA:** 権限と同意
Requirement: FR-A07 / WF-01 · **Actor:** Citizen · **Surface:** Access · **Layout:** form

Purpose: Choose how SmartAlarm supports you.
Key rule: Verify identity and organization scope before exposing data.
Entry: A06.
Output: Save choices → C01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Notifications	Enable	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location	While using app	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.

Field / component	Sample value	Validation / editing permission
Emergency sharing	Selected contacts	Select within granted permissions; the server blocks escalation beyond the actor's rights.
Camera	Ask when attaching	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C01 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A07 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For permissions & consent, what do Notifications, Location establish, and is "Save choices" available to Citizen in this scope?

Primary visual: Separate permission switches with purpose pictograms

Primary action: Save choices → C01

Progressive disclosure: Keep Notifications, Location in the decision layer. Put the supporting details for Emergency sharing, Camera, full source/version history and FR-A07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a permissions & consent fixture for Citizen; change Notifications and make Camera unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save choices" reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A08 — Session & device security

EN: Session & device security · **JA:** セッションと端末
Requirement: FR-A08 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** list

Purpose: Review devices with access.
Key rule: Verify identity and organization scope before exposing data.
Entry: A09.
Output: Review session → A09.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
This device	Active · Berlin	Read-only within object scope; display source/time when they affect decisions.
Work tablet	Last active 2 hours ago	Read-only within object scope; display source/time when they affect decisions.
Old phone	Revoked	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A09 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A08 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For session & device security, what do This device, Work tablet establish, and is “Review session” available to All roles in this scope?

Primary visual: Device silhouettes with current-session badge and last activity

Primary action: Review session → A09

Progressive disclosure: Keep This device, Work tablet in the decision layer. Put the supporting details for Old phone, full source/version history and FR-A08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a session & device security fixture for All roles; change This device and make Old phone unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review session” reaches A09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A09 — Revoke session

EN: Revoke session · **JA:** セッションの取り消し
Requirement: FR-A09 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: End access on the selected device.
Key rule: Verify identity and organization scope before exposing data.
Entry: A08.
Output: Revoke selected session → A08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Device	Work tablet	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Effect	Sign-in required again	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reason	Device replaced	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A08 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A09 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For revoke session, what do Device, Effect establish, and is “Revoke selected session” available to All roles in this scope?

Primary visual: Selected device and revocation impact confirmation

Primary action: Revoke selected session → A08

Progressive disclosure: Keep Device, Effect in the decision layer. Put the supporting details for Reason, full source/version history and FR-A09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a revoke session fixture for All roles; change Device and make Reason unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Revoke selected session” reaches A08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

A10 — Profile & accessibility

EN: Profile & accessibility · **JA:** プロフィールとアクセシビリティ
Requirement: FR-A10 / WF-01 · **Actor:** All roles · **Surface:** Access · **Layout:** form

Purpose: Make the interface work for you.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Save profile → C01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Name	Alex Morgan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Language	English	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Text size	Large	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reduce motion	On	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Timezone	Europe/Berlin	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-A10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C01 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

A10 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For profile & accessibility, what do Name, Language establish, and is “Save profile” available to All roles in this scope?

Primary visual: Live accessibility preview beside language and motion settings

Primary action: Save profile → C01

Progressive disclosure: Keep Name, Language in the decision layer. Put the supporting details for Text size, Reduce motion, Timezone, full source/version history and FR-A10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not reveal the selected workspace before identity and membership are verified; retain the chosen method without implying successful access.

V2 acceptance scenario: Use a profile & accessibility fixture for All roles; change Name and make Timezone unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save profile” reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C01 — Safety home

EN: Safety home · **JA:** あなたの安全
Requirement: FR-C01 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** home

Purpose: A clear next step, when it matters.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: A07, A10, C23, C34.
Output: View active alert → C02.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Your area	Harbor Campus	Read-only within object scope; display source/time when they affect decisions.
Active alert	Fire · Building A	Read-only within object scope; display source/time when they affect decisions.
Family	3 checked in · 1 unknown	Read-only within object scope; display source/time when they affect decisions.
Connection	Online	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-C01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C02 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C01 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For safety home, what do Your area, Active alert establish, and is “View active alert” available to Citizen in this scope?

Primary visual: Local hazard summary above family check-in and connection state

Primary action: View active alert → C02

Progressive disclosure: Keep Your area, Active alert in the decision layer. Put the supporting details for Family, Connection, full source/version history and FR-C01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a safety home fixture for Citizen; change Your area and make Connection unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View active alert” reaches C02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C02 — Alert detail

EN: Alert detail · **JA:** 警報の詳細
Requirement: FR-C02 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: Fire reported in Building A.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: C01, C13.
Output: View guidance → C03.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Severity	Severe	Read-only within object scope; display source/time when they affect decisions.
Source	Campus command · Verified	Read-only within object scope; display source/time when they affect decisions.
Updated	10:42:08 local	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Instruction	Follow the approved site evacuation procedure	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C03 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C02 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For alert detail, what do Severity, Source establish, and is “View guidance” available to Citizen in this scope?

Primary visual: Hazard pictogram, affected place and one authoritative instruction

Primary action: View guidance → C03

Progressive disclosure: Keep Severity, Source in the decision layer. Put the supporting details for Updated, Instruction, full source/version history and FR-C02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a alert detail fixture for Citizen; change Severity and make Instruction unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View guidance” reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C03 — Evacuation guidance

EN: Evacuation guidance · **JA:** 避難案内
Requirement: FR-C03 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** floor

Purpose: Use the published evacuation plan.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: C02, C04, C18.

Output: I need assistance → C05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Location	Building A · Floor 05	Read-only within object scope; display source/time when they affect decisions.
Plan	Revision 12 · Approved	Read-only within object scope; display source/time when they affect decisions.
Route status	Not verified in real time	Read-only within object scope; display source/time when they affect decisions.
Accessible route	Refuge zone B	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C05 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C03 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For evacuation guidance, what do Location, Plan establish, and is “I need assistance” available to Citizen in this scope?

Primary visual: Floor schematic with current position uncertainty, blocked corridor and accessible refuge

Primary action: I need assistance → C05

Progressive disclosure: Keep Location, Plan in the decision layer. Put the supporting details for Route status, Accessible route, full source/version history and FR-C03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: If route approval, floor or hazard freshness becomes invalid, withdraw the route line and show assistance/refuge options validated for the site.

V2 acceptance scenario: Use a evacuation guidance fixture for Citizen; change Location and make Accessible route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “I need assistance” reaches C05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C04 — Voice & accessible guidance

EN: Voice & accessible guidance · **JA:** 音声案内
Requirement: FR-C04 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Read or listen to the same instruction.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: Role-based menu / deep link after authorization.
Output: Return to plan → C03.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Language	English	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Audio	Manual play	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Transcript	Follow the approved evacuation plan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Haptics	Enabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C03 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C04 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For voice & accessible guidance, what do Language, Audio establish, and is “Return to plan” available to Citizen in this scope?

Primary visual: Large audio controls paired with synchronized readable transcript

Primary action: Return to plan → C03

Progressive disclosure: Keep Language, Audio in the decision layer. Put the supporting details for Transcript, Haptics, full source/version history and FR-C04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a voice & accessible guidance fixture for Citizen; change Language and make Haptics unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Return to plan" reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C05 — SOS type

EN: SOS type · **JA:** 救援の種類を選択
Requirement: FR-C05 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** choices

Purpose: What kind of help do you need?.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: C03, C38.
Output: Continue with selected need → C06.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Fire or trapped	Urgent assistance	Read-only within object scope; display source/time when they affect decisions.
Medical or fall	Care assistance	Read-only within object scope; display source/time when they affect decisions.
Road collision	Traffic assistance	Read-only within object scope; display source/time when they affect decisions.
Personal threat	Discreet help	Read-only within object scope; display source/time when they affect decisions.
Other need	Describe your situation	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C06 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C05 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For sos type, what do Fire or trapped, Medical or fall establish, and is “Continue with selected need” available to Citizen in this scope?

Primary visual: Distinct fire, medical, collision and threat pictograms with short labels

Primary action: Continue with selected need → C06

Progressive disclosure: Keep Fire or trapped, Medical or fall in the decision layer. Put the supporting details for Road collision, Personal threat, Other need, full source/version history and FR-C05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a sos type fixture for Citizen; change Fire or trapped and make Other need unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Continue with selected need” reaches C06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C06 — SOS location & details

EN: SOS location & details · **JA:** SOSの位置と詳細
Requirement: FR-C06 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Tell responders where to find you.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: C05, X02.
Output: Review SOS → C07.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Location	Building A / 05 / Room 505	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Accuracy	User-entered · unverified	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
People needing help	1	Integer within policy limits; unknown is a separate value, not coerced to 0.
Note	Cannot reach the corridor	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C07 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C06 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For sos location & details, what do Location, Accuracy establish, and is “Review SOS” available to Citizen in this scope?

Primary visual: Location pin with accuracy boundary and manual correction

Primary action: Review SOS → C07

Progressive disclosure: Keep Location, Accuracy in the decision layer. Put the supporting details for People needing help, Note, full source/version history and FR-C06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a sos location & details fixture for Citizen; change Location and make Note unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review SOS” reaches C07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C07 — Confirm SOS

EN: Confirm SOS · **JA:** SOS送信の確認
Requirement: FR-C07 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: Send this request to campus command?.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: C06, C26.
Output: Send SOS — demo → C08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Category	Fire or trapped	Read-only within object scope; display source/time when they affect decisions.
Location	Room 505	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Recipients	Campus command	Read-only within object scope; display source/time when they affect decisions.
Sharing	Location and note	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C08 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C07 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For confirm sos, what do Category, Location establish, and is “Send SOS — demo” available to Citizen in this scope?

Primary visual: SOS review showing need, location and recipients before sending

Primary action: Send SOS — demo → C08

Progressive disclosure: Keep Category, Location in the decision layer. Put the supporting details for Recipients, Sharing, full source/version history and FR-C07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Sending may remain local pending; display server receipt separately and keep a safe alternate contact path, without repeated duplicate SOS.

V2 acceptance scenario: Use a confirm sos fixture for Citizen; change Category and make Sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Send SOS — demo” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C08 — SOS tracking

EN: SOS tracking · **JA:** 救援要請の状況
Requirement: FR-C08 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** timeline

Purpose: Your request has been received.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: C07, C10, C25, C27, C28.

Output: Contact assigned team → C09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
10:42:12	Received by server	Read-only within object scope; display source/time when they affect decisions.
10:42:18	Operator acknowledged	Read-only within object scope; display source/time when they affect decisions.
10:43:02	Team assigned	Read-only within object scope; display source/time when they affect decisions.
Current	Awaiting team contact	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C09 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C08 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For sos tracking, what do 10:42:12, 10:42:18 establish, and is “Contact assigned team” available to Citizen in this scope?

Primary visual: Receipt ladder distinguishing queued, server-received and human-accepted

Primary action: Contact assigned team → C09

Progressive disclosure: Keep 10:42:12, 10:42:18 in the decision layer. Put the supporting details for 10:43:02, Current, full source/version history and FR-C08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a sos tracking fixture for Citizen; change 10:42:12 and make Current unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Contact assigned team” reaches C09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C09 — Incident conversation

EN: Incident conversation · **JA:** 対応チームとの会話
Requirement: FR-C09 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** chat

Purpose: Stay connected to the response team.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: C08, C09.
Output: Send demo message → C09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Operator	We received your location	Read-only within object scope; display source/time when they affect decisions.
You	I am in room 505	Read-only within object scope; display source/time when they affect decisions.
System	Attachments are visible to assigned responders	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C09 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C09 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For incident conversation, what do Operator, You establish, and is “Send demo message” available to Citizen in this scope?

Primary visual: Incident conversation showing sender role and delivery state per message

Primary action: Send demo message → C09

Progressive disclosure: Keep Operator, You in the decision layer. Put the supporting details for System, full source/version history and FR-C09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a incident conversation fixture for Citizen; change Operator and make System unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send demo message” reaches C09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C10 — Update or cancel SOS

EN: Update or cancel SOS · **JA:** SOSの更新・取り消し申請
Requirement: FR-C10 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Update your situation.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: Role-based menu / deep link after authorization.
Output: Submit update → C08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Request	SOS-041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Action	Request cancellation	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reason	Help arrived	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Notice	Operator will review the change	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C08 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C10 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For update or cancel sos, what do Request, Action establish, and is “Submit update” available to Citizen in this scope?

Primary visual: Existing SOS beside proposed update and cancellation effect

Primary action: Submit update → C08

Progressive disclosure: Keep Request, Action in the decision layer. Put the supporting details for Reason, Notice, full source/version history and FR-C10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a update or cancel sos fixture for Citizen; change Request and make Notice unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit update” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C11 — Confirm personal safety

EN: Confirm personal safety · **JA:** 安全状況を報告
Requirement: FR-C11 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Let your circle know your status.
Key rule: A personal safety report does not close the incident.
Entry: Role-based menu / deep link after authorization.
Output: Confirm my safety → C12.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Status	I am safe	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location	Assembly point North	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Share with	Family + Campus group	Select within granted permissions; the server blocks escalation beyond the actor’s rights.
Precision	Status only outside response team	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C12 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C11 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For confirm personal safety, what do Status, Location establish, and is “Confirm my safety” available to Citizen in this scope?

Primary visual: Three explicit personal-status choices with sharing precision

Primary action: Confirm my safety → C12

Progressive disclosure: Keep Status, Location in the decision layer. Put the supporting details for Share with, Precision, full source/version history and FR-C11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Personal confirmation is self-reported, audience-scoped and timestamped; it must not mark an entire household or incident safe.

V2 acceptance scenario: Use a confirm personal safety fixture for Citizen; change Status and make Precision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Confirm my safety” reaches C12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C12 — Safety receipt

EN: Safety receipt · **JA:** 安全報告の受領確認
Requirement: FR-C12 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** success

Purpose: Your safety update was saved.
Key rule: A personal safety report does not close the incident.
Entry: C11.
Output: View delivery details → C13.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Personal status	Self-reported safe	Read-only within object scope; display source/time when they affect decisions.
Groups	2 updates queued	Read-only within object scope; display source/time when they affect decisions.
Contacts	3 delivery attempts	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Incident	Still active	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C13 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C12 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For safety receipt, what do Personal status, Groups establish, and is “View delivery details” available to Citizen in this scope?

Primary visual: Personal report and separate delivery receipts for each audience

Primary action: View delivery details → C13

Progressive disclosure: Keep Personal status, Groups in the decision layer. Put the supporting details for Contacts, Incident, full source/version history and FR-C12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a safety receipt fixture for Citizen; change Personal status and make Incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View delivery details” reaches C13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C13 — Notification inbox

EN: Notification inbox · **JA:** 通知一覧
Requirement: FR-C13 / WF-10 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** list

Purpose: Updates that need your attention.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: C12.
Output: Open alert → C02.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Fire alert	Read · 10:42	Read-only within object scope; display source/time when they affect decisions.
Safety confirmation	Saved · 10:45	Read-only within object scope; display source/time when they affect decisions.
Family update	Delivered · 10:45	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C13: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C02 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C13 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For notification inbox, what do Fire alert, Safety confirmation establish, and is “Open alert” available to Citizen in this scope?

Primary visual: Alert inbox ordered by urgency with source and expiration

Primary action: Open alert → C02

Progressive disclosure: Keep Fire alert, Safety confirmation in the decision layer. Put the supporting details for Family update, full source/version history and FR-C13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a notification inbox fixture for Citizen; change Fire alert and make Family update unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open alert” reaches C02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C14 — Nearby hazards

EN: Nearby hazards · **JA:** 周辺の危険情報
Requirement: FR-C14 / WF-08 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** map

Purpose: Know what is happening nearby.
Key rule: Preserve issuer, validity and warning update or cancellation history.
Entry: Role-based menu / deep link after authorization.
Output: View hazard → C15.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Flood watch	2 km · official feed	Read-only within object scope; display source/time when they affect decisions.
Road closure	500 m · verified operator	Read-only within object scope; display source/time when they affect decisions.
Fire	Campus A · active	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C14: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C15 with a valid payload/version, and verify that it is blocked when the guard for WF-08 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C14 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For nearby hazards, what do Flood watch, Road closure establish, and is “View hazard” available to Citizen in this scope?

Primary visual: Nearby hazard map with list equivalent and measured distance uncertainty

Primary action: View hazard → C15

Progressive disclosure: Keep Flood watch, Road closure in the decision layer. Put the supporting details for Fire, full source/version history and FR-C14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a nearby hazards fixture for Citizen; change Flood watch and make Fire unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “View hazard” reaches C15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C15 — Disaster advisory

EN: Disaster advisory · **JA:** 災害情報
Requirement: FR-C15 / WF-08 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: Flood watch for Riverside district.
Key rule: Preserve issuer, validity and warning update or cancellation history.
Entry: C14.
Output: View shelter information → C16.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Source	Approved authority feed	Read-only within object scope; display source/time when they affect decisions.
Certainty	Likely	Read-only within object scope; display source/time when they affect decisions.
Valid until	18:00 local	Read-only within object scope; display source/time when they affect decisions.
Area	Riverside polygon	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C15: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C16 with a valid payload/version, and verify that it is blocked when the guard for WF-08 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C15 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For disaster advisory, what do Source, Certainty establish, and is “View shelter information” available to Citizen in this scope?

Primary visual: Advisory boundary and validity band beside source certainty

Primary action: View shelter information → C16

Progressive disclosure: Keep Source, Certainty in the decision layer. Put the supporting details for Valid until, Area, full source/version history and FR-C15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a disaster advisory fixture for Citizen; change Source and make Area unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "View shelter information" reaches C16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C16 — Shelters & assembly points

EN: Shelters & assembly points · **JA:** 避難所と集合場所
Requirement: FR-C16 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** map

Purpose: Find a designated assistance point.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: C15.
Output: Open shelter details → C17.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
North shelter	Open · updated 5 min ago	Read-only within object scope; display source/time when they affect decisions.
West center	Capacity unknown	Read-only within object scope; display source/time when they affect decisions.
Campus North	Assembly point	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C16: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C17 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C16 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For shelters & assembly points, what do North shelter, West center establish, and is "Open shelter details" available to Citizen in this scope?

Primary visual: Shelter cards comparing accessibility, current availability and distance

Primary action: Open shelter details → C17

Progressive disclosure: Keep North shelter, West center in the decision layer. Put the supporting details for Campus North, full source/version history and FR-C16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a shelters & assembly points fixture for Citizen; change North shelter and make Campus North unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and "Open shelter details" reaches C17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C17 — Shelter details

EN: Shelter details · **JA:** 避難所の詳細
Requirement: FR-C17 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** detail

Purpose: North shelter.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: C16.
Output: Save shelter offline → C18.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Status	Open · confirmed by operator	Read-only within object scope; display source/time when they affect decisions.
Accessibility	Step-free entrance	Read-only within object scope; display source/time when they affect decisions.
Capacity	80 available · estimate	Read-only within object scope; display source/time when they affect decisions.
Address	12 Harbor Road	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C17: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C18 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C17 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For shelter details, what do Status, Accessibility establish, and is “Save shelter offline” available to Citizen in this scope?

Primary visual: Shelter entrance diagram and verified access/capacity status

Primary action: Save shelter offline → C18

Progressive disclosure: Keep Status, Accessibility in the decision layer. Put the supporting details for Capacity, Address, full source/version history and FR-C17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a shelter details fixture for Citizen; change Status and make Address unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save shelter offline” reaches C18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C18 — Offline safety pack

EN: Offline safety pack · **JA:** オフライン安全情報
Requirement: FR-C18 / WF-13 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** list

Purpose: Essential information on your device.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: C17, X05.
Output: Review cached plan → C03.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Campus plan	Revision 12 · downloaded	Read-only within object scope; display source/time when they affect decisions.
Contacts	3 saved	Read-only within object scope; display source/time when they affect decisions.
Guidance	English · version 4	Read-only within object scope; display source/time when they affect decisions.
Last sync	Today 09:00	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C18: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C03 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C18 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For offline safety pack, what do Campus plan, Contacts establish, and is “Review cached plan” available to Citizen in this scope?

Primary visual: Offline pack contents with version and last synchronization age

Primary action: Review cached plan → C03

Progressive disclosure: Keep Campus plan, Contacts in the decision layer. Put the supporting details for Guidance, Last sync, full source/version history and FR-C18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a offline safety pack fixture for Citizen; change Campus plan and make Last sync unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review cached plan” reaches C03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C19 — My circles

EN: My circles · **JA:** マイグループ
Requirement: FR-C19 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** cards

Purpose: The people who should hear from you.
Key rule: A personal safety report does not close the incident.
Entry: C21, C33.
Output: Open family → C20.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Family	4 members	Read-only within object scope; display source/time when they affect decisions.
Campus	24 members	Read-only within object scope; display source/time when they affect decisions.
Travel team	6 members	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C19: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C20 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C19 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For my circles, what do Family, Campus establish, and is “Open family” available to Citizen in this scope?

Primary visual: Circle membership summary with outstanding check-in count

Primary action: Open family → C20

Progressive disclosure: Keep Family, Campus in the decision layer. Put the supporting details for Travel team, full source/version history and FR-C19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a my circles fixture for Citizen; change Family and make Travel team unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open family” reaches C20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C20 — Circle conversation

EN: Circle conversation · **JA:** グループチャット
Requirement: FR-C20 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** chat

Purpose: Family.
Key rule: A personal safety report does not close the incident.
Entry: C19, C22.
Output: View member status → C22.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
System	Alex reported safe at 10:45	Read-only within object scope; display source/time when they affect decisions.
Jamie	Thank you for letting us know	Read-only within object scope; display source/time when they affect decisions.
Morgan	My status is unknown	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C20: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C22 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C20 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For circle conversation, what do System, Jamie establish, and is “View member status” available to Citizen in this scope?

Primary visual: Group conversation with member check-in context

Primary action: View member status → C22

Progressive disclosure: Keep System, Jamie in the decision layer. Put the supporting details for Morgan, full source/version history and FR-C20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a circle conversation fixture for Citizen; change System and make Morgan unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “View member status” reaches C22 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C21 — Create or edit circle

EN: Create or edit circle · **JA:** グループの作成・編集
Requirement: FR-C21 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Choose members and sharing rules.
Key rule: A personal safety report does not close the incident.
Entry: Role-based menu / deep link after authorization.
Output: Save circle → C19.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Name	Family	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Membership	Invite only	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Safety updates	Enabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location sharing	Off by default	Select within granted permissions; the server blocks escalation beyond the actor's rights.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C21: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C19 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C21 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For create or edit circle, what do Name, Membership establish, and is "Save circle" available to Citizen in this scope?

Primary visual: Circle membership and sharing scope preview

Primary action: Save circle → C19

Progressive disclosure: Keep Name, Membership in the decision layer. Put the supporting details for Safety updates, Location sharing, full source/version history and FR-C21 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a create or edit circle fixture for Citizen; change Name and make Location sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save circle" reaches C19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C22 — Circle member status

EN: Circle member status · **JA:** メンバーの状況
Requirement: FR-C22 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** list

Purpose: A status is a report, not a guarantee.
Key rule: A personal safety report does not close the incident.
Entry: C20.
Output: Send check-in request → C20.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	Self-reported safe · 10:45	Read-only within object scope; display source/time when they affect decisions.
Jamie	Self-reported safe · 10:40	Read-only within object scope; display source/time when they affect decisions.
Morgan	Unknown · no recent report	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C22: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C20 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C22 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For circle member status, what do Alex, Jamie establish, and is “Send check-in request” available to Citizen in this scope?

Primary visual: Member status board separating reported safe from unknown

Primary action: Send check-in request → C20

Progressive disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, full source/version history and FR-C22 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a circle member status fixture for Citizen; change Alex and make Morgan unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send check-in request” reaches C20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C23 — Emergency contacts

EN: Emergency contacts · **JA:** 緊急連絡先
Requirement: FR-C23 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Who should receive your updates?.
Key rule: A personal safety report does not close the incident.
Entry: Role-based menu / deep link after authorization.
Output: Save contacts → C01.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Primary	Jamie · verified	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Secondary	Morgan · pending	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Channel	Push then SMS	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Share	Status only	Select within granted permissions; the server blocks escalation beyond the actor’s rights.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C23: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C01 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C23 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For emergency contacts, what do Primary, Secondary establish, and is “Save contacts” available to Citizen in this scope?

Primary visual: Contact order with channel pictograms and disclosure preview

Primary action: Save contacts → C01

Progressive disclosure: Keep Primary, Secondary in the decision layer. Put the supporting details for Channel, Share, full source/version history and FR-C23 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a emergency contacts fixture for Citizen; change Primary and make Share unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save contacts" reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C24 — Care & fall detection

EN: Care & fall detection · **JA:** 見守りと転倒検知
Requirement: FR-C24 / WF-06 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Configure a supported fall detector.
Key rule: No response requires escalation, not an automatic medical conclusion.
Entry: Role-based menu / deep link after authorization.
Output: Review test alert → C25.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Wearable	WR-401 · paired	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Detection	Enabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Check-in window	30 seconds	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.
Escalation	Care team then campus command	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C24: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C25 with a valid payload/version, and verify that it is blocked when the guard for WF-06 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C24 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For care & fall detection, what do Wearable, Detection establish, and is “Review test alert” available to Citizen in this scope?

Primary visual: Wearable connection and fall-check-in escalation sequence

Primary action: Review test alert → C25

Progressive disclosure: Keep Wearable, Detection in the decision layer. Put the supporting details for Check-in window, Escalation, full source/version history and FR-C24 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a care & fall detection fixture for Citizen; change Wearable and make Escalation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review test alert” reaches C25 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C25 — Possible fall check-in

EN: Possible fall check-in · **JA:** 転倒の可能性を確認
Requirement: FR-C25 / WF-06 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** countdown

Purpose: A possible fall was detected.
Key rule: No response requires escalation, not an automatic medical conclusion.
Entry: C24.
Output: I need help now → C08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Evidence	Wearable signal · unverified	Read-only within object scope; display source/time when they affect decisions.
Check-in window	30 seconds	Read-only within object scope; display source/time when they affect decisions.
No response	Notify care team	Read-only within object scope; display source/time when they affect decisions.
Alternative	Get help now	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C25: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C08 with a valid payload/version, and verify that it is blocked when the guard for WF-06 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C25 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For possible fall check-in, what do Evidence, Check-in window establish, and is “I need help now” available to Citizen in this scope?

Primary visual: Fall check-in with large help and dismiss controls and time remaining

Primary action: I need help now → C08

Progressive disclosure: Keep Evidence, Check-in window in the decision layer. Put the supporting details for No response, Alternative, full source/version history and FR-C25 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: No response remains unconfirmed; a stopped timer is not a help receipt, and accessibility must not depend on a press-and-hold gesture.

V2 acceptance scenario: Use a possible fall check-in fixture for Citizen; change Evidence and make Alternative unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “I need help now” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C26 — Road incident report

EN: Road incident report · **JA:** 交通事故の報告
Requirement: FR-C26 / WF-07 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Report a road incident safely.
Key rule: Verify location and road direction; injury status may remain unknown.
Entry: Role-based menu / deep link after authorization.
Output: Review SOS → C07.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Position	User-confirmed map pin	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Direction	Northbound	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Vehicles	2	Integer within policy limits; unknown is a separate value, not coerced to 0.
Injuries	Unknown	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Attachment	Optional photo	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C26: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C07 with a valid payload/version, and verify that it is blocked when the guard for WF-07 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C26 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For road incident report, what do Position, Direction establish, and is “Review SOS” available to Citizen in this scope?

Primary visual: Road orientation sketch with manual direction and injury entry

Primary action: Review SOS → C07

Progressive disclosure: Keep Position, Direction in the decision layer. Put the supporting details for Vehicles, Injuries, Attachment, full source/version history and FR-C26 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a road incident report fixture for Citizen; change Position and make Attachment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review SOS” reaches C07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C27 — Discreet assistance

EN: Discreet assistance · **JA:** 目立たない救援要請
Requirement: FR-C27 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: Request help with a quiet interface.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: Role-based menu / deep link after authorization.
Output: Send discreet request — demo → C08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Sound	Off	Read-only within object scope; display source/time when they affect decisions.
Haptics	Off	Read-only within object scope; display source/time when they affect decisions.
Sharing	Assigned operator only	Read-only within object scope; display source/time when they affect decisions.
Location	Approximate · 40 m	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C27: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C08 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C27 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For discreet assistance, what do Sound, Haptics establish, and is “Send discreet request — demo” available to Citizen in this scope?

Primary visual: Quiet high-contrast assistance controls showing sound and vibration state

Primary action: Send discreet request — demo → C08

Progressive disclosure: Keep Sound, Haptics in the decision layer. Put the supporting details for Sharing, Location, full source/version history and FR-C27 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Never autoplay speech or vibration against the discreet selection; reveal pending versus received without a conspicuous success animation.

V2 acceptance scenario: Use a discreet assistance fixture for Citizen; change Sound and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Send discreet request — demo” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C28 — My reports & history

EN: My reports & history · **JA:** 要請履歴
Requirement: FR-C28 / WF-05 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** list

Purpose: Your previous requests.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: Role-based menu / deep link after authorization.
Output: View request → C08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
SOS-041	Active · fire assistance	Read-only within object scope; display source/time when they affect decisions.
SOS-028	Closed · road incident	Read-only within object scope; display source/time when they affect decisions.
Check-in	Safe · yesterday	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C28: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C08 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C28 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For my reports & history, what do SOS-041, SOS-028 establish, and is “View request” available to Citizen in this scope?

Primary visual: Personal request history showing open, resolved and pending receipts

Primary action: View request → C08

Progressive disclosure: Keep SOS-041, SOS-028 in the decision layer. Put the supporting details for Check-in, full source/version history and FR-C28 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a my reports & history fixture for Citizen; change SOS-041 and make Check-in unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “View request” reaches C08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C29 — Privacy & data requests

EN: Privacy & data requests · **JA:** プライバシーとデータ
Requirement: FR-C29 / WF-19 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Control optional sharing.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: C30.
Output: Review data request → C30.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Live location	Off	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Circle visibility	Status only	Select within granted permissions; the server blocks escalation beyond the actor’s rights.
Export	Request a copy	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Deletion	Subject to incident retention review	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C29: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C30 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C29 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For privacy & data requests, what do Live location, Circle visibility establish, and is “Review data request” available to Citizen in this scope?

Primary visual: Privacy scope preview showing live location and export/deletion choices

Primary action: Review data request → C30

Progressive disclosure: Keep Live location, Circle visibility in the decision layer. Put the supporting details for Export, Deletion, full source/version history and FR-C29 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a privacy & data requests fixture for Citizen; change Live location and make Deletion unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review data request” reaches C30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C30 — Data request receipt

EN: Data request receipt · **JA:** データ申請の受領確認
Requirement: FR-C30 / WF-19 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** detail

Purpose: Request DR-202 received.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: C29, T10.
Output: Back to privacy → C29.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Type	Export personal data	Read-only within object scope; display source/time when they affect decisions.
Scope	Profile and eligible records	Read-only within object scope; display source/time when they affect decisions.
Status	Verification pending	Read-only within object scope; display source/time when they affect decisions.
Delivery	Secure download link after review	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-C30: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C29 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C30 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For data request receipt, what do Type, Scope establish, and is “Back to privacy” available to Citizen in this scope?

Primary visual: Data request progress and verified delivery destination

Primary action: Back to privacy → C29

Progressive disclosure: Keep Type, Scope in the decision layer. Put the supporting details for Status, Delivery, full source/version history and FR-C30 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a data request receipt fixture for Citizen; change Type and make Delivery unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Back to privacy” reaches C29 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R01 — Shift & readiness

EN: Shift & readiness · **JA:** 勤務と対応準備
Requirement: FR-R01 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** home

Purpose: Ready for your next assignment.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R08.
Output: View assignments → R02.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Unit	Rescue 02	Read-only within object scope; display source/time when they affect decisions.
Shift	10:00–18:00	Read-only within object scope; display source/time when they affect decisions.
Capabilities	Fire · assisted evacuation	Read-only within object scope; display source/time when they affect decisions.
Readiness	Available	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R02 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R01 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For shift & readiness, what do Unit, Shift establish, and is “View assignments” available to Responder in this scope?

Primary visual: Unit readiness checklist with capability and shift status

Primary action: View assignments → R02

Progressive disclosure: Keep Unit, Shift in the decision layer. Put the supporting details for Capabilities, Readiness, full source/version history and FR-R01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a shift & readiness fixture for Responder; change Unit and make Readiness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View assignments” reaches R02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R02 — Assignment inbox

EN: Assignment inbox · **JA:** 担当任務
Requirement: FR-R02 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** list

Purpose: Assignments for Rescue 02.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R01, R12.
Output: Open assignment → R03.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
INC-2041	Fire · Building A · P1	Read-only within object scope; display source/time when they affect decisions.
INC-2037	Welfare check · P2	Read-only within object scope; display source/time when they affect decisions.
Coverage	Campus east	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R03 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R02 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For assignment inbox, what do INC-2041, INC-2037 establish, and is “Open assignment” available to Responder in this scope?

Primary visual: Assignment queue ranked by urgency and acceptance deadline

Primary action: Open assignment → R03

Progressive disclosure: Keep INC-2041, INC-2037 in the decision layer. Put the supporting details for Coverage, full source/version history and FR-R02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a assignment inbox fixture for Responder; change INC-2041 and make Coverage unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open assignment” reaches R03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R03 — Accept or decline task

EN: Accept or decline task · **JA:** 任務の受諾
Requirement: FR-R03 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** detail

Purpose: Assist person at Room 505.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R02.

Output: Accept assignment → R04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Incident	INC-2041	Read-only within object scope; display source/time when they affect decisions.
Person	Alex · SOS-041	Read-only within object scope; display source/time when they affect decisions.
Location	User-reported · 05 / 505	Read-only within object scope; display source/time when they affect decisions.
Hazards	Smoke report · not confirmed	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R04 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R03 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For accept or decline task, what do Incident, Person establish, and is “Accept assignment” available to Responder in this scope?

Primary visual: Task identity, person location and hazard summary above accept/decline

Primary action: Accept assignment → R04

Progressive disclosure: Keep Incident, Person in the decision layer. Put the supporting details for Location, Hazards, full source/version history and FR-R03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a accept or decline task fixture for Responder; change Incident and make Hazards unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Accept assignment” reaches R04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R04 — Navigation & arrival

EN: Navigation & arrival · **JA:** 進入経路と現場到着
Requirement: FR-R04 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** map

Purpose: Building A · East access.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R03.
Output: Report arrival → R05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Assignment	Accepted	Read-only within object scope; display source/time when they affect decisions.
Destination	Entrance E2	Read-only within object scope; display source/time when they affect decisions.
Location freshness	8 seconds	Read-only within object scope; display source/time when they affect decisions.
Route	Operator-approved access plan	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R05 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R04 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For navigation & arrival, what do Assignment, Destination establish, and is “Report arrival” available to Responder in this scope?

Primary visual: Destination and approach map showing route validity and arrival action

Primary action: Report arrival → R05

Progressive disclosure: Keep Assignment, Destination in the decision layer. Put the supporting details for Location freshness, Route, full source/version history and FR-R04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

V2 acceptance scenario: Use a navigation & arrival fixture for Responder; change Assignment and make Route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Report arrival" reaches R05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R05 — Field situation

EN: Field situation · **JA:** 現場の状況
Requirement: FR-R05 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** floor

Purpose: Floor 05 response context.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R04.
Output: Open assistance record → R06.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Open SOS	1	Read-only within object scope; display source/time when they affect decisions.
Blocked area	East corridor · reported	Read-only within object scope; display source/time when they affect decisions.
Plan	Revision 12	Read-only within object scope; display source/time when they affect decisions.
Team	Rescue 02	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R06 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R05 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For field situation, what do Open SOS, Blocked area establish, and is "Open assistance record" available to Responder in this scope?

Primary visual: Field floor map with blocked zones, open SOS and teammate locations

Primary action: Open assistance record → R06

Progressive disclosure: Keep Open SOS, Blocked area in the decision layer. Put the supporting details for Plan, Team, full source/version history and FR-R05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

V2 acceptance scenario: Use a field situation fixture for Responder; change Open SOS and make Team unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Open assistance record" reaches R06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R06 — Assistance & triage record

EN: Assistance & triage record · **JA:** 現場支援の記録
Requirement: FR-R06 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** form

Purpose: Record observations, not diagnosis.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R05, H12.
Output: Save observation → R07.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Person	Alex Morgan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Contact	Established	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Condition	Needs mobility assistance	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Action	Assisting to assembly point	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R07 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R06 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For assistance & triage record, what do Person, Contact establish, and is “Save observation” available to Responder in this scope?

Primary visual: Person-centered observation form with condition and assistance actions

Primary action: Save observation → R07

Progressive disclosure: Keep Person, Contact in the decision layer. Put the supporting details for Condition, Action, full source/version history and FR-R06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a assistance & triage record fixture for Responder; change Person and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save observation” reaches R07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R07 — Task status & handover

EN: Task status & handover · **JA:** 任務状況と引き継ぎ
Requirement: FR-R07 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** form

Purpose: Hand over with a clear record.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R06.
Output: Submit handover → R08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Task	Rescue-112	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
State	Assistance completed	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Receiver	Assembly team North	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Evidence	10:48 · responder report	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R08 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R07 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For task status & handover, what do Task, State establish, and is “Submit handover” available to Responder in this scope?

Primary visual: Task handover timeline showing receiver acceptance separately

Primary action: Submit handover → R08

Progressive disclosure: Keep Task, State in the decision layer. Put the supporting details for Receiver, Evidence, full source/version history and FR-R07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a task status & handover fixture for Responder; change Task and make Evidence unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Submit handover” reaches R08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R08 — Task completion

EN: Task completion · **JA:** 任務の完了
Requirement: FR-R08 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** success

Purpose: Assignment completed.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R07.
Output: Back to shift → R01.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Person	Responder-confirmed assisted	Read-only within object scope; display source/time when they affect decisions.
Handover	Assembly team North	Read-only within object scope; display source/time when they affect decisions.
Incident	Still active	Read-only within object scope; display source/time when they affect decisions.
Next	Return to availability	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R01 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R08 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For task completion, what do Person, Handover establish, and is “Back to shift” available to Responder in this scope?

Primary visual: Completion evidence and receiving team acknowledgment

Primary action: Back to shift → R01

Progressive disclosure: Keep Person, Handover in the decision layer. Put the supporting details for Incident, Next, full source/version history and FR-R08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a task completion fixture for Responder; change Person and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Back to shift” reaches R01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R09 — Team communications

EN: Team communications · **JA:** チーム連絡
Requirement: FR-R09 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** chat

Purpose: Incident channel · INC-2041.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R09.
Output: Send demo message → R09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Command	Use entrance E2	Read-only within object scope; display source/time when they affect decisions.
Rescue 02	At floor 05	Read-only within object scope; display source/time when they affect decisions.
System	Only assigned teams can view this channel	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R09 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R09 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For team communications, what do Command, Rescue 02 establish, and is “Send demo message” available to Responder in this scope?

Primary visual: Team channel with command identity and pending message markers

Primary action: Send demo message → R09

Progressive disclosure: Keep Command, Rescue 02 in the decision layer. Put the supporting details for System, full source/version history and FR-R09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a team communications fixture for Responder; change Command and make System unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Send demo message” reaches R09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed

results separately from this specification.

R10 — Evidence capture

EN: Evidence capture · **JA:** 証拠の記録
Requirement: FR-R10 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** form

Purpose: Attach a field observation.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: X10.
Output: Queue attachment → R11.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Type	Photo	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Incident	INC-2041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Caption	Corridor condition at arrival	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Privacy	Restricted incident evidence	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R11 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R10 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For evidence capture, what do Type, Incident establish, and is “Queue attachment” available to Responder in this scope?

Primary visual: Evidence preview with incident binding and privacy redaction

Primary action: Queue attachment → R11

Progressive disclosure: Keep Type, Incident in the decision layer. Put the supporting details for Caption, Privacy, full source/version history and FR-R10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor's receipt.

V2 acceptance scenario: Use a evidence capture fixture for Responder; change Type and make Privacy unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Queue attachment" reaches R11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R11 — Offline outbox

EN: Offline outbox · **JA:** 同期待ちデータ
Requirement: FR-R11 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** list

Purpose: Updates waiting to sync.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R10, X01.
Output: Resolve sync conflict → R12.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Observation	Queued · 10:47	Read-only within object scope; display source/time when they affect decisions.
Photo	Queued · 2 MB	Read-only within object scope; display source/time when they affect decisions.
Status update	Conflict requires review	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R12 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R11 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For offline outbox, what do Observation, Photo establish, and is “Resolve sync conflict” available to Responder in this scope?

Primary visual: Outbox grouped by queued, retryable and conflicting items

Primary action: Resolve sync conflict → R12

Progressive disclosure: Keep Observation, Photo in the decision layer. Put the supporting details for Status update, full source/version history and FR-R11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a offline outbox fixture for Responder; change Observation and make Status update unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Resolve sync conflict” reaches R12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

R12 — Resolve field conflict

EN: Resolve field conflict · **JA:** 同期競合の解決
Requirement: FR-R12 / WF-11 · **Actor:** Responder · **Surface:** Responder · **Layout:** detail

Purpose: A newer assignment state exists.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: R11.
Output: Keep note and refresh → R02.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Local	Task in progress · 10:47	Read-only within object scope; display source/time when they affect decisions.
Server	Task reassigned · 10:48	Read-only within object scope; display source/time when they affect decisions.
Your note	Preserved as observation	Read-only within object scope; display source/time when they affect decisions.
Action	Refresh task ownership	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-R12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R02 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

R12 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For resolve field conflict, what do Local, Server establish, and is “Keep note and refresh” available to Responder in this scope?

Primary visual: Local observation beside server change with preserve-note option

Primary action: Keep note and refresh → R02

Progressive disclosure: Keep Local, Server in the decision layer. Put the supporting details for Your note, Action, full source/version history and FR-R12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain the field observation locally with its incident binding; do not show task acceptance or handover completion without the receiving actor’s receipt.

V2 acceptance scenario: Use a resolve field conflict fixture for Responder; change Local and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Keep note and refresh” reaches R02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O01 — Command overview

EN: Command overview · **JA:** 指令ダッシュボード
Requirement: FR-O01 / WF-20 · **Actor:** Operator · **Surface:** Command · **Layout:** dashboard

Purpose: Every signal. One coordinated response..
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: A04, O25, X09.
Output: Open priority incident → O04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Active incidents	12	Read-only within object scope; display source/time when they affect decisions.
Unacknowledged P1	2	Read-only within object scope; display source/time when they affect decisions.
Open SOS	7	Read-only within object scope; display source/time when they affect decisions.
Available units	18	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O04 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O01 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For command overview, what do Active incidents, Unacknowledged P1 establish, and is “Open priority incident” available to Operator in this scope?

Primary visual: Priority incident strip, capacity board and site situation map

Primary action: Open priority incident → O04

Progressive disclosure: Keep Active incidents, Unacknowledged P1 in the decision layer. Put the supporting details for Open SOS, Available units, full source/version history and FR-O01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a command overview fixture for Operator; change Active incidents and make Available units unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open priority incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O02 — Live situation map

EN: Live situation map · **JA:** 状況マップ
Requirement: FR-O02 / WF-20 · **Actor:** Operator · **Surface:** Command · **Layout:** map

Purpose: See incidents across your authorized region.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Open incident → O04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Fire	Building A · P1	Read-only within object scope; display source/time when they affect decisions.
Flood	River district · P2	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Road collision	Harbor Road · P1	Read-only within object scope; display source/time when they affect decisions.
Map freshness	12 seconds	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O04 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O02 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For live situation map, what do Fire, Flood establish, and is “Open incident” available to Operator in this scope?

Primary visual: Multi-incident geographic map with event freshness by layer

Primary action: Open incident → O04

Progressive disclosure: Keep Fire, Flood in the decision layer. Put the supporting details for Road collision, Map freshness, full source/version history and FR-O02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a live situation map fixture for Operator; change Fire and make Map freshness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O03 — Signal triage queue

EN: Signal triage queue · **JA:** 信号の確認待ち
Requirement: FR-O03 / WF-03 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Separate evidence from assumptions.
Key rule: Preserve source, time and ID; duplicates do not create new effects.
Entry: S14.

Output: Inspect signal → O05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
EV-3101	Camera flame · confidence 0.82	Read-only within object scope; display source/time when they affect decisions.
EV-3102	Smoke sensor · threshold exceeded	Read-only within object scope; display source/time when they affect decisions.
EV-3103	Wearable fall · unverified	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O05 with a valid payload/version, and verify that it is blocked when the guard for WF-03 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O03 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For signal triage queue, what do EV-3101, EV-3102 establish, and is “Inspect signal” available to Operator in this scope?

Primary visual: Signal evidence queue showing duplicate suspicion and source quality

Primary action: Inspect signal → O05

Progressive disclosure: Keep EV-3101, EV-3102 in the decision layer. Put the supporting details for EV-3103, full source/version history and FR-O03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a signal triage queue fixture for Operator; change EV-3101 and make EV-3103 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect signal” reaches O05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O04 — Incident workspace

EN: Incident workspace · **JA:** インシデント対応
Requirement: FR-O04 / WF-20 · **Actor:** Operator · **Surface:** Command · **Layout:** incident

Purpose: Fire · Building A / Floor 05.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: O01, O02, O06, O07, O32.
Output: Open dispatch → O08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
INC-2041	P1 · response active	Read-only within object scope; display source/time when they affect decisions.
Open SOS	1	Read-only within object scope; display source/time when they affect decisions.
Self-reported safe	20	Read-only within object scope; display source/time when they affect decisions.
Unconfirmed people	3	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O08 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O04 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For incident workspace, what do INC-2041, Open SOS establish, and is “Open dispatch” available to Operator in this scope?

Primary visual: Incident map beside open SOS, unconfirmed people and action ownership

Primary action: Open dispatch → O08

Progressive disclosure: Keep INC-2041, Open SOS in the decision layer. Put the supporting details for Self-reported safe, Unconfirmed people, full source/version history and FR-O04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a incident workspace fixture for Operator; change INC-2041 and make Unconfirmed people unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Open dispatch" reaches O08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O05 — Signal evidence review

EN: Signal evidence review · **JA:** 証拠の確認
Requirement: FR-O05 / WF-04 · **Actor:** Operator · **Surface:** Command · **Layout:** evidence

Purpose: Camera and sensor evidence.
Key rule: Detection evidence is not the same as a verified incident.
Entry: O03, S09.
Output: Create incident from signal → O06.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Source	CAM-A05-03	Read-only within object scope; display source/time when they affect decisions.
Observed	10:40:12 UTC+02	Read-only within object scope; display source/time when they affect decisions.
Confidence	0.82 · model v3	Read-only within object scope; display source/time when they affect decisions.
Location	Building A / Floor 05	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O06 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O05 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For signal evidence review, what do Source, Observed establish, and is "Create incident from signal" available to Operator in this scope?

Primary visual: Evidence frame with observed time, confidence and mapped location

Primary action: Create incident from signal → O06

Progressive disclosure: Keep Source, Observed in the decision layer. Put the supporting details for Confidence, Location, full source/version history and FR-O05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a signal evidence review fixture for Operator; change Source and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Create incident from signal" reaches O06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O06 — Create incident

EN: Create incident · **JA:** インシデントの作成
Requirement: FR-O06 / WF-04 · **Actor:** Operator · **Surface:** Command · **Layout:** form

Purpose: Establish the incident record.
Key rule: Detection evidence is not the same as a verified incident.
Entry: O05, O29.
Output: Create demo incident → O04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Category	Fire	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Severity	P1	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location	Building A / Floor 05	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Evidence	EV-3101	Read-only from the authoritative source/version; cannot be edited in this form.
Owner	Campus command	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-O06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O04 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O06 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For create incident, what do Category, Severity establish, and is “Create demo incident” available to Operator in this scope?

Primary visual: Incident creation preview linking category, severity and owner

Primary action: Create demo incident → O04

Progressive disclosure: Keep Category, Severity in the decision layer. Put the supporting details for Location, Evidence, Owner, full source/version history and FR-O06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a create incident fixture for Operator; change Category and make Owner unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Create demo incident” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O07 — Correlate or split incidents

EN: Correlate or split incidents · **JA:** インシデントの関連付け・分割
Requirement: FR-O07 / WF-04 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Review possible duplicate reports.
Key rule: Detection evidence is not the same as a verified incident.
Entry: Role-based menu / deep link after authorization.
Output: Save correlation → O04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Primary	INC-2041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Related	INC-2042	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Evidence	Same zone within 60 seconds	Read-only from the authoritative source/version; cannot be edited in this form.

Field / component	Sample value	Validation / editing permission
Decision	Link without deleting source	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O04 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O07 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For correlate or split incidents, what do Primary, Related establish, and is “Save correlation” available to Supervisor in this scope?

Primary visual: Side-by-side incident lineage with merge/split impact

Primary action: Save correlation → O04

Progressive disclosure: Keep Primary, Related in the decision layer. Put the supporting details for Evidence, Decision, full source/version history and FR-O07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a correlate or split incidents fixture for Supervisor; change Primary and make Decision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save correlation” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O08 — Dispatch board

EN: Dispatch board · **JA:** 配備ボード
Requirement: FR-O08 / WF-11 · **Actor:** Dispatcher · **Surface:** Command · **Layout:** dispatch

Purpose: Match the task to a capable team.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: O04.

Output: Review assignment → O09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Rescue 02	Available · fire capable	Read-only within object scope; display source/time when they affect decisions.
Medical 01	Assigned · care capable	Read-only within object scope; display source/time when they affect decisions.
Security 03	Available · perimeter	Read-only within object scope; display source/time when they affect decisions.
Task	Assist SOS-041	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O09 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O08 — V2 visual composition

Context: command · **Role:** Dispatcher

Primary question: For dispatch board, what do Rescue 02, Medical 01 establish, and is “Review assignment” available to Dispatcher in this scope?

Primary visual: Unit lanes showing capabilities, tasks and acceptance state

Primary action: Review assignment → O09

Progressive disclosure: Keep Rescue 02, Medical 01 in the decision layer. Put the supporting details for Security 03, Task, full source/version history and FR-O08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a dispatch board fixture for Dispatcher; change Rescue 02 and make Task unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review assignment” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O09 — Assign responder

EN: Assign responder · **JA:** 対応チームの割り当て
Requirement: FR-O09 / WF-11 · **Actor:** Dispatcher · **Surface:** Command · **Layout:** form

Purpose: Confirm ownership of the task.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: O08, O12, O34, H09.
Output: Assign team — demo → O10.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Incident	INC-2041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Unit	Rescue 02	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Task	Assist person in Room 505	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Acceptance deadline	60 seconds	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O10 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O09 — V2 visual composition

Context: command · **Role:** Dispatcher

Primary question: For assign responder, what do Incident, Unit establish, and is “Assign team — demo” available to Dispatcher in this scope?

Primary visual: Assignment preview binding exact unit, task and response deadline

Primary action: Assign team — demo → O10

Progressive disclosure: Keep Incident, Unit in the decision layer. Put the supporting details for Task, Acceptance deadline, full source/version history and FR-O09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a assign responder fixture for Dispatcher; change Incident and make Acceptance deadline unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign team — demo” reaches O10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O10 — Response timeline

EN: Response timeline · **JA:** 対応タイムライン
Requirement: FR-O10 / WF-11 · **Actor:** Operator · **Surface:** Command · **Layout:** timeline

Purpose: A shared record of decisions.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: O09.
Output: Review people status → O11.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
10:40:12	Signal received	Read-only within object scope; display source/time when they affect decisions.
10:40:19	Incident confirmed	Read-only within object scope; display source/time when they affect decisions.
10:42:18	SOS acknowledged	Read-only within object scope; display source/time when they affect decisions.
10:43:02	Rescue 02 assigned	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O11 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O10 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For response timeline, what do 10:40:12, 10:40:19 establish, and is “Review people status” available to Operator in this scope?

Primary visual: Chronological response milestones with source-specific receipts

Primary action: Review people status → O11

Progressive disclosure: Keep 10:40:12, 10:40:19 in the decision layer. Put the supporting details for 10:42:18, 10:43:02, full source/version history and FR-O10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a response timeline fixture for Operator; change 10:40:12 and make 10:43:02 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review people status” reaches O11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O11 — People accountability

EN: People accountability · **JA:** 安否確認一覧
Requirement: FR-O11 / WF-12 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Know what is reported and what is unknown.
Key rule: A personal safety report does not close the incident.
Entry: O10.
Output: Open person → O12.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	SOS · Room 505 · 10:42	Read-only within object scope; display source/time when they affect decisions.
Jamie	Self-reported safe · 10:45	Read-only within object scope; display source/time when they affect decisions.
Morgan	Unknown · last seen 09:50	Read-only within object scope; display source/time when they affect decisions.
Scope	24 registered occupants	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O12 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O11 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For people accountability, what do Alex, Jamie establish, and is “Open person” available to Operator in this scope?

Primary visual: People board separating unconfirmed, assisted and self-reported safe

Primary action: Open person → O12

Progressive disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Scope, full source/version history and FR-O11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a people accountability fixture for Operator; change Alex and make Scope unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open person” reaches O12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O12 — Person assistance detail

EN: Person assistance detail · **JA:** 支援要請の詳細
Requirement: FR-O12 / WF-12 · **Actor:** Operator · **Surface:** Command · **Layout:** detail

Purpose: Alex Morgan · SOS-041.
Key rule: A personal safety report does not close the incident.
Entry: O11.
Output: Assign assistance → O09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Location	User-submitted room 505	Read-only within object scope; display source/time when they affect decisions.
Consent	Response team only	Read-only within object scope; display source/time when they affect decisions.
Status	Assistance requested	Read-only within object scope; display source/time when they affect decisions.
Freshness	3 minutes old	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O09 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O12 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For person assistance detail, what do Location, Consent establish, and is “Assign assistance” available to Operator in this scope?

Primary visual: Assistance need, consent scope and last-known location

Primary action: Assign assistance → O09

Progressive disclosure: Keep Location, Consent in the decision layer. Put the supporting details for Status, Freshness, full source/version history and FR-O12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a person assistance detail fixture for Operator; change Location and make Freshness unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign assistance” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O13 — Compose warning

EN: Compose warning · **JA:** 警報の作成
Requirement: FR-O13 / WF-09 · **Actor:** Warning author · **Surface:** Command · **Layout:** form

Purpose: Prepare a targeted warning.
Key rule: Approval binds to an exact content version and recipient scope.
Entry: Role-based menu / deep link after authorization.
Output: Preview audience → O14.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Category	Fire	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Field / component	Sample value	Validation / editing permission
Area	Building A	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Language	English	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Instruction	Follow approved evacuation plan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Expiry	11:30 local	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O13: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O14 with a valid payload/version, and verify that it is blocked when the guard for WF-09 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O13 — V2 visual composition

Context: command · **Role:** Warning author

Primary question: For compose warning, what do Category, Area establish, and is “Preview audience” available to Warning author in this scope?

Primary visual: Warning composer beside affected-area and instruction preview

Primary action: Preview audience → O14

Progressive disclosure: Keep Category, Area in the decision layer. Put the supporting details for Language, Instruction, Expiry, full source/version history and FR-O13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a compose warning fixture for Warning author; change Category and make Expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview audience” reaches O14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O14 — Warning preview & audience

EN: Warning preview & audience · **JA:** 内容と対象者のプレビュー
Requirement: FR-O14 / WF-09 · **Actor:** Warning author · **Surface:** Command · **Layout:** detail

Purpose: Review message and recipients.
Key rule: Approval binds to an exact content version and recipient scope.
Entry: O13, T06.
Output: Submit for approval → O15.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Audience	480 eligible subscriptions	Read-only within object scope; display source/time when they affect decisions.
Channels	Push · SMS · PA	Read-only within object scope; display source/time when they affect decisions.
Translations	English approved · Japanese pending	Read-only within object scope; display source/time when they affect decisions.
Exclusions	Outside Building A	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O14: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O15 with a valid payload/version, and verify that it is blocked when the guard for WF-09 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O14 — V2 visual composition

Context: command · **Role:** Warning author

Primary question: For warning preview & audience, what do Audience, Channels establish, and is “Submit for approval” available to Warning author in this scope?

Primary visual: Audience footprint and channel/language coverage comparison

Primary action: Submit for approval → O15

Progressive disclosure: Keep Audience, Channels in the decision layer. Put the supporting details for Translations, Exclusions, full source/version history and FR-O14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a warning preview & audience fixture for Warning author; change Audience and make Exclusions unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Submit for approval" reaches O15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O15 — Approve warning

EN: Approve warning · **JA:** 警報の承認
Requirement: FR-O15 / WF-09 · **Actor:** Warning approver · **Surface:** Command · **Layout:** form

Purpose: Independent approval before broadcast.
Key rule: Approval binds to an exact content version and recipient scope.
Entry: O14.
Output: Approve demo warning → O16.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Author	Operator A	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Approver	Supervisor B	Read-only from the authoritative source/version; cannot be edited in this form.
Version	3	Read-only from the authoritative source/version; cannot be edited in this form.
Reason	Verified incident in occupied zone	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O15: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O16 with a valid payload/version, and verify that it is blocked when the guard for WF-09 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O15 — V2 visual composition

Context: command · **Role:** Warning approver

Primary question: For approve warning, what do Author, Approver establish, and is “Approve demo warning” available to Warning approver in this scope?

Primary visual: Exact warning version and author/approver separation

Primary action: Approve demo warning → O16

Progressive disclosure: Keep Author, Approver in the decision layer. Put the supporting details for Version, Reason, full source/version history and FR-O15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a approve warning fixture for Warning approver; change Author and make Reason unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve demo warning” reaches O16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O16 — Delivery & escalation

EN: Delivery & escalation · **JA:** 配信とエスカレーション
Requirement: FR-O16 / WF-10 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Sending does not mean received.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: O15, O17.
Output: Review failed deliveries → O17.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Push	402 delivered · 78 unknown	Read-only within object scope; display source/time when they affect decisions.
SMS	60 delivered · 18 pending	Read-only within object scope; display source/time when they affect decisions.
Human acknowledgments	211	Read-only within object scope; display source/time when they affect decisions.
Next escalation	10:46 local	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O16: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O17 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O16 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For delivery & escalation, what do Push, SMS establish, and is “Review failed deliveries” available to Operator in this scope?

Primary visual: Delivery funnel separating provider receipt and human acknowledgment

Primary action: Review failed deliveries → O17

Progressive disclosure: Keep Push, SMS in the decision layer. Put the supporting details for Human acknowledgments, Next escalation, full source/version history and FR-O16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a delivery & escalation fixture for Operator; change Push and make Next escalation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review failed deliveries” reaches O17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O17 — Delivery failure review

EN: Delivery failure review · **JA:** 配信失敗への対応
Requirement: FR-O17 / WF-10 · **Actor:** Operator · **Surface:** Command · **Layout:** detail

Purpose: Resolve notification gaps.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: O16.
Output: Queue fallback — demo → O16.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Provider	SMS provider B	Read-only within object scope; display source/time when they affect decisions.
Failure	Rate limit · retryable	Read-only within object scope; display source/time when they affect decisions.
Affected	18 recipients	Read-only within object scope; display source/time when they affect decisions.
Fallback	Voice provider C	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O17: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O16 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O17 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For delivery failure review, what do Provider, Failure establish, and is “Queue fallback — demo” available to Operator in this scope?

Primary visual: Failed-recipient cohort beside fallback-channel proposal

Primary action: Queue fallback — demo → O16

Progressive disclosure: Keep Provider, Failure in the decision layer. Put the supporting details for Affected, Fallback, full source/version history and FR-O17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a delivery failure review fixture for Operator; change Provider and make Fallback unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Queue fallback — demo” reaches O16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O18 — Physical command approval

EN: Physical command approval · **JA:** 機器コマンドの承認
Requirement: FR-O18 / WF-15 · **Actor:** Authorized controller · **Surface:** Command · **Layout:** form

Purpose: Review a proposed equipment action.
Key rule: Accepted commands are not completed actions; enforce TTL and interlocks.
Entry: Role-based menu / deep link after authorization.
Output: Approve simulated command → O19.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Target	PA controller · zone A05	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Command	Play approved evacuation message	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Interlocks	Test passed	Read-only from the authoritative source/version; cannot be edited in this form.
Expires	30 seconds after approval	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O18: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O19 with a valid payload/version, and verify that it is blocked when the guard for WF-15 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O18 — V2 visual composition

Context: command · **Role:** Authorized controller

Primary question: For physical command approval, what do Target, Command establish, and is “Approve simulated command” available to Authorized controller in this scope?

Primary visual: Equipment target schematic with interlocks and expiring command digest

Primary action: Approve simulated command → O19

Progressive disclosure: Keep Target, Command in the decision layer. Put the supporting details for Interlocks, Expires, full source/version history and FR-O18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Expired command, failed interlock or changed target invalidates approval; require review of the exact command version.

V2 acceptance scenario: Use a physical command approval fixture for Authorized controller; change Target and make Expires unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve simulated command” reaches O19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O19 — Physical command status

EN: Physical command status · **JA:** コマンドの実行状況
Requirement: FR-O19 / WF-15 · **Actor:** Authorized controller · **Surface:** Command · **Layout:** timeline

Purpose: Track controller evidence.
Key rule: Accepted commands are not completed actions; enforce TTL and interlocks.
Entry: O18, O19.
Output: Refresh command evidence → O19.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
10:44:00	Requested	Read-only within object scope; display source/time when they affect decisions.
10:44:03	Approved	Read-only within object scope; display source/time when they affect decisions.
10:44:04	Gateway accepted	Read-only within object scope; display source/time when they affect decisions.
Current	Completion not yet confirmed	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O19: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O19 with a valid payload/version, and verify that it is blocked when the guard for WF-15 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O19 — V2 visual composition

Context: command · **Role:** Authorized controller

Primary question: For physical command status, what do 10:44:00, 10:44:03 establish, and is “Refresh command evidence” available to Authorized controller in this scope?

Primary visual: Command timeline separating transport receipt and physical effect

Primary action: Refresh command evidence → O19

Progressive disclosure: Keep 10:44:00, 10:44:03 in the decision layer. Put the supporting details for 10:44:04, Current, full source/version history and FR-O19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Controller acknowledgment without physical feedback remains unknown; show reconciliation rather than repeat-command success.

V2 acceptance scenario: Use a physical command status fixture for Authorized controller; change 10:44:00 and make Current unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Refresh command evidence" reaches O19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O20 — Incident closure review

EN: Incident closure review · **JA:** 終結前の確認
Requirement: FR-O20 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Resolve outstanding obligations.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Close demo incident → O21.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Open SOS	0	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Unconfirmed people	3 · handover documented	Integer within policy limits; unknown is a separate value, not coerced to 0.
Units	Released	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reason	Transferred to authority	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Approver	Supervisor	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O20: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O21 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O20 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For incident closure review, what do Open SOS, Unconfirmed people establish, and is “Close demo incident” available to Supervisor in this scope?

Primary visual: Closure blockers highlighting open SOS and unresolved people

Primary action: Close demo incident → O21

Progressive disclosure: Keep Open SOS, Unconfirmed people in the decision layer. Put the supporting details for Units, Reason, Approver, full source/version history and FR-O20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Open SOS and unresolved people remain visible blockers or require explicit governed disposition; closing the view is not incident closure.

V2 acceptance scenario: Use a incident closure review fixture for Supervisor; change Open SOS and make Approver unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Close demo incident” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O21 — After-action review

EN: After-action review · **JA:** 対応後の振り返り
Requirement: FR-O21 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** detail

Purpose: Learn from the response.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: O20, O22, O24.
Output: Create improvement action → O22.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
First acknowledgment	7 seconds	Read-only within object scope; display source/time when they affect decisions.
Assignment acceptance	42 seconds	Read-only within object scope; display source/time when they affect decisions.
Delivery gaps	18 SMS attempts	Read-only within object scope; display source/time when they affect decisions.
Improvement	Review provider capacity	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-O21: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O22 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O21 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For after-action review, what do First acknowledgment, Assignment acceptance establish, and is "Create improvement action" available to Supervisor in this scope?

Primary visual: Response interval chart with delivery gaps and improvement owners

Primary action: Create improvement action → O22

Progressive disclosure: Keep First acknowledgment, Assignment acceptance in the decision layer. Put the supporting details for Delivery gaps, Improvement, full source/version history and FR-O21 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a after-action review fixture for Supervisor; change First acknowledgment and make Improvement unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Create improvement action" reaches O22 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O22 — Improvement action

EN: Improvement action · **JA:** 改善アクション
Requirement: FR-O22 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Give follow-up work an owner.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: O21.
Output: Save action → O21.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Action	Increase tested SMS capacity	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Owner	Platform operations	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Due date	2026-11-06	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Field / component	Sample value	Validation / editing permission
Linked incident	INC-2041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O22: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O21 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O22 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For improvement action, what do Action, Owner establish, and is “Save action” available to Supervisor in this scope?

Primary visual: Improvement task with accountable owner and due-date progress

Primary action: Save action → O21

Progressive disclosure: Keep Action, Owner in the decision layer. Put the supporting details for Due date, Linked incident, full source/version history and FR-O22 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a improvement action fixture for Supervisor; change Action and make Linked incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save action” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O23 — Drill scheduler

EN: Drill scheduler · **JA:** 訓練の計画
Requirement: FR-O23 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Practice without contacting live recipients.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.

Output: Open drill control → O24.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Scenario	Building A evacuation	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Environment	Exercise only	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Participants	Test cohort	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Start	2026-11-02 09:00 local	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O23: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O24 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O23 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For drill scheduler, what do Scenario, Environment establish, and is “Open drill control” available to Supervisor in this scope?

Primary visual: Drill scenario schedule clearly separated from live environment

Primary action: Open drill control → O24

Progressive disclosure: Keep Scenario, Environment in the decision layer. Put the supporting details for Participants, Start, full source/version history and FR-O23 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a drill scheduler fixture for Supervisor; change Scenario and make Start unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open drill control” reaches O24 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately

from this specification.

O24 — Drill control room

EN: Drill control room · **JA:** 訓練の指揮
Requirement: FR-O24 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** dashboard

Purpose: EXERCISE · Building A evacuation.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: O23.
Output: Review drill report → O21.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Mode	Exercise	Read-only within object scope; display source/time when they affect decisions.
Live actuators	Disconnected	Read-only within object scope; display source/time when they affect decisions.
Test recipients	24	Read-only within object scope; display source/time when they affect decisions.
Completed steps	3 of 5	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O24: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O21 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O24 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For drill control room, what do Mode, Live actuators establish, and is “Review drill report” available to Supervisor in this scope?

Primary visual: Drill progress board with explicit actuator isolation state

Primary action: Review drill report → O21

Progressive disclosure: Keep Mode, Live actuators in the decision layer. Put the supporting details for Test recipients, Completed steps, full source/version history and FR-O24 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a drill control room fixture for Supervisor; change Mode and make Completed steps unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review drill report” reaches O21 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O25 — Shift handover

EN: Shift handover · **JA:** 勤務の引き継ぎ
Requirement: FR-O25 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Transfer active operational responsibility.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Record handover → O01.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Outgoing	Operator A	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Incoming	Operator B	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Active incidents	12	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Outstanding actions	4	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Acknowledgment	Required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O25: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O01 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O25 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For shift handover, what do Outgoing, Incoming establish, and is “Record handover” available to Supervisor in this scope?

Primary visual: Outgoing/incoming responsibility transfer with unresolved actions

Primary action: Record handover → O01

Progressive disclosure: Keep Outgoing, Incoming in the decision layer. Put the supporting details for Open incidents, Outstanding actions, Acknowledgment, full source/version history and FR-O25 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a shift handover fixture for Supervisor; change Outgoing and make Acknowledgment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record handover” reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O26 — Shelter operations

EN: Shelter operations · **JA:** 避難所の運営
Requirement: FR-O26 / WF-13 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Publish fresh shelter information.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: O27.
Output: Edit shelter → O27.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
North shelter	Open · 80 spaces	Read-only within object scope; display source/time when they affect decisions.
West center	Status unverified	Read-only within object scope; display source/time when they affect decisions.
South shelter	Closed	Read-only within object scope; display source/time when they affect decisions.
Last refresh	10:40 local	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O26: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O27 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O26 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For shelter operations, what do North shelter, West center establish, and is “Edit shelter” available to Operator in this scope?

Primary visual: Shelter capacity board including unknown and expired reports

Primary action: Edit shelter → O27

Progressive disclosure: Keep North shelter, West center in the decision layer. Put the supporting details for South shelter, Last refresh, full source/version history and FR-O26 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a shelter operations fixture for Operator; change North shelter and make Last refresh unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit shelter” reaches O27 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O27 — Edit shelter status

EN: Edit shelter status · **JA:** 避難所情報の更新
Requirement: FR-O27 / WF-13 · **Actor:** Operator · **Surface:** Command · **Layout:** form

Purpose: Update a designated shelter.
Key rule: Guidance and destinations expose source, version and freshness.
Entry: O26.
Output: Publish demo update → O26.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Shelter	North	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Status	Open	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Available capacity	80	Integer within policy limits; unknown is a separate value, not coerced to 0.
Accessibility	Step-free	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Valid until	12:00 local	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O27: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O26 with a valid payload/version, and verify that it is blocked when the guard for WF-13 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O27 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For edit shelter status, what do Shelter, Status establish, and is “Publish demo update” available to Operator in this scope?

Primary visual: Shelter status editor beside capacity and validity preview

Primary action: Publish demo update → O26

Progressive disclosure: Keep Shelter, Status in the decision layer. Put the supporting details for Available capacity, Accessibility, Valid until, full source/version history and FR-O27 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a edit shelter status fixture for Operator; change Shelter and make Valid until unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish demo update” reaches O26 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O28 — Regional disaster feed

EN: Regional disaster feed · **JA:** 災害警報フィード
Requirement: FR-O28 / WF-08 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Review trusted public advisories.
Key rule: Preserve issuer, validity and warning update or cancellation history.
Entry: Role-based menu / deep link after authorization.
Output: Inspect advisory → O29.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
CAP-041	Flood watch · Update	Read-only within object scope; display source/time when they affect decisions.
CAP-039	Wind alert · expired	Read-only within object scope; display source/time when they affect decisions.
CAP-038	Storm warning · canceled	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O28: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O29 with a valid payload/version, and verify that it is blocked when the guard for WF-08 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O28 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For regional disaster feed, what do CAP-041, CAP-039 establish, and is “Inspect advisory” available to Operator in this scope?

Primary visual: Regional advisory queue with source identity and supersession

Primary action: Inspect advisory → O29

Progressive disclosure: Keep CAP-041, CAP-039 in the decision layer. Put the supporting details for CAP-038, full source/version history and FR-O28 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a regional disaster feed fixture for Operator; change CAP-041 and make CAP-038 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect advisory” reaches O29 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O29 — CAP advisory review

EN: CAP advisory review · **JA:** CAP情報の確認
Requirement: FR-O29 / WF-08 · **Actor:** Operator · **Surface:** Command · **Layout:** detail

Purpose: Preserve the issuing authority's context.
Key rule: Preserve issuer, validity and warning update or cancellation history.
Entry: O28.
Output: Create local response → O06.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Sender	Configured authority	Read-only within object scope; display source/time when they affect decisions.
Identifier	CAP-041	Read-only within object scope; display source/time when they affect decisions.
Message type	Update	Read-only within object scope; display source/time when they affect decisions.
References	CAP-040	Read-only within object scope; display source/time when they affect decisions.
Expires	18:00 local	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O29: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O06 with a valid payload/version, and verify that it is blocked when the guard for WF-08 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O29 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For cap advisory review, what do Sender, Identifier establish, and is “Create local response” available to Operator in this scope?

Primary visual: CAP message lineage with update/cancel references and expiry

Primary action: Create local response → O06

Progressive disclosure: Keep Sender, Identifier in the decision layer. Put the supporting details for Message type, References, Expires, full source/version history and FR-O29 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a cap advisory review fixture for Operator; change Sender and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Create local response" reaches O06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O30 — Search & audit timeline

EN: Search & audit timeline · **JA:** 検索と監査履歴
Requirement: FR-O30 / WF-19 · **Actor:** Auditor · **Surface:** Command · **Layout:** list

Purpose: Find an event without changing history.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: T11, X07.
Output: Inspect audit entry → T11.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Query	INC-2041	Read-only within object scope; display source/time when they affect decisions.
Create	Operator A · 10:40	Read-only within object scope; display source/time when they affect decisions.
Dispatch	Dispatcher B · 10:43	Read-only within object scope; display source/time when they affect decisions.
Export	Auditor C · 11:20	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O30: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T11 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O30 — V2 visual composition

Context: command · **Role:** Auditor

Primary question: For search & audit timeline, what do Query, Create establish, and is “Inspect audit entry” available to Auditor in this scope?

Primary visual: Audit timeline with actor, scope and immutable event details

Primary action: Inspect audit entry → T11

Progressive disclosure: Keep Query, Create in the decision layer. Put the supporting details for Dispatch, Export, full source/version history and FR-O30 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a search & audit timeline fixture for Auditor; change Query and make Export unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect audit entry” reaches T11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S01 — Site directory

EN: Site directory · **JA:** 拠点一覧
Requirement: FR-S01 / WF-02 · **Actor:** Site admin · **Surface:** Sites · **Layout:** cards

Purpose: One location model across industries.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: Role-based menu / deep link after authorization.
Output: Open site → S02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Harbor Campus	Buildings · EU	Read-only within object scope; display source/time when they affect decisions.
Riverside	Geofence · EU	Read-only within object scope; display source/time when they affect decisions.
North highway	Road segment · EU	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S02 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S01 — V2 visual composition

Context: configuration · **Role:** Site admin

Primary question: For site directory, what do Harbor Campus, Riverside establish, and is “Open site” available to Site admin in this scope?

Primary visual: Site map and directory with region boundary

Primary action: Open site → S02

Progressive disclosure: Keep Harbor Campus, Riverside in the decision layer. Put the supporting details for North highway, full source/version history and FR-S01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a site directory fixture for Site admin; change Harbor Campus and make North highway unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Open site” reaches S02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S02 — Site structure

EN: Site structure · **JA:** 拠点の構造
Requirement: FR-S02 / WF-02 · **Actor:** Site admin · **Surface:** Sites · **Layout:** tree

Purpose: Harbor Campus / Building A.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: S01, S03.
Output: Edit location → S03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Building A	Floors 01–08	Read-only within object scope; display source/time when they affect decisions.
Floor 05	Zones A05-01 to A05-06	Read-only within object scope; display source/time when they affect decisions.
Zone A05-03	Camera + smoke detector	Read-only within object scope; display source/time when they affect decisions.
Assembly	North court	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S03 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S02 — V2 visual composition

Context: configuration · **Role:** Site admin

Primary question: For site structure, what do Building A, Floor 05 establish, and is “Edit location” available to Site admin in this scope?

Primary visual: Building-floor-zone tree with selected spatial context

Primary action: Edit location → S03

Progressive disclosure: Keep Building A, Floor 05 in the decision layer. Put the supporting details for Zone A05-03, Assembly, full source/version history and FR-S02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a site structure fixture for Site admin; change Building A and make Assembly unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit location” reaches S03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S03 — Location editor

EN: Location editor · **JA:** 位置情報の編集
Requirement: FR-S03 / WF-02 · **Actor:** Site admin · **Surface:** Sites · **Layout:** form

Purpose: Link physical and geographic context.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: S02.
Output: Save location → S02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Name	East corridor	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Parent	Floor 05	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Coordinates	52.5200 / 13.4050	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Floor reference	05	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Timezone	Europe/Berlin	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S02 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S03 — V2 visual composition

Context: configuration · **Role:** Site admin

Primary question: For location editor, what do Name, Parent establish, and is “Save location” available to Site admin in this scope?

Primary visual: Parent-location path beside coordinate and floor preview

Primary action: Save location → S02

Progressive disclosure: Keep Name, Parent in the decision layer. Put the supporting details for Coordinates, Floor reference, Timezone, full source/version history and FR-S03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a location editor fixture for Site admin; change Name and make Timezone unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save location” reaches S02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S04 — Floor plan editor

EN: Floor plan editor · **JA:** フロア図の編集
Requirement: FR-S04 / WF-02 · **Actor:** Site admin · **Surface:** Sites · **Layout:** floor

Purpose: Publish a reviewed evacuation map.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: S05.
Output: Review plan publication → S05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Plan	Building A / Floor 05	Read-only within object scope; display source/time when they affect decisions.
Revision	13 draft	Read-only within object scope; display source/time when they affect decisions.
Exits	2	Read-only within object scope; display source/time when they affect decisions.
Accessible refuge	Zone B	Read-only within object scope; display source/time when they affect decisions.
Hazard overlay	Operator controlled	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S05 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S04 — V2 visual composition

Context: configuration · **Role:** Site admin

Primary question: For floor plan editor, what do Plan, Revision establish, and is “Review plan publication” available to Site admin in this scope?

Primary visual: Plan canvas showing exits, refuges and revision differences

Primary action: Review plan publication → S05

Progressive disclosure: Keep Plan, Revision in the decision layer. Put the supporting details for Exits, Accessible refuge, Hazard overlay, full source/version history and FR-S04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a floor plan editor fixture for Site admin; change Plan and make Hazard overlay unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review plan publication" reaches S05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S05 — Plan approval & versions

EN: Plan approval & versions · **JA:** 図面の承認と版管理
Requirement: FR-S05 / WF-02 · **Actor:** Supervisor · **Surface:** Sites · **Layout:** list

Purpose: Keep published guidance traceable.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: S04, X06.
Output: Publish reviewed draft — demo → S04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Revision 13	Draft · accessibility review pending	Read-only within object scope; display source/time when they affect decisions.
Revision 12	Published · 2026-09-02	Read-only within object scope; display source/time when they affect decisions.
Revision 11	Archived	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S04 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S05 — V2 visual composition

Context: configuration · **Role:** Supervisor

Primary question: For plan approval & versions, what do Revision 13, Revision 12 establish, and is "Publish reviewed draft — demo" available to Supervisor in this scope?

Primary visual: Plan revision comparison with published/draft state

Primary action: Publish reviewed draft — demo → S04

Progressive disclosure: Keep Revision 13, Revision 12 in the decision layer. Put the supporting details for Revision 11, full source/version history and FR-S05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a plan approval & versions fixture for Supervisor; change Revision 13 and make Revision 11 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and "Publish reviewed draft — demo" reaches S04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S06 — Device inventory

EN: Device inventory · **JA:** 機器一覧
Requirement: FR-S06 / WF-14 · **Actor:** Device admin · **Surface:** Sites · **Layout:** list

Purpose: Know which devices can be trusted.
Key rule: Online status does not prove area safety or physical device health.
Entry: S10, X03.
Output: Open device → S07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
CAM-A05-03	Camera · healthy	Read-only within object scope; display source/time when they affect decisions.
SMK-A05-03	Smoke sensor · healthy	Read-only within object scope; display source/time when they affect decisions.
GW-012	Gateway · stale	Read-only within object scope; display source/time when they affect decisions.
WR-401	Wearable · paired	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S07 with a valid payload/version, and verify that it is blocked when the guard for WF-14 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S06 — V2 visual composition

Context: configuration · **Role:** Device admin

Primary question: For device inventory, what do CAM-A05-03, SMK-A05-03 establish, and is “Open device” available to Device admin in this scope?

Primary visual: Device inventory grouped by physical function and health

Primary action: Open device → S07

Progressive disclosure: Keep CAM-A05-03, SMK-A05-03 in the decision layer. Put the supporting details for GW-012, WR-401, full source/version history and FR-S06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a device inventory fixture for Device admin; change CAM-A05-03 and make WR-401 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open device” reaches S07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S07 — Device detail & health

EN: Device detail & health · **JA:** 機器の詳細と稼働状態
Requirement: FR-S07 / WF-14 · **Actor:** Device admin · **Surface:** Sites · **Layout:** detail

Purpose: CAM-A05-03.
Key rule: Online status does not prove area safety or physical device health.
Entry: S06, S08.
Output: Review device configuration → S08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Certificate	Valid · 48 days remaining	Read-only within object scope; display source/time when they affect decisions.
Last heartbeat	8 seconds ago	Read-only within object scope; display source/time when they affect decisions.
Firmware	3.2 · approved	Read-only within object scope; display source/time when they affect decisions.
Location	Zone A05-03	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-S07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S08 with a valid payload/version, and verify that it is blocked when the guard for WF-14 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S07 — V2 visual composition

Context: configuration · **Role:** Device admin

Primary question: For device detail & health, what do Certificate, Last heartbeat establish, and is “Review device configuration” available to Device admin in this scope?

Primary visual: Device diagram with heartbeat, certificate and firmware evidence

Primary action: Review device configuration → S08

Progressive disclosure: Keep Certificate, Last heartbeat in the decision layer. Put the supporting details for Firmware, Location, full source/version history and FR-S07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a device detail & health fixture for Device admin; change Certificate and make Location unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review device configuration” reaches S08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S08 — Provision device

EN: Provision device · **JA:** 機器の登録
Requirement: FR-S08 / WF-14 · **Actor:** Device admin · **Surface:** Sites · **Layout:** form

Purpose: Bind a device to an authorized tenant.
Key rule: Online status does not prove area safety or physical device health.
Entry: S07.
Output: Enroll simulated device → S07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Serial	CAM-009182	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Enrollment token	Single-use · masked	Read-only from the authoritative source/version; cannot be edited in this form.
Location	Zone A05-03	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.

Field / component	Sample value	Validation / editing permission
Capabilities	Video + detection metadata	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S07 with a valid payload/version, and verify that it is blocked when the guard for WF-14 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S08 — V2 visual composition

Context: configuration · **Role:** Device admin

Primary question: For provision device, what do Serial, Enrollment token establish, and is “Enroll simulated device” available to Device admin in this scope?

Primary visual: Enrollment sequence tying serial, site and capability claim

Primary action: Enroll simulated device → S07

Progressive disclosure: Keep Serial, Enrollment token in the decision layer. Put the supporting details for Location, Capabilities, full source/version history and FR-S08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a provision device fixture for Device admin; change Serial and make Capabilities unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Enroll simulated device” reaches S07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S09 — Camera wall

EN: Camera wall · **JA:** カメラ監視
Requirement: FR-S09 / WF-14 · **Actor:** Operator · **Surface:** Sites · **Layout:** evidence

Purpose: Review selected camera sources.
Key rule: Online status does not prove area safety or physical device health.
Entry: Role-based menu / deep link after authorization.

Output: Open detection evidence → O05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
CAM-A05-01	Lobby · online	Read-only within object scope; display source/time when they affect decisions.
CAM-A05-02	West corridor · online	Read-only within object scope; display source/time when they affect decisions.
CAM-A05-03	East corridor · stale frame	Read-only within object scope; display source/time when they affect decisions.
Access	Incident-scoped	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O05 with a valid payload/version, and verify that it is blocked when the guard for WF-14 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S09 — V2 visual composition

Context: configuration · **Role:** Operator

Primary question: For camera wall, what do CAM-A05-01, CAM-A05-02 establish, and is “Open detection evidence” available to Operator in this scope?

Primary visual: Permission-gated camera tiles with timestamp and missing-feed state

Primary action: Open detection evidence → O05

Progressive disclosure: Keep CAM-A05-01, CAM-A05-02 in the decision layer. Put the supporting details for CAM-A05-03, Access, full source/version history and FR-S09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a camera wall fixture for Operator; change CAM-A05-01 and make Access unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open detection evidence” reaches O05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S10 — Maintenance & calibration

EN: Maintenance & calibration · **JA:** 保守と校正
Requirement: FR-S10 / WF-14 · **Actor:** Device admin · **Surface:** Sites · **Layout:** form

Purpose: Keep suppression visible during maintenance.
Key rule: Online status does not prove area safety or physical device health.
Entry: Role-based menu / deep link after authorization.
Output: Schedule maintenance → S06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Device	SMK-A05-03	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Window	14:00–14:30 local	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.
Backup coverage	Patrol confirmed	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Approver	Site supervisor	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S06 with a valid payload/version, and verify that it is blocked when the guard for WF-14 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S10 — V2 visual composition

Context: configuration · **Role:** Device admin

Primary question: For maintenance & calibration, what do Device, Window establish, and is “Schedule maintenance” available to Device admin in this scope?

Primary visual: Maintenance window alongside backup-coverage map

Primary action: Schedule maintenance → S06

Progressive disclosure: Keep Device, Window in the decision layer. Put the supporting details for Backup coverage, Approver, full source/version history and FR-S10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a maintenance & calibration fixture for Device admin; change Device and make Approver unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Schedule maintenance" reaches S06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S11 — Rule library

EN: Rule library · **JA:** ルールー覧
Requirement: FR-S11 / WF-04 · **Actor:** Rule author · **Surface:** Sites · **Layout:** list

Purpose: Detection policies by hazard and site.
Key rule: Detection evidence is not the same as a verified incident.
Entry: Role-based menu / deep link after authorization.
Output: Edit rule → S12.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Fire rule v4	Published	Read-only within object scope; display source/time when they affect decisions.
Fall rule v2	Published	Read-only within object scope; display source/time when they affect decisions.
Road collision v1	Draft	Read-only within object scope; display source/time when they affect decisions.
Flood threshold v3	Published	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S12 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S11 — V2 visual composition

Context: configuration · **Role:** Rule author

Primary question: For rule library, what do Fire rule v4, Fall rule v2 establish, and is “Edit rule” available to Rule author in this scope?

Primary visual: Versioned rule list showing hazard family and enabled scope

Primary action: Edit rule → S12

Progressive disclosure: Keep Fire rule v4, Fall rule v2 in the decision layer. Put the supporting details for Road collision v1, Flood threshold v3, full source/version history and FR-S11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a rule library fixture for Rule author; change Fire rule v4 and make Flood threshold v3 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit rule” reaches S12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S12 — Rule editor & simulation

EN: Rule editor & simulation · **JA:** ルール編集とシミュレーション
Requirement: FR-S12 / WF-04 · **Actor:** Rule author · **Surface:** Sites · **Layout:** form

Purpose: Test a rule against recorded events.
Key rule: Detection evidence is not the same as a verified incident.
Entry: S11, S13.
Output: View simulation results → S13.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Rule	Fire corroboration	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Condition	Camera event + smoke event	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Window	60 seconds	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.
Action	Create P1 incident	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Mode	Simulation	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S13 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S12 — V2 visual composition

Context: configuration · **Role:** Rule author

Primary question: For rule editor & simulation, what do Rule, Condition establish, and is “View simulation results” available to Rule author in this scope?

Primary visual: Condition-action flow beside simulation-only indicator

Primary action: View simulation results → S13

Progressive disclosure: Keep Rule, Condition in the decision layer. Put the supporting details for Window, Action, Mode, full source/version history and FR-S12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a rule editor & simulation fixture for Rule author; change Rule and make Mode unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “View simulation results” reaches S13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S13 — Rule test & approval

EN: Rule test & approval · **JA:** ルールのテストと承認
Requirement: FR-S13 / WF-04 · **Actor:** Rule approver · **Surface:** Sites · **Layout:** detail

Purpose: Review behavioral changes before release.
Key rule: Detection evidence is not the same as a verified incident.
Entry: S12.
Output: Return rule for changes → S12.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Fixture cases	24 passed · 2 need review	Read-only within object scope; display source/time when they affect decisions.
False merges	0 in test set	Read-only within object scope; display source/time when they affect decisions.
Version	5 draft	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Reviewer	Independent supervisor	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S13: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S12 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S13 — V2 visual composition

Context: configuration · **Role:** Rule approver

Primary question: For rule test & approval, what do Fixture cases, False merges establish, and is “Return rule for changes” available to Rule approver in this scope?

Primary visual: Test-case outcome grid with false-merge evidence

Primary action: Return rule for changes → S12

Progressive disclosure: Keep Fixture cases, False merges in the decision layer. Put the supporting details for Version, Reviewer, full source/version history and FR-S13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a rule test & approval fixture for Rule approver; change Fixture cases and make Reviewer unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Return rule for changes” reaches S12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S14 — AI model monitoring

EN: AI model monitoring · **JA:** AIモデルの監視
Requirement: FR-S14 / WF-04 · **Actor:** Model operator · **Surface:** Sites · **Layout:** detail

Purpose: Track detector behavior by site.
Key rule: Detection evidence is not the same as a verified incident.
Entry: Role-based menu / deep link after authorization.
Output: Open review queue → O03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Model	Flame v3	Read-only within object scope; display source/time when they affect decisions.
Labeled sample	120 events	Read-only within object scope; display source/time when they affect decisions.
False-positive review	8 cases	Read-only within object scope; display source/time when they affect decisions.
Drift	Lighting shift detected	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S14: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O03 with a valid payload/version, and verify that it is blocked when the guard for WF-04 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S14 — V2 visual composition

Context: configuration · **Role:** Model operator

Primary question: For ai model monitoring, what do Model, Labeled sample establish, and is “Open review queue” available to Model operator in this scope?

Primary visual: Model quality trend with sample provenance and drift

Primary action: Open review queue → O03

Progressive disclosure: Keep Model, Labeled sample in the decision layer. Put the supporting details for False-positive review, Drift, full source/version history and FR-S14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a ai model monitoring fixture for Model operator; change Model and make Drift unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open review queue” reaches O03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I01 — Connector catalog

EN: Connector catalog · **JA:** コネクター一覧
Requirement: FR-I01 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** cards

Purpose: Connect systems through declared capabilities.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: D06.
Output: Configure connector → I02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
MQTT sensors	Telemetry	Read-only within object scope; display source/time when they affect decisions.
CAP authority	Public warning	Read-only within object scope; display source/time when they affect decisions.
MAVLink gateway	Drone telemetry	Read-only within object scope; display source/time when they affect decisions.
IOC bridge	Incident exchange	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I02 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I01 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For connector catalog, what do MQTT sensors, CAP authority establish, and is “Configure connector” available to Integration admin in this scope?

Primary visual: Connector catalog grouped by event, advisory and control capability

Primary action: Configure connector → I02

Progressive disclosure: Keep MQTT sensors, CAP authority in the decision layer. Put the supporting details for MAVLink gateway, IOC bridge, full source/version history and FR-I01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a connector catalog fixture for Integration admin; change MQTT sensors and make IOC bridge unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Configure connector" reaches I02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I02 — Connector setup

EN: Connector setup · **JA:** コネクターの設定
Requirement: FR-I02 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** form

Purpose: Create a scoped connection.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I01.
Output: Test connection → I03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Type	MQTT 5	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Endpoint	mqtts://broker.example.test	HTTPS/mTLS profile; allowlist hosts and block loopback/private SSRF under connector policy.
Region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Auth	mTLS	Read-only from the authoritative source/version; cannot be edited in this form.
Capabilities	telemetry.read	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I03 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I02 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For connector setup, what do Type, Endpoint establish, and is “Test connection” available to Integration admin in this scope?

Primary visual: Connector setup flow with endpoint-region-capability preview

Primary action: Test connection → I03

Progressive disclosure: Keep Type, Endpoint in the decision layer. Put the supporting details for Region, Auth, Capabilities, full source/version history and FR-I02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a connector setup fixture for Integration admin; change Type and make Capabilities unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Test connection” reaches I03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I03 — Schema mapping

EN: Schema mapping · **JA:** データマッピング
Requirement: FR-I03 / WF-03 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** form

Purpose: Map source data into a versioned event.
Key rule: Preserve source, time and ID; duplicates do not create new effects.
Entry: I02.
Output: Run contract tests → I04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Source field	temperature_c	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Target	data.measurement.value	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Unit	Cel	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location	device registry binding	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Unknown fields	Quarantine	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I04 with a valid payload/version, and verify that it is blocked when the guard for WF-03 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I03 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For schema mapping, what do Source field, Target establish, and is “Run contract tests” available to Integration admin in this scope?

Primary visual: Source-to-target field mapping with units and unknown-field lane

Primary action: Run contract tests → I04

Progressive disclosure: Keep Source field, Target in the decision layer. Put the supporting details for Unit, Location, Unknown fields, full source/version history and FR-I03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a schema mapping fixture for Integration admin; change Source field and make Unknown fields unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Run contract tests” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I04 — Connector test console

EN: Connector test console · **JA:** 接続テスト
Requirement: FR-I04 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** detail

Purpose: Prove behavior before enabling traffic.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I03, I08, I10.
Output: Enable demo connector → I05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Valid event	Accepted	Read-only within object scope; display source/time when they affect decisions.
Duplicate event	Deduplicated	Read-only within object scope; display source/time when they affect decisions.
Wrong tenant	Rejected	Read-only within object scope; display source/time when they affect decisions.
Expired certificate	Rejected	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I05 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I04 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For connector test console, what do Valid event, Duplicate event establish, and is “Enable demo connector” available to Integration admin in this scope?

Primary visual: Contract-test matrix separating accepted and rejected fixtures

Primary action: Enable demo connector → I05

Progressive disclosure: Keep Valid event, Duplicate event in the decision layer. Put the supporting details for Wrong tenant, Expired certificate, full source/version history and FR-I04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a connector test console fixture for Integration admin; change Valid event and make Expired certificate unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Enable demo connector” reaches I05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I05 — Connector health

EN: Connector health · **JA:** 接続の稼働状態
Requirement: FR-I05 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** list

Purpose: Monitor transport and processing separately.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I04, I06.
Output: Inspect dead letters → I06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
MQTT bridge	Connected · 0 queue lag	Read-only within object scope; display source/time when they affect decisions.
IOC bridge	Degraded · retrying	Read-only within object scope; display source/time when they affect decisions.
CAP feed	Healthy · last poll 30 s	Read-only within object scope; display source/time when they affect decisions.
Drone bridge	Telemetry only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I06 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I05 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For connector health, what do MQTT bridge, IOC bridge establish, and is “Inspect dead letters” available to Integration admin in this scope?

Primary visual: Connector health lanes showing lag, failure and data age

Primary action: Inspect dead letters → I06

Progressive disclosure: Keep MQTT bridge, IOC bridge in the decision layer. Put the supporting details for CAP feed, Drone bridge, full source/version history and FR-I05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a connector health fixture for Integration admin; change MQTT bridge and make Drone bridge unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect dead letters” reaches I06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record

observed results separately from this specification.

I06 — Dead-letter & replay

EN: Dead-letter & replay · **JA:** 失敗イベントと再処理
Requirement: FR-I06 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** form

Purpose: Replay only after inspecting the failure.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I05.
Output: Replay simulated event → I05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Event	EV-3090	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Error	Unknown location mapping	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Correction	Bind device to zone	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Replay	Preserve original event ID	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I05 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I06 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For dead-letter & replay, what do Event, Error establish, and is “Replay simulated event” available to Integration admin in this scope?

Primary visual: Dead-letter payload diff and replay eligibility

Primary action: Replay simulated event → I05

Progressive disclosure: Keep Event, Error in the decision layer. Put the supporting details for Correction, Replay, full source/version history and FR-I06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a dead-letter & replay fixture for Integration admin; change Event and make Replay unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Replay simulated event" reaches I05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I07 — API clients & scopes

EN: API clients & scopes · **JA:** APIクライアントと権限
Requirement: FR-I07 / WF-17 · **Actor:** Tenant admin · **Surface:** Integrations · **Layout:** form

Purpose: Grant only required API access.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: D08.
Output: Save client → I08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Client	Partner IOC	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Scopes	incident.read + incident.ack	Select within granted permissions; the server blocks escalation beyond the actor's rights.
Region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Secret	Stored in vault · not shown	Read-only from the authoritative source/version; cannot be edited in this form.
Expiry	90 days	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I08 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I07 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For api clients & scopes, what do Client, Scopes establish, and is “Save client” available to Tenant admin in this scope?

Primary visual: API scope grid with masked credentials and expiry

Primary action: Save client → I08

Progressive disclosure: Keep Client, Scopes in the decision layer. Put the supporting details for Region, Secret, Expiry, full source/version history and FR-I07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a api clients & scopes fixture for Tenant admin; change Client and make Expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save client” reaches I08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I08 — Webhook subscriptions

EN: Webhook subscriptions · **JA:** Webhookの登録
Requirement: FR-I08 / WF-17 · **Actor:** Integration admin · **Surface:** Integrations · **Layout:** form

Purpose: Deliver filtered incident updates.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I07, D07.
Output: Send test payload → I04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
URL	https://partner.example.test/hooks	HTTPS/mTLS profile; allowlist hosts and block loopback/private SSRF under connector policy.
Events	incident.updated	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Signature	HMAC + timestamp	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Field / component	Sample value	Validation / editing permission
Retries	Exponential · max 24 h	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I04 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I08 — V2 visual composition

Context: configuration · **Role:** Integration admin

Primary question: For webhook subscriptions, what do URL, Events establish, and is “Send test payload” available to Integration admin in this scope?

Primary visual: Webhook route diagram showing signature and retry policy

Primary action: Send test payload → I04

Progressive disclosure: Keep URL, Events in the decision layer. Put the supporting details for Signature, Retries, full source/version history and FR-I08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a webhook subscriptions fixture for Integration admin; change URL and make Retries unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Send test payload” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I09 — Partner exchange agreements

EN: Partner exchange agreements · **JA:** データ共有契約
Requirement: FR-I09 / WF-17 · **Actor:** Tenant admin · **Surface:** Integrations · **Layout:** form

Purpose: Define cross-organization sharing.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: Role-based menu / deep link after authorization.

Output: Save agreement draft → I10.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Partner	City command	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Purpose	Incident response	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Fields	Location + pseudonymous SOS	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Retention	30 days	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Approval	Both organizations required	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I10 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I09 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For partner exchange agreements, what do Partner, Purpose establish, and is “Save agreement draft” available to Tenant admin in this scope?

Primary visual: Partner disclosure boundary with retention and approval state

Primary action: Save agreement draft → I10

Progressive disclosure: Keep Partner, Purpose in the decision layer. Put the supporting details for Fields, Retention, Approval, full source/version history and FR-I09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a partner exchange agreements fixture for Tenant admin; change Partner and make Approval unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Save agreement draft" reaches I10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I10 — IOC exchange history

EN: IOC exchange history · **JA:** IOC連携履歴
Requirement: FR-I10 / WF-17 · **Actor:** Operator · **Surface:** Integrations · **Layout:** list

Purpose: Verify what a partner actually received.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: I09, O33.
Output: Inspect payload → I04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
INC-2041	Sent · application ACK received	Read-only within object scope; display source/time when they affect decisions.
SOS-041	Sent · ACK pending	Read-only within object scope; display source/time when they affect decisions.
Update 4	Rejected · schema mismatch	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I04 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I10 — V2 visual composition

Context: configuration · **Role:** Operator

Primary question: For ioc exchange history, what do INC-2041, SOS-041 establish, and is "Inspect payload" available to Operator in this scope?

Primary visual: Exchange timeline with payload scope and receipt identity

Primary action: Inspect payload → I04

Progressive disclosure: Keep INC-2041, SOS-041 in the decision layer. Put the supporting details for Update 4, full source/version history and FR-I10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a ioc exchange history fixture for Operator; change INC-2041 and make Update 4 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Inspect payload” reaches I04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I11 — Drone fleet

EN: Drone fleet · **JA:** ドローン一覧
Requirement: FR-I11 / WF-16 · **Actor:** Drone operator · **Surface:** Integrations · **Layout:** cards

Purpose: Mission readiness at a glance.
Key rule: The local pilot and controller retain operational authority.
Entry: I15.
Output: Plan mission → I12.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
DR-01	Ready · battery 92%	Read-only within object scope; display source/time when they affect decisions.
DR-02	Maintenance	Read-only within object scope; display source/time when they affect decisions.
DR-03	Offline	Read-only within object scope; display source/time when they affect decisions.
Control	Certified local operator	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I12 with a valid payload/version, and verify that it is blocked when the guard for WF-16 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I11 — V2 visual composition

Context: configuration · **Role:** Drone operator

Primary question: For drone fleet, what do DR-01, DR-02 establish, and is “Plan mission” available to Drone operator in this scope?

Primary visual: Fleet readiness cards showing battery, availability and pilot control

Primary action: Plan mission → I12

Progressive disclosure: Keep DR-01, DR-02 in the decision layer. Put the supporting details for DR-03, Control, full source/version history and FR-I11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a drone fleet fixture for Drone operator; change DR-01 and make Control unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Plan mission” reaches I12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I12 — Drone mission planning

EN: Drone mission planning · **JA:** ドローン任務の計画
Requirement: FR-I12 / WF-16 · **Actor:** Drone operator · **Surface:** Integrations · **Layout:** map

Purpose: Observe the perimeter of Building A.
Key rule: The local pilot and controller retain operational authority.
Entry: I11.
Output: Review mission approval → I13.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Purpose	Situation assessment	Read-only within object scope; display source/time when they affect decisions.
Geofence	Campus perimeter	Read-only within object scope; display source/time when they affect decisions.
Pilot	Assigned	Read-only within object scope; display source/time when they affect decisions.
Weather	Requires local verification	Read-only within object scope; display source/time when they affect decisions.
Authority	Pending clearance	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I13 with a valid payload/version, and verify that it is blocked when the guard for WF-16 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I12 — V2 visual composition

Context: configuration · **Role:** Drone operator

Primary question: For drone mission planning, what do Purpose, Geofence establish, and is "Review mission approval" available to Drone operator in this scope?

Primary visual: Mission geofence with pilot, weather and authority checklist

Primary action: Review mission approval → I13

Progressive disclosure: Keep Purpose, Geofence in the decision layer. Put the supporting details for Pilot, Weather, Authority, full source/version history and FR-I12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a drone mission planning fixture for Drone operator; change Purpose and make Authority unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review mission approval" reaches I13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I13 — Drone preflight & approval

EN: Drone preflight & approval · **JA:** 飛行前確認と承認
Requirement: FR-I13 / WF-16 · **Actor:** Drone supervisor · **Surface:** Integrations · **Layout:** form

Purpose: Local authority remains with the pilot.
Key rule: The local pilot and controller retain operational authority.
Entry: I12, P07.
Output: Approve simulated mission → I14.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Pilot	Verified role	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Battery	92%	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Geofence	Validated	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Field / component	Sample value	Validation / editing permission
Clearance	Evidence attached	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Lost-link plan	Local controller policy	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I13: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I14 with a valid payload/version, and verify that it is blocked when the guard for WF-16 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I13 — V2 visual composition

Context: configuration · **Role:** Drone supervisor

Primary question: For drone preflight & approval, what do Pilot, Battery establish, and is “Approve simulated mission” available to Drone supervisor in this scope?

Primary visual: Preflight gate showing lost-link plan and explicit pilot approval

Primary action: Approve simulated mission → I14

Progressive disclosure: Keep Pilot, Battery in the decision layer. Put the supporting details for Geofence, Clearance, Lost-link plan, full source/version history and FR-I13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a drone preflight & approval fixture for Drone supervisor; change Pilot and make Lost-link plan unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve simulated mission” reaches I14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I14 — Drone mission telemetry

EN: Drone mission telemetry · **JA:** ドローン任務の監視
Requirement: FR-I14 / WF-16 · **Actor:** Drone operator · **Surface:** Integrations · **Layout:** map

Purpose: DR-01 · observation mission.
Key rule: The local pilot and controller retain operational authority.
Entry: I13.
Output: Review mission record → I15.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Flight state	Simulated · in progress	Read-only within object scope; display source/time when they affect decisions.
Altitude	30 m AGL	Read-only within object scope; display source/time when they affect decisions.
Battery	86%	Read-only within object scope; display source/time when they affect decisions.
Telemetry age	2 seconds	Read-only within object scope; display source/time when they affect decisions.
Command	Accepted · completion unconfirmed	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I14: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I15 with a valid payload/version, and verify that it is blocked when the guard for WF-16 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I14 — V2 visual composition

Context: command · **Role:** Drone operator

Primary question: For drone mission telemetry, what do Flight state, Altitude establish, and is “Review mission record” available to Drone operator in this scope?

Primary visual: Telemetry strip with stale-data warning and local-control ownership

Primary action: Review mission record → I15

Progressive disclosure: Keep Flight state, Altitude in the decision layer. Put the supporting details for Battery, Telemetry age, Command, full source/version history and FR-I14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a drone mission telemetry fixture for Drone operator; change Flight state and make Command unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review mission record” reaches I15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

I15 — Drone mission record

EN: Drone mission record · **JA:** ドローン任務の記録
Requirement: FR-I15 / WF-16 · **Actor:** Drone operator · **Surface:** Integrations · **Layout:** detail

Purpose: Preserve mission evidence.
Key rule: The local pilot and controller retain operational authority.
Entry: I14.
Output: Return to fleet → I11.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Mission	MS-0041	Read-only within object scope; display source/time when they affect decisions.
Incident	INC-2041	Read-only within object scope; display source/time when they affect decisions.
Flight log	Stored in regional evidence vault	Read-only within object scope; display source/time when they affect decisions.
Outcome	Observation complete	Read-only within object scope; display source/time when they affect decisions.
Pilot	Signed off	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-I15: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I11 with a valid payload/version, and verify that it is blocked when the guard for WF-16 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

I15 — V2 visual composition

Context: configuration · **Role:** Drone operator

Primary question: For drone mission record, what do Mission, Incident establish, and is “Return to fleet” available to Drone operator in this scope?

Primary visual: Mission evidence timeline with flight log and accountable pilot

Primary action: Return to fleet → I11

Progressive disclosure: Keep Mission, Incident in the decision layer. Put the supporting details for Flight log, Outcome, Pilot, full source/version history and FR-I15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Treat transport acknowledgment as transport evidence only; show rejected payloads and stale telemetry without optimistic control success.

V2 acceptance scenario: Use a drone mission record fixture for Drone operator; change Mission and make Pilot unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Return to fleet" reaches I11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T01 — Tenant overview

EN: Tenant overview · **JA:** 組織の概要
Requirement: FR-T01 / WF-20 · **Actor:** Tenant admin · **Surface:** Administration · **Layout:** dashboard

Purpose: Operate across regions with clear boundaries.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T02, T05.
Output: Manage organization → T02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Region	EU	Read-only within object scope; display source/time when they affect decisions.
Sites	24	Read-only within object scope; display source/time when they affect decisions.
Licensed devices	480	Read-only within object scope; display source/time when they affect decisions.
Operational health	Degraded connector	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T02 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T01 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For tenant overview, what do Region, Sites establish, and is “Manage organization” available to Tenant admin in this scope?

Primary visual: Tenant health summary with region and capacity allocation

Primary action: Manage organization → T02

Progressive disclosure: Keep Region, Sites in the decision layer. Put the supporting details for Licensed devices, Operational health, full source/version history and FR-T01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a tenant overview fixture for Tenant admin; change Region and make Operational health unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Manage organization” reaches T02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T02 — Organization & regions

EN: Organization & regions · **JA:** 組織とデータ地域
Requirement: FR-T02 / WF-18 · **Actor:** Tenant admin · **Surface:** Administration · **Layout:** form

Purpose: Choose residency and local defaults.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: T01.
Output: Save settings → T01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Organization	Northstar	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Home region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Locale	en-GB	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Timezone	Europe/Berlin	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Cross-region transfer	Disabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T01 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T02 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For organization & regions, what do Organization, Home region establish, and is “Save settings” available to Tenant admin in this scope?

Primary visual: Organization region boundary beside locale and timezone controls

Primary action: Save settings → T01

Progressive disclosure: Keep Organization, Home region in the decision layer. Put the supporting details for Locale, Timezone, Cross-region transfer, full source/version history and FR-T02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a organization & regions fixture for Tenant admin; change Organization and make Cross-region transfer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save settings” reaches T01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T03 — Users & invitations

EN: Users & invitations · **JA:** ユーザーと招待
Requirement: FR-T03 / WF-01 · **Actor:** Tenant admin · **Surface:** Administration · **Layout:** list

Purpose: Review access across the organization.
Key rule: Verify identity and organization scope before exposing data.
Entry: T04.
Output: Edit membership → T04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	Operator · active	Read-only within object scope; display source/time when they affect decisions.
Jamie	Responder · active	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Morgan	Invite pending	Read-only within object scope; display source/time when they affect decisions.
Service client	IOC · scoped	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T04 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T03 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For users & invitations, what do Alex, Jamie establish, and is “Edit membership” available to Tenant admin in this scope?

Primary visual: Membership roster with invitation lifecycle and role badges

Primary action: Edit membership → T04

Progressive disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Service client, full source/version history and FR-T03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a users & invitations fixture for Tenant admin; change Alex and make Service client unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit membership” reaches T04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T04 — Role & scope editor

EN: Role & scope editor · **JA:** ロールと権限範囲
Requirement: FR-T04 / WF-01 · **Actor:** Tenant admin · **Surface:** Administration · **Layout:** form

Purpose: Separate operational duties.
Key rule: Verify identity and organization scope before exposing data.
Entry: T03.

Output: Save role → T03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Role	Warning approver	Select within granted permissions; the server blocks escalation beyond the actor's rights.
Sites	Harbor Campus	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Permission	warning.approve	Select within granted permissions; the server blocks escalation beyond the actor's rights.
Restriction	Cannot approve own draft	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T03 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T04 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For role & scope editor, what do Role, Sites establish, and is "Save role" available to Tenant admin in this scope?

Primary visual: Permission-by-site matrix with explicit denied capabilities

Primary action: Save role → T03

Progressive disclosure: Keep Role, Sites in the decision layer. Put the supporting details for Permission, Restriction, full source/version history and FR-T04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a role & scope editor fixture for Tenant admin; change Role and make Restriction unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save role" reaches T03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T05 — On-call & escalation

EN: On-call & escalation · **JA:** 当番とエスカレーション
Requirement: FR-T05 / WF-10 · **Actor:** Supervisor · **Surface:** Administration · **Layout:** form

Purpose: Escalate when acknowledgment is missing.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: Role-based menu / deep link after authorization.
Output: Save escalation → T01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Primary	Campus duty operator	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
After 60 seconds	Shift supervisor	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
After 120 seconds	Regional backup	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Calendar	Europe/Berlin	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T01 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T05 — V2 visual composition

Context: configuration · **Role:** Supervisor

Primary question: For on-call & escalation, what do Primary, After 60 seconds establish, and is “Save escalation” available to Supervisor in this scope?

Primary visual: Escalation ladder with timed owner changes and calendar coverage

Primary action: Save escalation → T01

Progressive disclosure: Keep Primary, After 60 seconds in the decision layer. Put the supporting details for After 120 seconds, Calendar, full source/version history and FR-T05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a on-call & escalation fixture for Supervisor; change Primary and make Calendar unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save escalation” reaches T01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T06 — Message templates & languages

EN: Message templates & languages · **JA:** メッセージと言語
Requirement: FR-T06 / WF-09 · **Actor:** Content editor · **Surface:** Administration · **Layout:** form

Purpose: Publish reviewed instructions per locale.
Key rule: Approval binds to an exact content version and recipient scope.
Entry: Role-based menu / deep link after authorization.
Output: Preview template → O14.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Template	Fire evacuation	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
English	Approved v4	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Japanese	Review required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Fallback	Approved English	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Variables	site_name + issued_at	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O14 with a valid payload/version, and verify that it is blocked when the guard for WF-09 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T06 — V2 visual composition

Context: configuration · **Role:** Content editor

Primary question: For message templates & languages, what do Template, English establish, and is “Preview template” available to Content editor in this scope?

Primary visual: Multilingual message preview with variable and fallback status

Primary action: Preview template → O14

Progressive disclosure: Keep Template, English in the decision layer. Put the supporting details for Japanese, Fallback, Variables, full source/version history and FR-T06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a message templates & languages fixture for Content editor; change Template and make Variables unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview template” reaches O14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T07 — Hazard taxonomy & playbooks

EN: Hazard taxonomy & playbooks · **JA:** 危険分類と対応手順
Requirement: FR-T07 / WF-20 · **Actor:** Tenant admin · **Surface:** Administration · **Layout:** form

Purpose: Extend categories without changing the core.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T08.
Output: Save draft playbook → T08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Category	Personal safety / welfare check	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Severity default	P2	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Playbook	Contact then dispatch	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Field / component	Sample value	Validation / editing permission
Sensitivity	Restricted	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T08 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T07 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For hazard taxonomy & playbooks, what do Category, Severity default establish, and is “Save draft playbook” available to Tenant admin in this scope?

Primary visual: Hazard taxonomy tree connected to draft playbook steps

Primary action: Save draft playbook → T08

Progressive disclosure: Keep Category, Severity default in the decision layer. Put the supporting details for Playbook, Sensitivity, full source/version history and FR-T07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a hazard taxonomy & playbooks fixture for Tenant admin; change Category and make Sensitivity unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Save draft playbook” reaches T08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T08 — Playbook approval

EN: Playbook approval · **JA:** 対応手順の承認
Requirement: FR-T08 / WF-20 · **Actor:** Supervisor · **Surface:** Administration · **Layout:** detail

Purpose: Approve the operational sequence.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T07.

Output: Publish demo playbook → T07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Version	Welfare v2	Read-only within object scope; display source/time when they affect decisions.
Trigger	Manual SOS or detector	Read-only within object scope; display source/time when they affect decisions.
Steps	Check-in → operator → dispatch	Read-only within object scope; display source/time when they affect decisions.
Timeout	60 seconds	Read-only within object scope; display source/time when they affect decisions.
Owner	Care supervisor	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T07 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T08 — V2 visual composition

Context: configuration · **Role:** Supervisor

Primary question: For playbook approval, what do Version, Trigger establish, and is “Publish demo playbook” available to Supervisor in this scope?

Primary visual: Versioned playbook step sequence and timeout responsibilities

Primary action: Publish demo playbook → T07

Progressive disclosure: Keep Version, Trigger in the decision layer. Put the supporting details for Steps, Timeout, Owner, full source/version history and FR-T08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a playbook approval fixture for Supervisor; change Version and make Owner unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish demo playbook” reaches T07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T09 — Retention & legal holds

EN: Retention & legal holds · **JA:** 保存とリーガルホールド
Requirement: FR-T09 / WF-19 · **Actor:** Privacy officer · **Surface:** Administration · **Layout:** form

Purpose: Apply policies by data class and region.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: Role-based menu / deep link after authorization.
Output: Save policy draft → T10.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Precise location	30 days proposed	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Incident record	365 days proposed	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Evidence	90 days proposed	Read-only from the authoritative source/version; cannot be edited in this form.
Legal hold	Overrides deletion	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T10 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T09 — V2 visual composition

Context: configuration · **Role:** Privacy officer

Primary question: For retention & legal holds, what do Precise location, Incident record establish, and is “Save policy draft” available to Privacy officer in this scope?

Primary visual: Retention timeline with legal-hold overrides

Primary action: Save policy draft → T10

Progressive disclosure: Keep Precise location, Incident record in the decision layer. Put the supporting details for Evidence, Legal hold, full source/version history and FR-T09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a retention & legal holds fixture for Privacy officer; change Precise location and make Legal hold unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save policy draft" reaches T10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T10 — Privacy request review

EN: Privacy request review · **JA:** プライバシー申請の審査
Requirement: FR-T10 / WF-19 · **Actor:** Privacy officer · **Surface:** Administration · **Layout:** detail

Purpose: Review request DR-202.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: T09.
Output: Approve demo export → C30.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Identity	Verified	Read-only within object scope; display source/time when they affect decisions.
Data classes	Profile + eligible history	Read-only within object scope; display source/time when they affect decisions.
Incident evidence	Hold applied	Read-only within object scope; display source/time when they affect decisions.
Decision	Export allowed subset	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C30 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T10 — V2 visual composition

Context: configuration · **Role:** Privacy officer

Primary question: For privacy request review, what do Identity, Data classes establish, and is “Approve demo export” available to Privacy officer in this scope?

Primary visual: Data export preview showing identity proof and evidence exclusions

Primary action: Approve demo export → C30

Progressive disclosure: Keep Identity, Data classes in the decision layer. Put the supporting details for Incident evidence, Decision, full source/version history and FR-T10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a privacy request review fixture for Privacy officer; change Identity and make Decision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve demo export” reaches C30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T11 — Audit record

EN: Audit record · **JA:** 監査記録
Requirement: FR-T11 / WF-19 · **Actor:** Auditor · **Surface:** Administration · **Layout:** detail

Purpose: Immutable decision evidence.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: O30.
Output: Return to audit search → O30.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Actor	Supervisor B	Read-only within object scope; display source/time when they affect decisions.
Action	warning.approved	Read-only within object scope; display source/time when they affect decisions.
Object	WARN-03 v3	Read-only within object scope; display source/time when they affect decisions.
Time	2026-10-23T08:44:03Z	Read-only within object scope; display source/time when they affect decisions.
Trace	tr-9041	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O30 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T11 — V2 visual composition

Context: configuration · **Role:** Auditor

Primary question: For audit record, what do Actor, Action establish, and is “Return to audit search” available to Auditor in this scope?

Primary visual: Immutable audit event linking actor, object and trace

Primary action: Return to audit search → O30

Progressive disclosure: Keep Actor, Action in the decision layer. Put the supporting details for Object, Time, Trace, full source/version history and FR-T11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a audit record fixture for Auditor; change Actor and make Trace unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Return to audit search” reaches O30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T12 — Service health & recovery

EN: Service health & recovery · **JA:** サービス状態と復旧
Requirement: FR-T12 / WF-20 · **Actor:** Platform operator · **Surface:** Administration · **Layout:** dashboard

Purpose: Separate degraded service from silent failure.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T13.
Output: Open recovery runbook → T13.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
EU ingestion	Healthy	Read-only within object scope; display source/time when they affect decisions.
Notification provider	Degraded	Read-only within object scope; display source/time when they affect decisions.
Database replica	Lag 0.4 s	Read-only within object scope; display source/time when they affect decisions.
Last restore drill	Passed in test environment	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T13 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T12 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For service health & recovery, what do EU ingestion, Notification provider establish, and is "Open recovery runbook" available to Platform operator in this scope?

Primary visual: Dependency-health topology with recovery readiness

Primary action: Open recovery runbook → T13

Progressive disclosure: Keep EU ingestion, Notification provider in the decision layer. Put the supporting details for Database replica, Last restore drill, full source/version history and FR-T12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a service health & recovery fixture for Platform operator; change EU ingestion and make Last restore drill unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Open recovery runbook" reaches T13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T13 — Recovery runbook

EN: Recovery runbook · **JA:** 復旧手順
Requirement: FR-T13 / WF-20 · **Actor:** Platform operator · **Surface:** Administration · **Layout:** detail

Purpose: Regional incident data recovery.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T12.
Output: Record drill outcome → T12.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Step 1	Fence the failed writer	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Step 2	Promote verified replica	Read-only within object scope; display source/time when they affect decisions.
Step 3	Replay durable outbox	Read-only within object scope; display source/time when they affect decisions.
Step 4	Reconcile delivery and command receipts	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T13: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T12 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T13 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For recovery runbook, what do Step 1, Step 2 establish, and is “Record drill outcome” available to Platform operator in this scope?

Primary visual: Recovery checklist with verified completion evidence per step

Primary action: Record drill outcome → T12

Progressive disclosure: Keep Step 1, Step 2 in the decision layer. Put the supporting details for Step 3, Step 4, full source/version history and FR-T13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a recovery runbook fixture for Platform operator; change Step 1 and make Step 4 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Record drill outcome” reaches T12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T14 — Usage & subscription

EN: Usage & subscription · **JA:** 利用状況と契約
Requirement: FR-T14 / WF-20 · **Actor:** Billing admin · **Surface:** Administration · **Layout:** detail

Purpose: Capacity and commercial visibility.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T15.
Output: Review capacity policy → T15.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Plan	Enterprise regional	Read-only within object scope; display source/time when they affect decisions.
Sites	24 of 30	Read-only within object scope; display source/time when they affect decisions.
Devices	480 of 600	Read-only within object scope; display source/time when they affect decisions.
SMS budget	72% consumed	Read-only within object scope; display source/time when they affect decisions.
Emergency behavior	No silent incident drop	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T14: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T15 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T14 — V2 visual composition

Context: configuration · **Role:** Billing admin

Primary question: For usage & subscription, what do Plan, Sites establish, and is “Review capacity policy” available to Billing admin in this scope?

Primary visual: Capacity consumption chart with emergency reservation

Primary action: Review capacity policy → T15

Progressive disclosure: Keep Plan, Sites in the decision layer. Put the supporting details for Devices, SMS budget, Emergency behavior, full source/version history and FR-T14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a usage & subscription fixture for Billing admin; change Plan and make Emergency behavior unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review capacity policy” reaches T15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T15 — Capacity & quota policy

EN: Capacity & quota policy · **JA:** 容量と制限
Requirement: FR-T15 / WF-20 · **Actor:** Platform operator · **Surface:** Administration · **Layout:** form

Purpose: Protect emergency traffic during bursts.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T14.
Output: Save capacity draft → T14.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
P1 reserved queue	Enabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Telemetry sampling	Allowed under overload	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
SOS ingestion	Reserved capacity	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Warning threshold	80%	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Hard limit behavior	Visible degraded mode	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T15: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T14 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T15 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For capacity & quota policy, what do P1 reserved queue, Telemetry sampling establish, and is "Save capacity draft" available to Platform operator in this scope?

Primary visual: Queue policy comparison showing P1 reserves and hard-limit effects

Primary action: Save capacity draft → T14

Progressive disclosure: Keep P1 reserved queue, Telemetry sampling in the decision layer. Put the supporting details for SOS ingestion, Warning threshold, Hard limit behavior, full source/version history and FR-T15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a capacity & quota policy fixture for Platform operator; change P1 reserved queue and make Hard limit behavior unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Save capacity draft" reaches T14 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T16 — Analytics & export

EN: Analytics & export · **JA:** 分析とレポート出力
Requirement: FR-T16 / WF-20 · **Actor:** Analyst · **Surface:** Administration · **Layout:** detail

Purpose: Measure response without hiding gaps.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T17.
Output: Prepare demo report → T17.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Period	Last 30 days	Read-only within object scope; display source/time when they affect decisions.
Median acknowledgment	18 seconds · demo	Read-only within object scope; display source/time when they affect decisions.
P95 acknowledgment	61 seconds · demo	Read-only within object scope; display source/time when they affect decisions.
Unconfirmed delivery	3.4% · demo	Read-only within object scope; display source/time when they affect decisions.
Export	De-identified	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T16: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T17 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T16 — V2 visual composition

Context: configuration · **Role:** Analyst

Primary question: For analytics & export, what do Period, Median acknowledgment establish, and is “Prepare demo report” available to Analyst in this scope?

Primary visual: Distribution chart separating median, P95 and missing receipts

Primary action: Prepare demo report → T17

Progressive disclosure: Keep Period, Median acknowledgment in the decision layer. Put the supporting details for P95 acknowledgment, Unconfirmed delivery, Export, full source/version history and FR-T16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a analytics & export fixture for Analyst; change Period and make Export unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Prepare demo report” reaches T17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

T17 — Report schedule

EN: Report schedule · **JA:** レポートのスケジュール
Requirement: FR-T17 / WF-20 · **Actor:** Analyst · **Surface:** Administration · **Layout:** form

Purpose: Deliver an authorized operational report.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: T16.
Output: Save schedule draft → T16.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Report	Monthly readiness	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Audience	Tenant supervisors	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Timezone	Europe/Berlin	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Field / component	Sample value	Validation / editing permission
Schedule	First Monday · 09:00	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.
PII	Excluded	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-T17: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to T16 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

T17 — V2 visual composition

Context: configuration · **Role:** Analyst

Primary question: For report schedule, what do Report, Audience establish, and is “Save schedule draft” available to Analyst in this scope?

Primary visual: Report schedule preview with timezone and recipient disclosure

Primary action: Save schedule draft → T16

Progressive disclosure: Keep Report, Audience in the decision layer. Put the supporting details for Timezone, Schedule, PII, full source/version history and FR-T17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Preserve scope and draft policy on validation failure; never silently apply a tenant-wide or cross-region change.

V2 acceptance scenario: Use a report schedule fixture for Analyst; change Report and make PII unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save schedule draft” reaches T16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X01 — No connection / SOS pending

EN: No connection / SOS pending · **JA:** オフライン・SOS未送信
Requirement: FR-X01 / WF-05 · **Actor:** Citizen · **Surface:** System states · **Layout:** alert

Purpose: Your SOS has not reached the server.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: Role-based menu / deep link after authorization.
Output: View offline outbox → R11.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Local record	Saved on this device	Read-only within object scope; display source/time when they affect decisions.
Server receipt	Not received	Read-only within object scope; display source/time when they affect decisions.
Retry	When connection returns	Read-only within object scope; display source/time when they affect decisions.
Alternative	Use configured local emergency calling option	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R11 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X01 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For no connection / sos pending, what do Local record, Server receipt establish, and is “View offline outbox” available to Citizen in this scope?

Primary visual: Locally saved SOS receipt with server-receipt absence prominent

Primary action: View offline outbox → R11

Progressive disclosure: Keep Local record, Server receipt in the decision layer. Put the supporting details for Retry, Alternative, full source/version history and FR-X01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a no connection / sos pending fixture for Citizen; change Local record and make Alternative unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View offline outbox” reaches R11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X02 — Permission unavailable

EN: Permission unavailable · **JA:** 位置情報の権限なし
Requirement: FR-X02 / WF-05 · **Actor:** Citizen · **Surface:** System states · **Layout:** detail

Purpose: Location permission is off.
Key rule: Server receipt and human acknowledgment are separate states.
Entry: Role-based menu / deep link after authorization.
Output: Enter location manually → C06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Automatic location	Unavailable	Read-only within object scope; display source/time when they affect decisions.
Alternative	Enter building and room	Read-only within object scope; display source/time when they affect decisions.
Privacy	Permission remains your choice	Read-only within object scope; display source/time when they affect decisions.
SOS	Manual location supported	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C06 with a valid payload/version, and verify that it is blocked when the guard for WF-05 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X02 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For permission unavailable, what do Automatic location, Alternative establish, and is “Enter location manually” available to Citizen in this scope?

Primary visual: Manual location entry beside unavailable-permission explanation

Primary action: Enter location manually → C06

Progressive disclosure: Keep Automatic location, Alternative in the decision layer. Put the supporting details for Privacy, SOS, full source/version history and FR-X02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a permission unavailable fixture for Citizen; change Automatic location and make SOS unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Enter location manually" reaches C06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X03 — No active incidents

EN: No active incidents · **JA:** 対応中のインシデントなし
Requirement: FR-X03 / WF-20 · **Actor:** Operator · **Surface:** System states · **Layout:** success

Purpose: No active incidents in this view.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Review device health → S06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Filters	Harbor Campus · active	Read-only within object scope; display source/time when they affect decisions.
Coverage	Only authorized sites	Read-only within object scope; display source/time when they affect decisions.
Last sync	10 seconds ago	Read-only within object scope; display source/time when they affect decisions.
Unknown devices	Still listed in health	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S06 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X03 — V2 visual composition

Context: configuration · **Role:** Operator

Primary question: For no active incidents, what do Filters, Coverage establish, and is “Review device health” available to Operator in this scope?

Primary visual: Empty incident view with coverage and unknown-device evidence

Primary action: Review device health → S06

Progressive disclosure: Keep Filters, Coverage in the decision layer. Put the supporting details for Last sync, Unknown devices, full source/version history and FR-X03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a no active incidents fixture for Operator; change Filters and make Unknown devices unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review device health” reaches S06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X04 — Access denied

EN: Access denied · **JA:** アクセス権限なし
Requirement: FR-X04 / WF-01 · **Actor:** All roles · **Surface:** System states · **Layout:** detail

Purpose: This record is outside your access scope.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Choose workspace → A04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Request	Restricted incident	Read-only within object scope; display source/time when they affect decisions.
Data	Not disclosed	Read-only within object scope; display source/time when they affect decisions.
Next step	Ask your tenant administrator	Read-only within object scope; display source/time when they affect decisions.
Audit	Access attempt recorded	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A04 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X04 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For access denied, what do Request, Data establish, and is “Choose workspace” available to All roles in this scope?

Primary visual: Workspace selection recovery without disclosing denied object

Primary action: Choose workspace → A04

Progressive disclosure: Keep Request, Data in the decision layer. Put the supporting details for Next step, Audit, full source/version history and FR-X04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a access denied fixture for All roles; change Request and make Audit unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Choose workspace” reaches A04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X05 — Service unavailable

EN: Service unavailable · **JA:** サービスの一時停止
Requirement: FR-X05 / WF-20 · **Actor:** All roles · **Surface:** System states · **Layout:** alert

Purpose: This service is temporarily unavailable.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Open offline pack → C18.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Status	Regional API degraded	Read-only within object scope; display source/time when they affect decisions.
Cached content	May be stale	Read-only within object scope; display source/time when they affect decisions.
Submission	Not confirmed	Read-only within object scope; display source/time when they affect decisions.
Recovery	Retry without creating duplicates	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C18 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X05 — V2 visual composition

Context: emergency · **Role:** All roles

Primary question: For service unavailable, what do Status, Cached content establish, and is “Open offline pack” available to All roles in this scope?

Primary visual: Service outage panel with cached-content age and alternate path

Primary action: Open offline pack → C18

Progressive disclosure: Keep Status, Cached content in the decision layer. Put the supporting details for Submission, Recovery, full source/version history and FR-X05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a service unavailable fixture for All roles; change Status and make Recovery unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open offline pack” reaches C18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X06 — Unsaved changes & conflict

EN: Unsaved changes & conflict · **JA:** 未保存の変更と競合
Requirement: FR-X06 / WF-02 · **Actor:** Site admin · **Surface:** System states · **Layout:** detail

Purpose: A newer version was published.
Key rule: Location and plans are versioned; new revisions require approval.
Entry: Role-based menu / deep link after authorization.
Output: Review latest version → S05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Your version	12	Read-only within object scope; display source/time when they affect decisions.
Current version	13	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Your changes	Preserved locally	Read-only within object scope; display source/time when they affect decisions.
Resolution	Review before resubmitting	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S05 with a valid payload/version, and verify that it is blocked when the guard for WF-02 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X06 — V2 visual composition

Context: configuration · **Role:** Site admin

Primary question: For unsaved changes & conflict, what do Your version, Current version establish, and is “Review latest version” available to Site admin in this scope?

Primary visual: Local/server version comparison preserving unsaved edits

Primary action: Review latest version → S05

Progressive disclosure: Keep Your version, Current version in the decision layer. Put the supporting details for Your changes, Resolution, full source/version history and FR-X06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a unsaved changes & conflict fixture for Site admin; change Your version and make Resolution unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review latest version” reaches S05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X07 — Empty search & filter reset

EN: Empty search & filter reset · **JA:** 検索結果なし
Requirement: FR-X07 / WF-20 · **Actor:** Operator · **Surface:** System states · **Layout:** detail

Purpose: No matches for these filters.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.

Output: Reset search → O30.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Query	INC-9999	Read-only within object scope; display source/time when they affect decisions.
Scope	Harbor Campus	Read-only within object scope; display source/time when they affect decisions.
Period	Today	Read-only within object scope; display source/time when they affect decisions.
Suggestion	Check identifier or reset filters	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O30 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X07 — V2 visual composition

Context: configuration · **Role:** Operator

Primary question: For empty search & filter reset, what do Query, Scope establish, and is “Reset search” available to Operator in this scope?

Primary visual: Search query and active filters beside reset control

Primary action: Reset search → O30

Progressive disclosure: Keep Query, Scope in the decision layer. Put the supporting details for Period, Suggestion, full source/version history and FR-X07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a empty search & filter reset fixture for Operator; change Query and make Suggestion unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Reset search” reaches O30 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X08 — Expired session

EN: Expired session · **JA:** セッションの有効期限切れ
Requirement: FR-X08 / WF-01 · **Actor:** All roles · **Surface:** System states · **Layout:** alert

Purpose: Sign in again to continue.
Key rule: Verify identity and organization scope before exposing data.
Entry: Role-based menu / deep link after authorization.
Output: Sign in → A02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Draft	Preserved without submitting	Read-only within object scope; display source/time when they affect decisions.
Session	Expired	Read-only within object scope; display source/time when they affect decisions.
Sensitive data	Hidden	Read-only within object scope; display source/time when they affect decisions.
Next	Re-authenticate	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to A02 with a valid payload/version, and verify that it is blocked when the guard for WF-01 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X08 — V2 visual composition

Context: configuration · **Role:** All roles

Primary question: For expired session, what do Draft, Session establish, and is “Sign in” available to All roles in this scope?

Primary visual: Session-expired panel explaining protected draft recovery

Primary action: Sign in → A02

Progressive disclosure: Keep Draft, Session in the decision layer. Put the supporting details for Sensitive data, Next, full source/version history and FR-X08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a expired session fixture for All roles; change Draft and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Sign in” reaches A02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X09 — Loading & stale data

EN: Loading & stale data · **JA:** 読み込み中・古いデータ
Requirement: FR-X09 / WF-20 · **Actor:** Operator · **Surface:** System states · **Layout:** detail

Purpose: Refreshing the situation view.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Return to command → O01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Current view	Cached · 2 minutes old	Read-only within object scope; display source/time when they affect decisions.
Connection	Reconnecting	Read-only within object scope; display source/time when they affect decisions.
Commands	Disabled while stale	Read-only within object scope; display source/time when they affect decisions.
Next	Confirm fresh state	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O01 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X09 — V2 visual composition

Context: emergency · **Role:** Operator

Primary question: For loading & stale data, what do Current view, Connection establish, and is “Return to command” available to Operator in this scope?

Primary visual: Stale-data band showing source age and unavailable commands

Primary action: Return to command → O01

Progressive disclosure: Keep Current view, Connection in the decision layer. Put the supporting details for Commands, Next, full source/version history and FR-X09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a loading & stale data fixture for Operator; change Current view and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Return to command" reaches O01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

X10 — Attachment rejected

EN: Attachment rejected · **JA:** 添付ファイルの拒否
Requirement: FR-X10 / WF-11 · **Actor:** Responder · **Surface:** System states · **Layout:** detail

Purpose: This attachment could not be accepted.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: Role-based menu / deep link after authorization.
Output: Choose another attachment → R10.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
File	Observation clip	Read-only within object scope; display source/time when they affect decisions.
Reason	Exceeds 20 MB	Read-only within object scope; display source/time when they affect decisions.
Allowed	JPEG PNG MP4 PDF	Read-only within object scope; display source/time when they affect decisions.
Action	Choose a smaller file	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-X10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R10 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

X10 — V2 visual composition

Context: configuration · **Role:** Responder

Primary question: For attachment rejected, what do File, Reason establish, and is “Choose another attachment” available to Responder in this scope?

Primary visual: Rejected attachment preview with reason and allowed replacement

Primary action: Choose another attachment → R10

Progressive disclosure: Keep File, Reason in the decision layer. Put the supporting details for Allowed, Action, full source/version history and FR-X10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Show the exact unavailable capability and a valid recovery action while preserving safe local context; recovery navigation is not recovery evidence.

V2 acceptance scenario: Use a attachment rejected fixture for Responder; change File and make Action unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Choose another attachment” reaches R10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G01 — Regional operations

EN: Regional operations · **JA:** 地域別運用
Requirement: FR-G01 / WF-20 · **Actor:** Platform operator · **Surface:** Platform · **Layout:** dashboard

Purpose: Production cells, independently operated.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G04.
Output: Inspect regional cell → G02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
EU cell	Healthy · 3 availability zones	Read-only within object scope; display source/time when they affect decisions.
APAC cell	Healthy · 3 availability zones	Read-only within object scope; display source/time when they affect decisions.
Global directory	Healthy	Read-only within object scope; display source/time when they affect decisions.
Residency	Enforced	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G02 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G01 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For regional operations, what do EU cell, APAC cell establish, and is “Inspect regional cell” available to Platform operator in this scope?

Primary visual: Regional cell topology with residency boundaries

Primary action: Inspect regional cell → G02

Progressive disclosure: Keep EU cell, APAC cell in the decision layer. Put the supporting details for Global directory, Residency, full source/version history and FR-G01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a regional operations fixture for Platform operator; change EU cell and make Residency unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect regional cell” reaches G02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G02 — Regional cell detail

EN: Regional cell detail · **JA:** リージョナルセルの詳細
Requirement: FR-G02 / WF-20 · **Actor:** Platform operator · **Surface:** Platform · **Layout:** detail

Purpose: EU cell · operational readiness.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G01.
Output: Review recovery readiness → G03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Writer epoch	42	Read-only within object scope; display source/time when they affect decisions.
Broker lag	120 ms	Read-only within object scope; display source/time when they affect decisions.
Database	Primary + verified replicas	Read-only within object scope; display source/time when they affect decisions.
DR target	Approved EU destination	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G03 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G02 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For regional cell detail, what do Writer epoch, Broker lag establish, and is “Review recovery readiness” available to Platform operator in this scope?

Primary visual: Cell replication and writer-epoch status

Primary action: Review recovery readiness → G03

Progressive disclosure: Keep Writer epoch, Broker lag in the decision layer. Put the supporting details for Database, DR target, full source/version history and FR-G02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a regional cell detail fixture for Platform operator; change Writer epoch and make DR target unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review recovery readiness” reaches G03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G03 — Disaster recovery drill

EN: Disaster recovery drill · **JA:** 災害復旧訓練
Requirement: FR-G03 / WF-20 · **Actor:** Platform operator · **Surface:** Platform · **Layout:** form

Purpose: Validate recovery with recorded evidence.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G02.
Output: Record drill evidence → G04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Cell	EU	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Mode	Isolated recovery exercise	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Backup	Verified snapshot	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Target RPO	5 minutes	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Target RTO	60 minutes	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G04 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G03 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For disaster recovery drill, what do Cell, Mode establish, and is “Record drill evidence” available to Platform operator in this scope?

Primary visual: Recovery drill timeline with target RPO/RTO and evidence

Primary action: Record drill evidence → G04

Progressive disclosure: Keep Cell, Mode in the decision layer. Put the supporting details for Backup, Target RPO, Target RTO, full source/version history and FR-G03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a disaster recovery drill fixture for Platform operator; change Cell and make Target RTO unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record drill evidence” reaches G04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G04 — Recovery verification

EN: Recovery verification · **JA:** 復旧の検証
Requirement: FR-G04 / WF-20 · **Actor:** Platform operator · **Surface:** Platform · **Layout:** detail

Purpose: Verify data and ownership before routing.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G03.
Output: Review regional operations → G01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Writer fencing	Verified	Read-only within object scope; display source/time when they affect decisions.
SOS records	Reconciled	Read-only within object scope; display source/time when they affect decisions.
Outbox	Replayed without duplicate effects	Read-only within object scope; display source/time when they affect decisions.
Residency	EU only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G01 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G04 — V2 visual composition

Context: configuration · **Role:** Platform operator

Primary question: For recovery verification, what do Writer fencing, SOS records establish, and is “Review regional operations” available to Platform operator in this scope?

Primary visual: Recovery proof grid separating fencing and record durability

Primary action: Review regional operations → G01

Progressive disclosure: Keep Writer fencing, SOS records in the decision layer. Put the supporting details for Outbox, Residency, full source/version history and FR-G04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a recovery verification fixture for Platform operator; change Writer fencing and make Residency unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Review regional operations" reaches G01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G05 — Tenant provisioning

EN: Tenant provisioning · **JA:** テナントの作成
Requirement: FR-G05 / WF-18 · **Actor:** Platform admin · **Surface:** Platform · **Layout:** form

Purpose: Provision a production tenant boundary.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: Role-based menu / deep link after authorization.
Output: Review provisioning plan → G06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Organization	Northstar	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Home region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Isolation	Dedicated database	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Plan	Enterprise	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Policy set	Reviewed baseline	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G06 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G05 — V2 visual composition

Context: configuration · **Role:** Platform admin

Primary question: For tenant provisioning, what do Tenant, Home region establish, and is “Review provisioning plan” available to Platform admin in this scope?

Primary visual: Tenant provisioning dependency plan and isolation boundary

Primary action: Review provisioning plan → G06

Progressive disclosure: Keep Tenant, Home region in the decision layer. Put the supporting details for Isolation, Plan, Policy set, full source/version history and FR-G05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a tenant provisioning fixture for Platform admin; change Tenant and make Policy set unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review provisioning plan” reaches G06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G06 — Tenant lifecycle

EN: Tenant lifecycle · **JA:** テナントのライフサイクル
Requirement: FR-G06 / WF-18 · **Actor:** Platform admin · **Surface:** Platform · **Layout:** detail

Purpose: Northstar · lifecycle controls.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: G05, G08.
Output: Request support access → G07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Status	Active	Read-only within object scope; display source/time when they affect decisions.
Region migration	Review required	Read-only within object scope; display source/time when they affect decisions.
Deletion	Retention workflow required	Read-only within object scope; display source/time when they affect decisions.
Support access	Not granted	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The

prototype state selector is a shared review tool.

AC-G06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G07 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G06 — V2 visual composition

Context: configuration · **Role:** Platform admin

Primary question: For tenant lifecycle, what do Status, Region migration establish, and is “Request support access” available to Platform admin in this scope?

Primary visual: Tenant lifecycle state diagram with migration/deletion gates

Primary action: Request support access → G07

Progressive disclosure: Keep Status, Region migration in the decision layer. Put the supporting details for Deletion, Support access, full source/version history and FR-G06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a tenant lifecycle fixture for Platform admin; change Status and make Support access unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Request support access” reaches G07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G07 — Time-bound support access

EN: Time-bound support access · **JA:** 期限付きサポートアクセス
Requirement: FR-G07 / WF-19 · **Actor:** Security operator · **Surface:** Platform · **Layout:** form

Purpose: Request the smallest support scope.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: G06.
Output: Submit access request → G08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Organization	Northstar	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Scope	Connector diagnostics metadata	Select within granted permissions; the server blocks escalation beyond the actor’s rights.
Reason	Investigate IOC delay	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.

Field / component	Sample value	Validation / editing permission
Duration	30 minutes	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
PII access	Excluded	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G08 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G07 — V2 visual composition

Context: configuration · **Role:** Security operator

Primary question: For time-bound support access, what do Tenant, Scope establish, and is “Submit access request” available to Security operator in this scope?

Primary visual: Support request showing minimal scope and access expiry

Primary action: Submit access request → G08

Progressive disclosure: Keep Tenant, Scope in the decision layer. Put the supporting details for Reason, Duration, PII access, full source/version history and FR-G07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a time-bound support access fixture for Security operator; change Tenant and make PII access unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit access request” reaches G08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G08 — Support access approval

EN: Support access approval · **JA:** サポートアクセスの承認
Requirement: FR-G08 / WF-19 · **Actor:** Tenant admin · **Surface:** Platform · **Layout:** detail

Purpose: Approve a scoped support session.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: G07.
Output: Approve scoped access — sample → G06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Requester	Platform support	Read-only within object scope; display source/time when they affect decisions.
Scope	Diagnostics metadata	Read-only within object scope; display source/time when they affect decisions.
Expiry	30 minutes	Read-only within object scope; display source/time when they affect decisions.
Audit	All access recorded	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G06 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G08 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For support access approval, what do Requester, Scope establish, and is “Approve scoped access — sample” available to Tenant admin in this scope?

Primary visual: Support authorization comparing requester and approved scope

Primary action: Approve scoped access — sample → G06

Progressive disclosure: Keep Requester, Scope in the decision layer. Put the supporting details for Expiry, Audit, full source/version history and FR-G08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a support access approval fixture for Tenant admin; change Requester and make Audit unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Approve scoped access — sample” reaches G06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G09 — Release & schema compatibility

EN: Release & schema compatibility · **JA:** リリースとスキーマ互換性
Requirement: FR-G09 / WF-20 · **Actor:** Release manager · **Surface:** Platform · **Layout:** list

Purpose: Release only compatible contracts.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G10.
Output: Review release gate → G10.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Incident service	2.4 · canary	Read-only within object scope; display source/time when they affect decisions.
Event schema	v1.3 additive	Read-only within object scope; display source/time when they affect decisions.
Mobile API	v1 supported	Read-only within object scope; display source/time when they affect decisions.
Rollback	Previous image available	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G10 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G09 — V2 visual composition

Context: configuration · **Role:** Release manager

Primary question: For release & schema compatibility, what do Incident service, Event schema establish, and is “Review release gate” available to Release manager in this scope?

Primary visual: Release compatibility matrix across schema and mobile versions

Primary action: Review release gate → G10

Progressive disclosure: Keep Incident service, Event schema in the decision layer. Put the supporting details for Mobile API, Rollback, full source/version history and FR-G09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a release & schema compatibility fixture for Release manager; change Incident service and make Rollback unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Review release gate" reaches G10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G10 — Release gate & rollback

EN: Release gate & rollback · **JA:** リリース判定とロールバック
Requirement: FR-G10 / WF-20 · **Actor:** Release manager · **Surface:** Platform · **Layout:** detail

Purpose: Verify behavior before promotion.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: G09.
Output: Record release decision → G09.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Contract tests	Passed	Read-only within object scope; display source/time when they affect decisions.
Tenant isolation	Passed	Read-only within object scope; display source/time when they affect decisions.
SLO burn rate	Within gate	Read-only within object scope; display source/time when they affect decisions.
Migration	Expand phase only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G09 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G10 — V2 visual composition

Context: configuration · **Role:** Release manager

Primary question: For release gate & rollback, what do Contract tests, Tenant isolation establish, and is "Record release decision" available to Release manager in this scope?

Primary visual: Release gate checklist with rollback viability and SLO trend

Primary action: Record release decision → G09

Progressive disclosure: Keep Contract tests, Tenant isolation in the decision layer. Put the supporting details for SLO burn rate, Migration, full source/version history and FR-G10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a release gate & rollback fixture for Release manager; change Contract tests and make Migration unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Record release decision" reaches G09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G11 — Security incidents

EN: Security incidents · **JA:** セキュリティインシデント
Requirement: FR-G11 / WF-19 · **Actor:** Security operator · **Surface:** Platform · **Layout:** list

Purpose: Respond to platform threats.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: G12.
Output: Open security case → G12.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Device certificate	Revoked · investigation	Read-only within object scope; display source/time when they affect decisions.
Webhook replay	Blocked	Read-only within object scope; display source/time when they affect decisions.
Support session	Expired	Read-only within object scope; display source/time when they affect decisions.
Upload scan	Quarantined	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G12 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G11 — V2 visual composition

Context: configuration · **Role:** Security operator

Primary question: For security incidents, what do Device certificate, Webhook replay establish, and is “Open security case” available to Security operator in this scope?

Primary visual: Security signal queue with affected trust boundary

Primary action: Open security case → G12

Progressive disclosure: Keep Device certificate, Webhook replay in the decision layer. Put the supporting details for Support session, Upload scan, full source/version history and FR-G11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a security incidents fixture for Security operator; change Device certificate and make Upload scan unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open security case” reaches G12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

G12 — Security case & containment

EN: Security case & containment · **JA:** セキュリティ対応と封じ込め
Requirement: FR-G12 / WF-19 · **Actor:** Security operator · **Surface:** Platform · **Layout:** form

Purpose: Contain the source without hiding coverage gaps.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: G11.
Output: Record containment → G11.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Case	SEC-041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Source	GW-012	Read-only from the authoritative source/version; cannot be edited in this form.
Action	Revoke certificate	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Coverage	Site supervisor notified	Read-only from the authoritative source/version; cannot be edited in this form.
Evidence	Preserved	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-G12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to G11 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

G12 — V2 visual composition

Context: configuration · **Role:** Security operator

Primary question: For security case & containment, what do Case, Source establish, and is “Record containment” available to Security operator in this scope?

Primary visual: Containment timeline showing residual exposure and evidence

Primary action: Record containment → G11

Progressive disclosure: Keep Case, Source in the decision layer. Put the supporting details for Action, Coverage, Evidence, full source/version history and FR-G12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep tenant data boundaries and regional writer authority explicit; stale operational metadata cannot authorize recovery or support access.

V2 acceptance scenario: Use a security case & containment fixture for Security operator; change Case and make Evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Record containment” reaches G11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D01 — Developer workspace

EN: Developer workspace · **JA:** 開発者ワークスペース
Requirement: FR-D01 / WF-17 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** cards

Purpose: Build against explicit contracts.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: Role-based menu / deep link after authorization.
Output: Open API explorer → D02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
API reference	Version 1	Read-only within object scope; display source/time when they affect decisions.
Event schemas	CloudEvents + domain v1	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
SDK examples	HTTPS + MQTT	Read-only within object scope; display source/time when they affect decisions.
Sandbox	Exercise isolated	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to D02 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D01 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For developer workspace, what do API reference, Event schemas establish, and is “Open API explorer” available to Partner developer in this scope?

Primary visual: Developer resource map separating sandbox and production

Primary action: Open API explorer → D02

Progressive disclosure: Keep API reference, Event schemas in the decision layer. Put the supporting details for SDK examples, Sandbox, full source/version history and FR-D01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a developer workspace fixture for Partner developer; change API reference and make Sandbox unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open API explorer” reaches D02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D02 — API explorer

EN: API explorer · **JA:** APIエクスプローラー
Requirement: FR-D02 / WF-03 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** detail

Purpose: POST /v1/events.
Key rule: Preserve source, time and ID; duplicates do not create new effects.
Entry: D01.

Output: Run sandbox request → D03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Auth	event.ingest	Read-only within object scope; display source/time when they affect decisions.
Payload	CloudEvent v1	Read-only within object scope; display source/time when they affect decisions.
Idempotency	Source + event ID	Read-only within object scope; display source/time when they affect decisions.
Response	202 durable receipt	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to D03 with a valid payload/version, and verify that it is blocked when the guard for WF-03 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D02 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For api explorer, what do Auth, Payload establish, and is “Run sandbox request” available to Partner developer in this scope?

Primary visual: Request/response split pane with scope and idempotency preview

Primary action: Run sandbox request → D03

Progressive disclosure: Keep Auth, Payload in the decision layer. Put the supporting details for Idempotency, Response, full source/version history and FR-D02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a api explorer fixture for Partner developer; change Auth and make Response unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Run sandbox request” reaches D03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D03 — Sandbox request results

EN: Sandbox request results · **JA:** サンドボックスの実行結果
Requirement: FR-D03 / WF-03 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** detail

Purpose: Inspect the complete receipt.
Key rule: Preserve source, time and ID; duplicates do not create new effects.
Entry: D02.
Output: Open schema registry → D04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
HTTP status	202	Read-only within object scope; display source/time when they affect decisions.
Receipt	event-041	Read-only within object scope; display source/time when they affect decisions.
Durability	Committed	Read-only within object scope; display source/time when they affect decisions.
Incident	Not yet confirmed	Read-only within object scope; display source/time when they affect decisions.
Trace	tr-9041	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to D04 with a valid payload/version, and verify that it is blocked when the guard for WF-03 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D03 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For sandbox request results, what do HTTP status, Receipt establish, and is “Open schema registry” available to Partner developer in this scope?

Primary visual: Sandbox response showing HTTP, durable receipt and trace separately

Primary action: Open schema registry → D04

Progressive disclosure: Keep HTTP status, Receipt in the decision layer. Put the supporting details for Durability, Incident, Trace, full source/version history and FR-D03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a sandbox request results fixture for Partner developer; change HTTP status and make Trace unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Open schema registry" reaches D04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D04 — Schema registry

EN: Schema registry · **JA:** スキーマレジストリー
Requirement: FR-D04 / WF-03 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** list

Purpose: Track compatibility before upgrading.
Key rule: Preserve source, time and ID; duplicates do not create new effects.
Entry: D03.
Output: View schema contract → D05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
signal.fire.observed	v1 · active	Read-only within object scope; display source/time when they affect decisions.
incident.updated	v1 · active	Read-only within object scope; display source/time when they affect decisions.
command.result	v1 · restricted	Read-only within object scope; display source/time when they affect decisions.
Deprecated versions	None	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to D05 with a valid payload/version, and verify that it is blocked when the guard for WF-03 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D04 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For schema registry, what do signal.fire.observed, incident.updated establish, and is "View schema contract" available to Partner developer in this scope?

Primary visual: Schema version tree with compatibility and deprecation markers

Primary action: View schema contract → D05

Progressive disclosure: Keep signal.fire.observed, incident.updated in the decision layer. Put the supporting details for command.result, Deprecated versions, full source/version history and FR-D04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a schema registry fixture for Partner developer; change signal.fire.observed and make Deprecated versions unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "View schema contract" reaches D05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D05 — Connector manifest editor

EN: Connector manifest editor · **JA:** コネクターマニフェストの編集
Requirement: FR-D05 / WF-17 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** form

Purpose: Declare capabilities explicitly.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: D04.
Output: Submit for conformance → D06.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Connector	Partner sensor bridge	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Version	1.0.0	Read-only from the authoritative source/version; cannot be edited in this form.
Capabilities	telemetry.read	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Auth	mTLS	Read-only from the authoritative source/version; cannot be edited in this form.
Retry class	Idempotent ingest	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to D06 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D05 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For connector manifest editor, what do Connector, Version establish, and is “Submit for conformance” available to Partner developer in this scope?

Primary visual: Connector capability manifest with validation diff

Primary action: Submit for conformance → D06

Progressive disclosure: Keep Connector, Version in the decision layer. Put the supporting details for Capabilities, Auth, Retry class, full source/version history and FR-D05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a connector manifest editor fixture for Partner developer; change Connector and make Retry class unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit for conformance” reaches D06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D06 — Conformance certification

EN: Conformance certification · **JA:** 適合性の審査
Requirement: FR-D06 / WF-17 · **Actor:** Integration reviewer · **Surface:** Developer · **Layout:** detail

Purpose: Review adapter behavior under failure.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: D05.
Output: Approve adapter version → I01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Duplicate delivery	Passed	Read-only within object scope; display source/time when they affect decisions.
Tenant spoof	Rejected	Read-only within object scope; display source/time when they affect decisions.
Clock skew	Flagged	Read-only within object scope; display source/time when they affect decisions.
Expired credentials	Rejected	Read-only within object scope; display source/time when they affect decisions.
Backpressure	Preserved	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I01 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D06 — V2 visual composition

Context: configuration · **Role:** Integration reviewer

Primary question: For conformance certification, what do Duplicate delivery, Tenant spoof establish, and is "Approve adapter version" available to Integration reviewer in this scope?

Primary visual: Conformance scenario matrix with security and backpressure results

Primary action: Approve adapter version → I01

Progressive disclosure: Keep Duplicate delivery, Tenant spoof in the decision layer. Put the supporting details for Clock skew, Expired credentials, Backpressure, full source/version history and FR-D06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a conformance certification fixture for Integration reviewer; change Duplicate delivery and make Backpressure unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Approve adapter version" reaches I01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D07 — Webhook delivery trace

EN: Webhook delivery trace · **JA:** Webhook配信の追跡
Requirement: FR-D07 / WF-10 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** timeline

Purpose: Understand every delivery attempt.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: Role-based menu / deep link after authorization.
Output: Review subscription → I08.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
10:40:00	Queued	Read-only within object scope; display source/time when they affect decisions.
10:40:01	Partner returned 503	Read-only within object scope; display source/time when they affect decisions.
10:40:03	Retry scheduled	Read-only within object scope; display source/time when they affect decisions.
10:40:07	Application ACK received	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I08 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D07 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For webhook delivery trace, what do 10:40:00, 10:40:01 establish, and is “Review subscription” available to Partner developer in this scope?

Primary visual: Webhook retry timeline with exact delivery outcome

Primary action: Review subscription → I08

Progressive disclosure: Keep 10:40:00, 10:40:01 in the decision layer. Put the supporting details for 10:40:03, 10:40:07, full source/version history and FR-D07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a webhook delivery trace fixture for Partner developer; change 10:40:00 and make 10:40:07 unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review subscription” reaches I08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

D08 — Credential rotation

EN: Credential rotation · **JA:** 認証情報のローテーション
Requirement: FR-D08 / WF-17 · **Actor:** Partner developer · **Surface:** Developer · **Layout:** form

Purpose: Rotate credentials without exposing secrets.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: Role-based menu / deep link after authorization.
Output: Create rotation request → I07.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Client	Partner IOC	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Current key	Expires in 10 days	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Overlap window	24 hours	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.
New key	Vault reference only	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-D08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I07 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

D08 — V2 visual composition

Context: configuration · **Role:** Partner developer

Primary question: For credential rotation, what do Client, Current key establish, and is “Create rotation request” available to Partner developer in this scope?

Primary visual: Credential overlap timeline with secret-free rotation preview

Primary action: Create rotation request → I07

Progressive disclosure: Keep Client, Current key in the decision layer. Put the supporting details for Overlap window, New key, full source/version history and FR-D08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep sandbox requests distinguishable from production; never expose raw credentials in previews, traces or copied examples.

V2 acceptance scenario: Use a credential rotation fixture for Partner developer; change Client and make New key unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Create rotation request" reaches I07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L01 — Language coverage

EN: Language coverage · **JA:** 言語対応状況
Requirement: FR-L01 / WF-18 · **Actor:** Localization manager · **Surface:** Localization · **Layout:** dashboard

Purpose: Three languages, one product meaning.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: L03, L04, L05.
Output: Open translation workspace → L02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Vietnamese	vi-VN	Read-only within object scope; display source/time when they affect decisions.
English	en	Read-only within object scope; display source/time when they affect decisions.
Japanese	ja-JP	Read-only within object scope; display source/time when they affect decisions.
Critical templates	Review required before publish	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L02 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L01 — V2 visual composition

Context: configuration · **Role:** Localization manager

Primary question: For language coverage, what do Vietnamese, English establish, and is "Open translation workspace" available to Localization manager in this scope?

Primary visual: Language coverage grid highlighting critical missing content

Primary action: Open translation workspace → L02

Progressive disclosure: Keep Vietnamese, English in the decision layer. Put the supporting details for Japanese, Critical templates, full source/version history and FR-L01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a language coverage fixture for Localization manager; change Vietnamese and make Critical templates unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Open translation workspace" reaches L02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L02 — Translation editor

EN: Translation editor · **JA:** 翻訳エディター
Requirement: FR-L02 / WF-18 · **Actor:** Translator · **Surface:** Localization · **Layout:** form

Purpose: Translate a versioned message.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: L01, L06.
Output: Submit translation review → L03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Template	Fire evacuation v4	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Source language	English	Read-only from the authoritative source/version; cannot be edited in this form.
Target language	Japanese	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Variables	site_name + issued_at	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Review	Required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L03 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L02 — V2 visual composition

Context: configuration · **Role:** Translator

Primary question: For translation editor, what do Template, Source language establish, and is “Submit translation review” available to Translator in this scope?

Primary visual: Source/translation paired editor with protected variables

Primary action: Submit translation review → L03

Progressive disclosure: Keep Template, Source language in the decision layer. Put the supporting details for Target language, Variables, Review, full source/version history and FR-L02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a translation editor fixture for Translator; change Template and make Review unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Submit translation review” reaches L03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L03 — Translation review

EN: Translation review · **JA:** 翻訳のレビュー
Requirement: FR-L03 / WF-18 · **Actor:** Language reviewer · **Surface:** Localization · **Layout:** detail

Purpose: Review terminology and emergency meaning.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: L02.
Output: Approve translation version → L01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Source version	4	Read-only within object scope; display source/time when they affect decisions.
Target locale	ja-JP	Read-only within object scope; display source/time when they affect decisions.
Variable parity	Passed	Read-only within object scope; display source/time when they affect decisions.
Voice transcript	Matched	Read-only within object scope; display source/time when they affect decisions.
Reviewer	Independent language reviewer	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L01 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L03 — V2 visual composition

Context: configuration · **Role:** Language reviewer

Primary question: For translation review, what do Source version, Target locale establish, and is “Approve translation version” available to Language reviewer in this scope?

Primary visual: Version-parity review showing text, variables and voice transcript

Primary action: Approve translation version → L01

Progressive disclosure: Keep Source version, Target locale in the decision layer. Put the supporting details for Variable parity, Voice transcript, Reviewer, full source/version history and FR-L03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a translation review fixture for Language reviewer; change Source version and make Reviewer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve translation version” reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L04 — Voice & pronunciation

EN: Voice & pronunciation · **JA:** 音声と発音
Requirement: FR-L04 / WF-18 · **Actor:** Content editor · **Surface:** Localization · **Layout:** form

Purpose: Keep speech aligned with reviewed text.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: P08.
Output: Save voice configuration → L01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Locale	ja-JP	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Voice	Approved Japanese voice	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Transcript	Approved template v4	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Site pronunciation	Reviewed	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Fallback	Text transcript	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L01 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L04 — V2 visual composition

Context: configuration · **Role:** Content editor

Primary question: For voice & pronunciation, what do Locale, Voice establish, and is “Save voice configuration” available to Content editor in this scope?

Primary visual: Pronunciation samples paired with exact transcript and fallback

Primary action: Save voice configuration → L01

Progressive disclosure: Keep Locale, Voice in the decision layer. Put the supporting details for Transcript, Site pronunciation, Fallback, full source/version history and FR-L04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a voice & pronunciation fixture for Content editor; change Locale and make Fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save voice configuration” reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L05 — Locale fallback policy

EN: Locale fallback policy · **JA:** 言語フォールバック設定
Requirement: FR-L05 / WF-18 · **Actor:** Tenant admin · **Surface:** Localization · **Layout:** form

Purpose: Make missing translations visible.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: Role-based menu / deep link after authorization.
Output: Save language policy → L01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Supported	vi-VN / en / ja-JP	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
User preference	Explicit profile choice	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Missing critical content	Approved fallback only	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Fallback language	English	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L01 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L05 — V2 visual composition

Context: configuration · **Role:** Tenant admin

Primary question: For locale fallback policy, what do Supported, User preference establish, and is “Save language policy” available to Tenant admin in this scope?

Primary visual: Locale fallback tree showing missing-critical-content behavior

Primary action: Save language policy → L01

Progressive disclosure: Keep Supported, User preference in the decision layer. Put the supporting details for Missing critical content, Fallback language, full source/version history and FR-L05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a locale fallback policy fixture for Tenant admin; change Supported and make Fallback language unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Save language policy" reaches L01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

L06 — Emergency terminology

EN: Emergency terminology · **JA:** 緊急対応用語
Requirement: FR-L06 / WF-18 · **Actor:** Language reviewer · **Surface:** Localization · **Layout:** list

Purpose: Keep terms consistent across channels.
Key rule: Vietnamese, English and Japanese content is versioned with approved fallback.
Entry: Role-based menu / deep link after authorization.
Output: Review translation editor → L02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
SOS	Yêu cầu trợ giúp / Assistance request / 救援要請	Read-only within object scope; display source/time when they affect decisions.
Safe	Tự báo an toàn / Self-reported safe / 本人の安全報告	Read-only within object scope; display source/time when they affect decisions.
Unknown	Chưa rõ / Unknown / 不明	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-L06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L02 with a valid payload/version, and verify that it is blocked when the guard for WF-18 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

L06 — V2 visual composition

Context: configuration · **Role:** Language reviewer

Primary question: For emergency terminology, what do SOS, Safe establish, and is “Review translation editor” available to Language reviewer in this scope?

Primary visual: Emergency term comparison distinguishing safe, unknown and SOS

Primary action: Review translation editor → L02

Progressive disclosure: Keep SOS, Safe in the decision layer. Put the supporting details for Unknown, full source/version history and FR-L06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Block publication when critical variables, source version or approved voice meaning diverge; show missing translations explicitly.

V2 acceptance scenario: Use a emergency terminology fixture for Language reviewer; change SOS and make Unknown unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review translation editor” reaches L02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C31 — Circle membership & invitations

EN: Circle membership & invitations · **JA:** グループメンバーと招待
Requirement: FR-C31 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** list

Purpose: Control who receives your updates.
Key rule: A personal safety report does not close the incident.
Entry: C32.
Output: Manage invitation → C32.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	Owner	Read-only within object scope; display source/time when they affect decisions.
Jamie	Member	Read-only within object scope; display source/time when they affect decisions.
Morgan	Invite pending	Read-only within object scope; display source/time when they affect decisions.
Sharing	Status only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C31: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C32 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C31 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For circle membership & invitations, what do Alex, Jamie establish, and is “Manage invitation” available to Citizen in this scope?

Primary visual: Circle roster with invitation state and sharing privileges

Primary action: Manage invitation → C32

Progressive disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Sharing, full source/version history and FR-C31 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a circle membership & invitations fixture for Citizen; change Alex and make Sharing unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Manage invitation” reaches C32 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C32 — Invite circle member

EN: Invite circle member · **JA:** メンバーの招待
Requirement: FR-C32 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Invite someone you trust.
Key rule: A personal safety report does not close the incident.
Entry: C31.
Output: Create invitation — sample → C31.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Circle	Family	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Contact	jamie@example.test	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Role	Member	Select within granted permissions; the server blocks escalation beyond the actor’s rights.
Invitation expiry	7 days	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Field / component	Sample value	Validation / editing permission
Location access	Not included	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C32: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C31 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C32 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For invite circle member, what do Circle, Contact establish, and is “Create invitation — sample” available to Citizen in this scope?

Primary visual: Invitation preview with recipient, role and location access

Primary action: Create invitation — sample → C31

Progressive disclosure: Keep Circle, Contact in the decision layer. Put the supporting details for Role, Invitation expiry, Location access, full source/version history and FR-C32 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a invite circle member fixture for Citizen; change Circle and make Location access unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Create invitation — sample” reaches C31 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C33 — Report or block conversation

EN: Report or block conversation · **JA:** 会話の報告・ブロック
Requirement: FR-C33 / WF-12 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Control unwanted social contact.
Key rule: A personal safety report does not close the incident.
Entry: Role-based menu / deep link after authorization.

Output: Submit report — sample → C19.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Conversation	Travel team	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reason	Unwanted messages	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Action	Report and block sender	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Site warnings	Remain governed by safety policy	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C33: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C19 with a valid payload/version, and verify that it is blocked when the guard for WF-12 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C33 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For report or block conversation, what do Conversation, Reason establish, and is "Submit report — sample" available to Citizen in this scope?

Primary visual: Conversation report scope and block effect on safety warnings

Primary action: Submit report — sample → C19

Progressive disclosure: Keep Conversation, Reason in the decision layer. Put the supporting details for Action, Site warnings, full source/version history and FR-C33 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a report or block conversation fixture for Citizen; change Conversation and make Site warnings unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Submit report — sample" reaches C19 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record

observed results separately from this specification.

C34 — Notification preferences

EN: Notification preferences · **JA:** 通知設定
Requirement: FR-C34 / WF-10 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Choose channels and quiet hours.
Key rule: Sent does not mean delivered, read or acknowledged by a person.
Entry: Role-based menu / deep link after authorization.
Output: Save preferences → C01.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Language	Profile language	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Social messages	Quiet hours enabled	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Safety alerts	Site policy applies	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Discreet requests	No sound	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
SMS	Verified contact required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C34: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C01 with a valid payload/version, and verify that it is blocked when the guard for WF-10 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C34 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For notification preferences, what do Language, Social messages establish, and is “Save preferences” available to Citizen in this scope?

Primary visual: Separate social, safety and discreet notification preferences

Primary action: Save preferences → C01

Progressive disclosure: Keep Language, Social messages in the decision layer. Put the supporting details for Safety alerts, Discreet requests, SMS, full source/version history and FR-C34 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a notification preferences fixture for Citizen; change Language and make SMS unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Save preferences" reaches C01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O31 — Incident archive & reopen

EN: Incident archive & reopen · **JA:** 過去のインシデント
Requirement: FR-O31 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** list

Purpose: Review closed incidents with full history.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: Role-based menu / deep link after authorization.
Output: Review reopening → O32.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
INC-2032	Closed · documented handover	Read-only within object scope; display source/time when they affect decisions.
INC-2028	Dismissed · false detection	Read-only within object scope; display source/time when they affect decisions.
INC-2010	Closed · all tasks resolved	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O31: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O32 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O31 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For incident archive & reopen, what do INC-2032, INC-2028 establish, and is “Review reopening” available to Supervisor in this scope?

Primary visual: Closed-incident archive preserving original chronology

Primary action: Review reopening → O32

Progressive disclosure: Keep INC-2032, INC-2028 in the decision layer. Put the supporting details for INC-2010, full source/version history and FR-O31 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a incident archive & reopen fixture for Supervisor; change INC-2032 and make INC-2010 unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review reopening” reaches O32 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O32 — Reopen incident

EN: Reopen incident · **JA:** インシデントの再開
Requirement: FR-O32 / WF-20 · **Actor:** Supervisor · **Surface:** Command · **Layout:** form

Purpose: Record why response needs to resume.
Key rule: Every decision needs an owner, evidence, time and recovery path.
Entry: O31.
Output: Reopen incident — sample → O04.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Incident	INC-2032	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Reason	New related evidence	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Owner	Campus command	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Severity	P1	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
History	Preserved	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O32: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O04 with a valid payload/version, and verify that it is blocked when the guard for WF-20 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O32 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For reopen incident, what do Incident, Reason establish, and is “Reopen incident — sample” available to Supervisor in this scope?

Primary visual: Reopening preview showing new owner and retained evidence

Primary action: Reopen incident — sample → O04

Progressive disclosure: Keep Incident, Reason in the decision layer. Put the supporting details for Owner, Severity, History, full source/version history and FR-O32 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a reopen incident fixture for Supervisor; change Incident and make History unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Reopen incident — sample” reaches O04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O33 — Federated incident sharing

EN: Federated incident sharing · **JA:** 組織間のインシデント共有
Requirement: FR-O33 / WF-17 · **Actor:** Supervisor · **Surface:** Command · **Layout:** detail

Purpose: Share a controlled incident projection.
Key rule: Enable only tested declared capabilities; do not replay physical commands.
Entry: Role-based menu / deep link after authorization.
Output: Review partner exchange → I10.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Partner	City command	Read-only within object scope; display source/time when they affect decisions.
Fields	Incident location + pseudonymous SOS	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Authority	Local incident owner retained	Read-only within object scope; display source/time when they affect decisions.
Agreement	Version 3 approved	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O33: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I10 with a valid payload/version, and verify that it is blocked when the guard for WF-17 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O33 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For federated incident sharing, what do Partner, Fields establish, and is “Review partner exchange” available to Supervisor in this scope?

Primary visual: Partner exchange preview showing disclosed fields and authority

Primary action: Review partner exchange → I10

Progressive disclosure: Keep Partner, Fields in the decision layer. Put the supporting details for Authority, Agreement, full source/version history and FR-O33 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a federated incident sharing fixture for Supervisor; change Partner and make Agreement unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review partner exchange” reaches I10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O34 — Accessibility assistance queue

EN: Accessibility assistance queue · **JA:** 配慮が必要な方への支援
Requirement: FR-O34 / WF-11 · **Actor:** Dispatcher · **Surface:** Command · **Layout:** list

Purpose: Assign support based on reported needs.
Key rule: Each task has one accountable owner; preserve handover history.
Entry: Role-based menu / deep link after authorization.

Output: Assign capable unit → O09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
SOS-041	Mobility assistance · Room 505	Read-only within object scope; display source/time when they affect decisions.
SOS-044	Text communication preferred	Read-only within object scope; display source/time when they affect decisions.
SOS-046	Assistance type unknown	Read-only within object scope; display source/time when they affect decisions.
Source	Self-reported needs	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O34: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O09 with a valid payload/version, and verify that it is blocked when the guard for WF-11 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O34 — V2 visual composition

Context: command · **Role:** Dispatcher

Primary question: For accessibility assistance queue, what do SOS-041, SOS-044 establish, and is “Assign capable unit” available to Dispatcher in this scope?

Primary visual: Accessibility queue matching functional needs to capable units

Primary action: Assign capable unit → O09

Progressive disclosure: Keep SOS-041, SOS-044 in the decision layer. Put the supporting details for SOS-046, Source, full source/version history and FR-O34 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep incident scope, evidence age and accountable owner visible; block consequential action when required authority or current evidence is unavailable.

V2 acceptance scenario: Use a accessibility assistance queue fixture for Dispatcher; change SOS-041 and make Source unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Assign capable unit” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C35 — Voice evacuation navigation

EN: Voice evacuation navigation · **JA:** 音声による避難ナビゲーション
Requirement: FR-C35 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** navigation

Purpose: Follow reviewed steps away from the danger area.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: C37, C40.
Output: Show next instruction → C36.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Location	Building A · Floor 05	Read-only within object scope; display source/time when they affect decisions.
Next instruction	Proceed to the marked west stairwell	Read-only within object scope; display source/time when they affect decisions.
Position quality	QR anchor confirmed	Read-only within object scope; display source/time when they affect decisions.
Hazard data	Updated 8 seconds ago	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C35: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C36 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C35 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For voice evacuation navigation, what do Location, Next instruction establish, and is “Show next instruction” available to Citizen in this scope?

Primary visual: Next landmark instruction above floor route and position-quality band

Primary action: Show next instruction → C36

Progressive disclosure: Keep Location, Next instruction in the decision layer. Put the supporting details for Position quality, Hazard data, full source/version history and FR-C35 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Stop obsolete speech when guidance version changes; stale position must not advance the next landmark.

V2 acceptance scenario: Use a voice evacuation navigation fixture for Citizen; change Location and make Hazard data unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Show next instruction” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C36 — Navigation step & orientation

EN: Navigation step & orientation · **JA:** 経路案内と方向確認
Requirement: FR-C36 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** navigation

Purpose: Use the WEST STAIR sign as your landmark.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: C35, C36, C39.
Output: Repeat voice instruction → C36.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Step	2 of 5	Read-only within object scope; display source/time when they affect decisions.
Instruction	Continue to the WEST STAIR sign	Read-only within object scope; display source/time when they affect decisions.
Heading	Uncertain · landmark guidance	Read-only within object scope; display source/time when they affect decisions.
Accessible route	Assistance refuge B	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C36: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C36 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C36 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For navigation step & orientation, what do Step, Instruction establish, and is “Repeat voice instruction” available to Citizen in this scope?

Primary visual: Single navigation step with landmark picture and heading-confidence label

Primary action: Repeat voice instruction → C36

Progressive disclosure: Keep Step, Instruction in the decision layer. Put the supporting details for Heading, Accessible route, full source/version history and FR-C36 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Unknown heading removes directional arrows; use a verified landmark and manual orientation instead of fabricated left/right.

V2 acceptance scenario: Use a navigation step & orientation fixture for Citizen; change Step and make Accessible route unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Repeat voice instruction” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C37 — Route change notification

EN: Route change notification · **JA:** 経路変更のお知らせ
Requirement: FR-C37 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: The previous route is no longer recommended.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: Role-based menu / deep link after authorization.
Output: Review updated route → C35.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Reason	New hazard reported on corridor E	Read-only within object scope; display source/time when they affect decisions.
Old route	Withdrawn	Read-only within object scope; display source/time when they affect decisions.
New instruction	Follow the newly reviewed west route	Read-only within object scope; display source/time when they affect decisions.
Guidance version	18	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C37: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C35 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C37 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For route change notification, what do Reason, Old route establish, and is “Review updated route” available to Citizen in this scope?

Primary visual: Withdrawn old route beside approved replacement and change reason

Primary action: Review updated route → C35

Progressive disclosure: Keep Reason, Old route in the decision layer. Put the supporting details for New instruction, Guidance version, full source/version history and FR-C37 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Withdraw old-route visuals and queued speech atomically; do not display replacement as approved before its version is accepted.

V2 acceptance scenario: Use a route change notification fixture for Citizen; change Reason and make Guidance version unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review updated route” reaches C35 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C38 — Position uncertain / no route

EN: Position uncertain / no route · **JA:** 位置不明・経路なし
Requirement: FR-C38 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** alert

Purpose: We cannot verify a route from your position.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: Role-based menu / deep link after authorization.
Output: Request assistance → C05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Floor	Not confirmed	Read-only within object scope; display source/time when they affect decisions.
Heading	Unreliable indoors	Read-only within object scope; display source/time when they affect decisions.
Next step	Identify a marked location or request help	Read-only within object scope; display source/time when they affect decisions.
Voice	Location uncertainty announced	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require

permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C38: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C05 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C38 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For position uncertain / no route, what do Floor, Heading establish, and is “Request assistance” available to Citizen in this scope?

Primary visual: Unknown floor/heading panel with manual location and assistance controls

Primary action: Request assistance → C05

Progressive disclosure: Keep Floor, Heading in the decision layer. Put the supporting details for Next step, Voice, full source/version history and FR-C38 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Do not invent a route or heading. Keep location correction and assistance reachable without permission recovery.

V2 acceptance scenario: Use a position uncertain / no route fixture for Citizen; change Floor and make Voice unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Request assistance” reaches C05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C39 — Outdoor hazard avoidance

EN: Outdoor hazard avoidance · **JA:** 危険区域からの移動
Requirement: FR-C39 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** navigation

Purpose: Leave the affected area using reviewed guidance.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: Role-based menu / deep link after authorization.
Output: View next route step → C36.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Area	Riverside flood polygon	Read-only within object scope; display source/time when they affect decisions.
Destination	North assembly point	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Road condition	Verified segments only	Read-only within object scope; display source/time when they affect decisions.
Unknown segments	Excluded	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C39: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C36 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C39 — V2 visual composition

Context: emergency · **Role:** Citizen

Primary question: For outdoor hazard avoidance, what do Area, Destination establish, and is “View next route step” available to Citizen in this scope?

Primary visual: Outdoor hazard boundary, destination and unknown road segments

Primary action: View next route step → C36

Progressive disclosure: Keep Area, Destination in the decision layer. Put the supporting details for Road condition, Unknown segments, full source/version history and FR-C39 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a outdoor hazard avoidance fixture for Citizen; change Area and make Unknown segments unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “View next route step” reaches C36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

C40 — Voice language & guidance settings

EN: Voice language & guidance settings · **JA:** 音声案内の設定
Requirement: FR-C40 / WF-22 · **Actor:** Citizen · **Surface:** Citizen · **Layout:** form

Purpose: Choose how guidance reaches you.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: Role-based menu / deep link after authorization.

Output: Save guidance preferences → C35.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Language	Vietnamese / English / Japanese	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Voice	Device-supported reviewed voice	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Speech rate	Normal	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Haptics	On	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Discreet mode	Respected	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-C40: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to C35 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

C40 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For voice language & guidance settings, what do Language, Voice establish, and is “Save guidance preferences” available to Citizen in this scope?

Primary visual: Voice sample, rate control and discreet-mode preview

Primary action: Save guidance preferences → C35

Progressive disclosure: Keep Language, Voice in the decision layer. Put the supporting details for Speech rate, Haptics, Discreet mode, full source/version history and FR-C40 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Separate local intent from confirmed delivery and show the last verified source; an unavailable observation must not become a safe status.

V2 acceptance scenario: Use a voice language & guidance settings fixture for Citizen; change Language and make Discreet mode unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save guidance preferences” reaches C35 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P01 — Speaker zones & coverage

EN: Speaker zones & coverage · **JA:** 放送区域とカバレッジ
Requirement: FR-P01 / WF-21 · **Actor:** PA operator · **Surface:** Public address · **Layout:** map

Purpose: Reach people without a smartphone.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: P05, P09, O36.
Output: Select broadcast zones → P02.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Building A	Floors 01–08 · 24 speakers	Read-only within object scope; display source/time when they affect decisions.
Outdoor campus	6 speakers	Read-only within object scope; display source/time when they affect decisions.
Coverage gaps	West loading bay	Read-only within object scope; display source/time when they affect decisions.
Live incident	INC-2041	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P02 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P01 — V2 visual composition

Context: command · **Role:** PA operator

Primary question: For speaker zones & coverage, what do Building A, Outdoor campus establish, and is “Select broadcast zones” available to PA operator in this scope?

Primary visual: Speaker coverage map with uncovered zones and incident context

Primary action: Select broadcast zones → P02

Progressive disclosure: Keep Building A, Outdoor campus in the decision layer. Put the supporting details for Coverage gaps, Live incident, full source/version history and FR-P01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a speaker zones & coverage fixture for PA operator; change Building A and make Live incident unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Select broadcast zones” reaches P02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P02 — Compose rescue announcement

EN: Compose rescue announcement · **JA:** 救援放送の作成
Requirement: FR-P02 / WF-21 · **Actor:** PA author · **Surface:** Public address · **Layout:** form

Purpose: Prepare an announcement for the right area.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: P01.
Output: Preview announcement → P03.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Zones	A05 + stairwell W	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Priority	Emergency	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Languages	vi-VN / en / ja-JP	Supported locale vi-VN/en/ja-JP; critical content requires an approved version.
Source	Approved rescue template	Read-only from the authoritative source/version; cannot be edited in this form.
Expires	60 seconds after approval	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P03 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P02 — V2 visual composition

Context: command · **Role:** PA author

Primary question: For compose rescue announcement, what do Zones, Priority establish, and is “Preview announcement” available to PA author in this scope?

Primary visual: Announcement text/audio preview tied to zones and expiry

Primary action: Preview announcement → P03

Progressive disclosure: Keep Zones, Priority in the decision layer. Put the supporting details for Languages, Source, Expires, full source/version history and FR-P02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a compose rescue announcement fixture for PA author; change Zones and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Preview announcement” reaches P03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P03 — Preview & approve announcement

EN: Preview & approve announcement · **JA:** 放送の確認と承認
Requirement: FR-P03 / WF-21 · **Actor:** PA approver · **Surface:** Public address · **Layout:** detail

Purpose: Check message, zones and playback behavior.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: P02.
Output: Approve sample playback → P04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Message version	PA-04 v3	Read-only within object scope; display source/time when they affect decisions.
Selected speakers	8	Read-only within object scope; display source/time when they affect decisions.
Language sequence	Vietnamese → English → Japanese	Read-only within object scope; display source/time when they affect decisions.
Repetition	2 cycles	Read-only within object scope; display source/time when they affect decisions.
Approver	Independent supervisor	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P04 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P03 — V2 visual composition

Context: command · **Role:** PA approver

Primary question: For preview & approve announcement, what do Message version, Selected speakers establish, and is “Approve sample playback” available to PA approver in this scope?

Primary visual: Exact message version with selected speakers and language sequence

Primary action: Approve sample playback → P04

Progressive disclosure: Keep Message version, Selected speakers in the decision layer. Put the supporting details for Language sequence, Repetition, Approver, full source/version history and FR-P03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a preview & approve announcement fixture for PA approver; change Message version and make Approver unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve sample playback” reaches P04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P04 — Speaker playback monitoring

EN: Speaker playback monitoring · **JA:** 放送状況の監視
Requirement: FR-P04 / WF-21 · **Actor:** PA operator · **Surface:** Public address · **Layout:** timeline

Purpose: Track playback separately from command receipt.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: P03, P06.
Output: Inspect speaker health → P05.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Controller	Accepted	Read-only within object scope; display source/time when they affect decisions.
Speakers	7 playing · 1 unknown	Read-only within object scope; display source/time when they affect decisions.
Acoustic verification	Not available	Read-only within object scope; display source/time when they affect decisions.
Next	Inspect unavailable speaker	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P05 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P04 — V2 visual composition

Context: command · **Role:** PA operator

Primary question: For speaker playback monitoring, what do Controller, Speakers establish, and is “Inspect speaker health” available to PA operator in this scope?

Primary visual: Playback progress distinguishing controller receipt from audibility

Primary action: Inspect speaker health → P05

Progressive disclosure: Keep Controller, Speakers in the decision layer. Put the supporting details for Acoustic verification, Next, full source/version history and FR-P04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Speaker receipt cannot prove occupants heard the message; coverage gaps and unknown acoustic evidence remain prominent.

V2 acceptance scenario: Use a speaker playback monitoring fixture for PA operator; change Controller and make Next unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect speaker health” reaches P05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P05 — Speaker device & self-test

EN: Speaker device & self-test · **JA:** スピーカーと自己診断
Requirement: FR-P05 / WF-21 · **Actor:** Device admin · **Surface:** Public address · **Layout:** detail

Purpose: SPK-A05-08 · playback unknown.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: P04.
Output: Open coverage fallback → P01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Power	Online	Read-only within object scope; display source/time when they affect decisions.
Network	Connected	Read-only within object scope; display source/time when they affect decisions.
Amplifier	Fault detected	Read-only within object scope; display source/time when they affect decisions.
Last self-test	Yesterday	Read-only within object scope; display source/time when they affect decisions.
Coverage fallback	Patrol announcement	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P01 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P05 — V2 visual composition

Context: command · **Role:** Device admin

Primary question: For speaker device & self-test, what do Power, Network establish, and is “Open coverage fallback” available to Device admin in this scope?

Primary visual: Speaker power-network-amplifier chain with self-test age

Primary action: Open coverage fallback → P01

Progressive disclosure: Keep Power, Network in the decision layer. Put the supporting details for Amplifier, Last self-test, Coverage fallback, full source/version history and FR-P05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a speaker device & self-test fixture for Device admin; change Power and make Coverage fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open coverage fallback” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P06 — Live rescue microphone

EN: Live rescue microphone · **JA:** 救援用ライブマイク
Requirement: FR-P06 / WF-21 · **Actor:** Authorized announcer · **Surface:** Public address · **Layout:** livevoice

Purpose: Announce live to approved rescue zones.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: Role-based menu / deep link after authorization.
Output: Review live session — sample → P04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Zones	A05 + stairwell W	Read-only within object scope; display source/time when they affect decisions.
Microphone	Permission required	Read-only within object scope; display source/time when they affect decisions.
Priority	Emergency	Read-only within object scope; display source/time when they affect decisions.
Maximum session	120 seconds	Read-only within object scope; display source/time when they affect decisions.
Recording	Policy-controlled	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P04 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P06 — V2 visual composition

Context: command · **Role:** Authorized announcer

Primary question: For live rescue microphone, what do Zones, Microphone establish, and is “Review live session — sample” available to Authorized announcer in this scope?

Primary visual: Live microphone session with selected zones and explicit stop control

Primary action: Review live session — sample → P04

Progressive disclosure: Keep Zones, Microphone in the decision layer. Put the supporting details for Priority, Maximum session, Recording, full source/version history and FR-P06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a live rescue microphone fixture for Authorized announcer; change Zones and make Recording unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review live session — sample" reaches P04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P07 — Drone / vehicle speaker mission

EN: Drone / vehicle speaker mission · **JA:** ドローン・車両の拡声放送
Requirement: FR-P07 / WF-21 · **Actor:** Drone operator · **Surface:** Public address · **Layout:** form

Purpose: Coordinate mobile public-address support.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: Role-based menu / deep link after authorization.
Output: Review mission → I13.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Platform	DR-01 speaker payload	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Message	Approved rescue template	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Pilot	Assigned	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Broadcast area	Campus perimeter	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Playback	Requires payload feedback	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to I13 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P07 — V2 visual composition

Context: command · **Role:** Drone operator

Primary question: For drone / vehicle speaker mission, what do Platform, Message establish, and is “Review mission” available to Drone operator in this scope?

Primary visual: Mobile speaker mission footprint with pilot ownership

Primary action: Review mission → I13

Progressive disclosure: Keep Platform, Message in the decision layer. Put the supporting details for Pilot, Broadcast area, Playback, full source/version history and FR-P07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a drone / vehicle speaker mission fixture for Drone operator; change Platform and make Playback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review mission” reaches I13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P08 — Audio library & TTS review

EN: Audio library & TTS review · **JA:** 音声ライブラリーとTTS審査
Requirement: FR-P08 / WF-21 · **Actor:** Content editor · **Surface:** Public address · **Layout:** list

Purpose: Use approved speech in every supported language.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: Role-based menu / deep link after authorization.
Output: Edit voice content → L04.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
PA-04 vi-VN	Reviewed recording	Read-only within object scope; display source/time when they affect decisions.
PA-04 en	Reviewed recording	Read-only within object scope; display source/time when they affect decisions.
PA-04 ja-JP	TTS reviewed	Read-only within object scope; display source/time when they affect decisions.
Pronunciation	Site names verified	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to L04 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P08 — V2 visual composition

Context: configuration · **Role:** Content editor

Primary question: For audio library & tts review, what do PA-04 vi-VN, PA-04 en establish, and is “Edit voice content” available to Content editor in this scope?

Primary visual: Language audio library with pronunciation review and transcript

Primary action: Edit voice content → L04

Progressive disclosure: Keep PA-04 vi-VN, PA-04 en in the decision layer. Put the supporting details for PA-04 ja-JP, Pronunciation, full source/version history and FR-P08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a audio library & tts review fixture for Content editor; change PA-04 vi-VN and make Pronunciation unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Edit voice content” reaches L04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

P09 — PA priority & arbitration

EN: PA priority & arbitration · **JA:** 放送の優先順位と調整
Requirement: FR-P09 / WF-21 · **Actor:** PA admin · **Surface:** Public address · **Layout:** form

Purpose: Avoid competing announcements.
Key rule: Track each speaker; playback completion does not prove people heard it.
Entry: Role-based menu / deep link after authorization.
Output: Save arbitration policy → P01.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Highest priority	Local certified emergency controller	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Cloud rescue message	Approved priority tier	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Routine audio	Preemptible	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Lost link	Local controller policy	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Resume	Explicit decision	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-P09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P01 with a valid payload/version, and verify that it is blocked when the guard for WF-21 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

P09 — V2 visual composition

Context: configuration · **Role:** PA admin

Primary question: For pa priority & arbitration, what do Highest priority, Cloud rescue message establish, and is “Save arbitration policy” available to PA admin in this scope?

Primary visual: Priority arbitration stack showing local priority and lost-link behavior

Primary action: Save arbitration policy → P01

Progressive disclosure: Keep Highest priority, Cloud rescue message in the decision layer. Put the supporting details for Routine audio, Lost link, Resume, full source/version history and FR-P09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Distinguish controller receipt, playback feedback and acoustic verification; retain local priority and stop/expiry behavior on lost link.

V2 acceptance scenario: Use a pa priority & arbitration fixture for PA admin; change Highest priority and make Resume unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save arbitration policy” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S15 — Evacuation graph editor

EN: Evacuation graph editor · **JA:** 避難経路グラフの編集
Requirement: FR-S15 / WF-22 · **Actor:** Route designer · **Surface:** Sites · **Layout:** floor

Purpose: Build a versioned indoor and outdoor route graph.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: S16, Q03.
Output: Review route validation → S16.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Nodes	Doors · stairs · refuges · exits	Read-only within object scope; display source/time when they affect decisions.
Edges	Direction + width + accessibility	Read-only within object scope; display source/time when they affect decisions.
Elevators	Excluded unless approved evacuation use	Read-only within object scope; display source/time when they affect decisions.
Map revision	13 draft	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S15: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S16 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S15 — V2 visual composition

Context: configuration · **Role:** Route designer

Primary question: For evacuation graph editor, what do Nodes, Edges establish, and is “Review route validation” available to Route designer in this scope?

Primary visual: Evacuation graph with node/edge types and inaccessible links

Primary action: Review route validation → S16

Progressive disclosure: Keep Nodes, Edges in the decision layer. Put the supporting details for Elevators, Map revision, full source/version history and FR-S15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a evacuation graph editor fixture for Route designer; change Nodes and make Map revision unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Review route validation" reaches S16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

S16 — Route simulation & approval

EN: Route simulation & approval · **JA:** 経路の検証と承認
Requirement: FR-S16 / WF-22 · **Actor:** Safety reviewer · **Surface:** Sites · **Layout:** detail

Purpose: Test hazards, accessibility and uncertainty.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: S15, Q10.
Output: Publish reviewed graph → S15.
Data: Configuration/operations; do not expose credentials or secrets.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Blocked corridor	Excluded	Read-only within object scope; display source/time when they affect decisions.
Unknown floor	No turn-by-turn route	Read-only within object scope; display source/time when they affect decisions.
Wheelchair profile	No stair-only route	Read-only within object scope; display source/time when they affect decisions.
Expired hazard data	Guidance degraded	Read-only within object scope; display source/time when they affect decisions.
Reviewer	Site safety specialist	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-S16: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S15 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

S16 — V2 visual composition

Context: configuration · **Role:** Safety reviewer

Primary question: For route simulation & approval, what do Blocked corridor, Unknown floor establish, and is "Publish reviewed graph" available to Safety reviewer in this scope?

Primary visual: Scenario comparison showing blocked, unknown and wheelchair routes

Primary action: Publish reviewed graph → S15

Progressive disclosure: Keep Blocked corridor, Unknown floor in the decision layer. Put the supporting details for Wheelchair profile, Expired hazard data, Reviewer, full source/version history and FR-S16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep draft geometry/configuration separate from published operational data; invalidate dependent routes or rules when their referenced version changes.

V2 acceptance scenario: Use a route simulation & approval fixture for Safety reviewer; change Blocked corridor and make Reviewer unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Publish reviewed graph" reaches S15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O35 — Hazard zones & route restrictions

EN: Hazard zones & route restrictions · **JA:** 危険区域と移動制限
Requirement: FR-O35 / WF-22 · **Actor:** Incident commander · **Surface:** Command · **Layout:** map

Purpose: Publish time-bounded movement restrictions.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: Role-based menu / deep link after authorization.
Output: Review guidance impact → O36.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Hazard	East corridor smoke report	Read-only within object scope; display source/time when they affect decisions.
Geometry	Zone A05-03	Read-only within object scope; display source/time when they affect decisions.
Certainty	Operator verified	Read-only within object scope; display source/time when they affect decisions.
Expires	Recheck in 2 minutes	Read-only within object scope; display source/time when they affect decisions.
Affected routes	3 invalidated	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O35: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O36 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O35 — V2 visual composition

Context: command · **Role:** Incident commander

Primary question: For hazard zones & route restrictions, what do Hazard, Geometry establish, and is “Review guidance impact” available to Incident commander in this scope?

Primary visual: Hazard polygon editor with affected route segments and expiry

Primary action: Review guidance impact → O36

Progressive disclosure: Keep Hazard, Geometry in the decision layer. Put the supporting details for Certainty, Expires, Affected routes, full source/version history and FR-O35 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Expired geometry removes its authority; changed restrictions identify affected guidance versions before publication.

V2 acceptance scenario: Use a hazard zones & route restrictions fixture for Incident commander; change Hazard and make Affected routes unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review guidance impact” reaches O36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

O36 — Guidance distribution & receipts

EN: Guidance distribution & receipts · **JA:** 案内の配信と受領確認
Requirement: FR-O36 / WF-22 · **Actor:** Operator · **Surface:** Command · **Layout:** list

Purpose: Keep app and speaker instructions consistent.
Key rule: Use landmarks when heading is uncertain; withdraw expired guidance.
Entry: O35, Q05.
Output: Open speaker operations → P01.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Guidance version	18	Read-only within object scope; display source/time when they affect decisions.
Smartphones	312 updated · 28 unknown	Read-only within object scope; display source/time when they affect decisions.
Speaker controller	Version 18 accepted	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Old route	Withdrawn	Read-only within object scope; display source/time when they affect decisions.
Unreachable users	Fallback action required	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-O36: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to P01 with a valid payload/version, and verify that it is blocked when the guard for WF-22 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

O36 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For guidance distribution & receipts, what do Guidance version, Smartphones establish, and is “Open speaker operations” available to Operator in this scope?

Primary visual: Guidance version distribution separating withdrawal and delivery receipts

Primary action: Open speaker operations → P01

Progressive disclosure: Keep Guidance version, Smartphones in the decision layer. Put the supporting details for Speaker controller, Old route, Unreachable users, full source/version history and FR-O36 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Old-route withdrawal and new-version delivery have separate receipts; unreachable people remain visible and require alternate channels.

V2 acceptance scenario: Use a guidance distribution & receipts fixture for Operator; change Guidance version and make Unreachable users unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open speaker operations” reaches P01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H01 — My rescue assistance profile

EN: My rescue assistance profile · **JA:** 自分の救援支援プロフィール
Requirement: FR-H01 / WF-26 · **Actor:** Citizen · **Surface:** Care profiles · **Layout:** form

Purpose: Tell responders what support you may need.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H02.

Output: Review emergency sharing → H02.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Age group	65–74	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Mobility	Wheelchair user	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Stairs	Assistance required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Communication	Text preferred	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Source	Self-reported	Read-only from the authoritative source/version; cannot be edited in this form.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H02 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H01 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For my rescue assistance profile, what do Age group, Mobility establish, and is “Review emergency sharing” available to Citizen in this scope?

Primary visual: Functional assistance profile with mobility and communication pictograms

Primary action: Review emergency sharing → H02

Progressive disclosure: Keep Age group, Mobility in the decision layer. Put the supporting details for Stairs, Communication, Source, full source/version history and FR-H01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a my rescue assistance profile fixture for Citizen; change Age group and make Source unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review emergency sharing” reaches H02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H02 — Health & assistance sharing

EN: Health & assistance sharing · **JA:** 支援・健康情報の共有
Requirement: FR-H02 / WF-26 · **Actor:** Citizen · **Surface:** Care profiles · **Layout:** form

Purpose: Share only what the response team needs.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H01.
Output: Save sharing preferences → H01.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Functional needs	Assigned rescue team	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Medical notes	Authorized medical role only	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Family groups	Not shared	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Consent	Version 3	Read-only from the authoritative source/version; cannot be edited in this form.
Review date	2027-01-23	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H01 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H02 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For health & assistance sharing, what do Functional needs, Medical notes establish, and is “Save sharing preferences” available to Citizen in this scope?

Primary visual: Separate functional-needs and medical-note disclosure previews

Primary action: Save sharing preferences → H01

Progressive disclosure: Keep Functional needs, Medical notes in the decision layer. Put the supporting details for Family groups, Consent, Review date, full source/version history and FR-H02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a health & assistance sharing fixture for Citizen; change Functional needs and make Review date unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Save sharing preferences” reaches H01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H03 — Caregiver relationships

EN: Caregiver relationships · **JA:** 介護者・支援者の権限
Requirement: FR-H03 / WF-26 · **Actor:** Citizen · **Surface:** Care profiles · **Layout:** list

Purpose: Review who can help manage your profile.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H04.
Output: Review caregiver authority → H04.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Jamie	Verified caregiver · assistance profile only	Read-only within object scope; display source/time when they affect decisions.
Morgan	Emergency contact · no profile access	Read-only within object scope; display source/time when they affect decisions.
Pending request	None	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H04 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H03 — V2 visual composition

Context: configuration · **Role:** Citizen

Primary question: For caregiver relationships, what do Jamie, Morgan establish, and is “Review caregiver authority” available to Citizen in this scope?

Primary visual: Caregiver relationship graph with pending and expired authority

Primary action: Review caregiver authority → H04

Progressive disclosure: Keep Jamie, Morgan in the decision layer. Put the supporting details for Pending request, full source/version history and FR-H03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a caregiver relationships fixture for Citizen; change Jamie and make Pending request unavailable. Verify the visual and warning update, all 3 source fields remain reachable, and “Review caregiver authority” reaches H04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H04 — Caregiver authorization

EN: Caregiver authorization · **JA:** 支援者権限の確認
Requirement: FR-H04 / WF-26 · **Actor:** Privacy officer · **Surface:** Care profiles · **Layout:** form

Purpose: Verify authority before granting access.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H03.
Output: Approve scoped relationship → H03.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Subject	Alex Morgan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Caregiver	Jamie	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Authority	Verified authorization record	Read-only from the authoritative source/version; cannot be edited in this form.
Scope	Assistance profile	Select within granted permissions; the server blocks escalation beyond the actor’s rights.

Field / component	Sample value	Validation / editing permission
Expires	2027-01-23	UTC datetime + IANA timezone; validate future time, TTL and DST according to the workflow.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H03 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H04 — V2 visual composition

Context: configuration · **Role:** Privacy officer

Primary question: For caregiver authorization, what do Subject, Caregiver establish, and is “Approve scoped relationship” available to Privacy officer in this scope?

Primary visual: Subject-caregiver scope comparison with authorization expiry

Primary action: Approve scoped relationship → H03

Progressive disclosure: Keep Subject, Caregiver in the decision layer. Put the supporting details for Authority, Scope, Expires, full source/version history and FR-H04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a caregiver authorization fixture for Privacy officer; change Subject and make Expires unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Approve scoped relationship” reaches H03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H05 — Dependent person profile

EN: Dependent person profile · **JA:** 支援対象者のプロフィール
Requirement: FR-H05 / WF-26 · **Actor:** Authorized caregiver · **Surface:** Care profiles · **Layout:** detail

Purpose: Review the assistance plan with the person.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H06.

Output: Request profile update → H06.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Person	Alex Morgan	Read-only within object scope; display source/time when they affect decisions.
Mobility	Assisted evacuation	Read-only within object scope; display source/time when they affect decisions.
Device	Wheelchair	Read-only within object scope; display source/time when they affect decisions.
Emergency note	Restricted medical record	Read-only within object scope; display source/time when they affect decisions.
Last reviewed	2026-10-01	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H06 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H05 — V2 visual composition

Context: configuration · **Role:** Authorized caregiver

Primary question: For dependent person profile, what do Person, Mobility establish, and is “Request profile update” available to Authorized caregiver in this scope?

Primary visual: Dependent profile with device link and review-age indicator

Primary action: Request profile update → H06

Progressive disclosure: Keep Person, Mobility in the decision layer. Put the supporting details for Device, Emergency note, Last reviewed, full source/version history and FR-H05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a dependent person profile fixture for Authorized caregiver; change Person and make Last reviewed unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Request profile update” reaches H06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H06 — Profile update & review

EN: Profile update & review · **JA:** プロフィールの更新と確認
Requirement: FR-H06 / WF-26 · **Actor:** Authorized caregiver · **Surface:** Care profiles · **Layout:** form

Purpose: Keep assistance information current.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: H05.
Output: Submit profile update → H05.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Profile version	4	Read-only from the authoritative source/version; cannot be edited in this form.
Change	Updated mobility assistance need	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.
Source	Authorized caregiver	Read-only from the authoritative source/version; cannot be edited in this form.
Review status	Subject review pending	Read-only from the authoritative source/version; cannot be edited in this form.
Reason	Support needs changed	Unicode, maximum 2,000 characters; escape output; a reason is mandatory for state-changing decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H05 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H06 — V2 visual composition

Context: configuration · **Role:** Authorized caregiver

Primary question: For profile update & review, what do Profile version, Change establish, and is “Submit profile update” available to Authorized caregiver in this scope?

Primary visual: Profile version diff identifying source and review responsibility

Primary action: Submit profile update → H05

Progressive disclosure: Keep Profile version, Change in the decision layer. Put the supporting details for Source, Review status, Reason, full source/version history and FR-H06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a profile update & review fixture for Authorized caregiver; change Profile version and make Reason unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Submit profile update" reaches H05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H07 — People & assistance registry

EN: People & assistance registry · **JA:** 支援対象者一覧
Requirement: FR-H07 / WF-26 · **Actor:** Care coordinator · **Surface:** Care profiles · **Layout:** list

Purpose: Find people who may need proactive help.
Key rule: Assistance profiles have provenance, review dates and field-level access.
Entry: Role-based menu / deep link after authorization.
Output: Open assistance case → H09.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	Mobility assistance · profile reviewed	Read-only within object scope; display source/time when they affect decisions.
Jamie	No assistance data	Read-only within object scope; display source/time when they affect decisions.
Morgan	Communication assistance · review overdue	Read-only within object scope; display source/time when they affect decisions.
Visibility	Functional needs only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H09 with a valid payload/version, and verify that it is blocked when the guard for WF-26 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H07 — V2 visual composition

Context: command · **Role:** Care coordinator

Primary question: For people & assistance registry, what do Alex, Jamie establish, and is “Open assistance case” available to Care coordinator in this scope?

Primary visual: Assistance registry with role-limited functional-needs columns

Primary action: Open assistance case → H09

Progressive disclosure: Keep Alex, Jamie in the decision layer. Put the supporting details for Morgan, Visibility, full source/version history and FR-H07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a people & assistance registry fixture for Care coordinator; change Alex and make Visibility unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Open assistance case” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H08 — Proactive rescue queue

EN: Proactive rescue queue · **JA:** 先行的な救援対応
Requirement: FR-H08 / WF-27 · **Actor:** Dispatcher · **Surface:** Care profiles · **Layout:** priority

Purpose: No SOS does not mean no need for help.
Key rule: Prioritize exposure and ability to self-evacuate, even without an SOS.
Entry: H10.
Output: Review priority reasons → H09.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Alex	Exposed zone · mobility support · no SOS	Read-only within object scope; display source/time when they affect decisions.
Morgan	Possible exposure · stale position	Read-only within object scope; display source/time when they affect decisions.
Jamie	Presence unknown · check-in needed	Read-only within object scope; display source/time when they affect decisions.
Priority policy	Version 3 · explainable	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H09 with a valid payload/version, and verify that it is blocked when the guard for WF-27 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H08 — V2 visual composition

Context: command · **Role:** Dispatcher

Primary question: For proactive rescue queue, what do Alex, Morgan establish, and is “Review priority reasons” available to Dispatcher in this scope?

Primary visual: Proactive rescue queue with explainable priority reasons

Primary action: Review priority reasons → H09

Progressive disclosure: Keep Alex, Morgan in the decision layer. Put the supporting details for Jamie, Priority policy, full source/version history and FR-H08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a proactive rescue queue fixture for Dispatcher; change Alex and make Priority policy unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review priority reasons” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H09 — Priority evidence & decision

EN: Priority evidence & decision · **JA:** 優先度の根拠と判断
Requirement: FR-H09 / WF-27 · **Actor:** Dispatcher · **Surface:** Care profiles · **Layout:** detail

Purpose: Review why assistance was recommended.
Key rule: Prioritize exposure and ability to self-evacuate, even without an SOS.
Entry: H07, H08, H11, Q02.
Output: Assign proactive assistance → O09.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Exposure	Inside impacted zone · QR 2 min ago	Read-only within object scope; display source/time when they affect decisions.
Self-evacuation	Assistance required	Read-only within object scope; display source/time when they affect decisions.
SOS	None received	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Reason	Approved assistance policy	Read-only within object scope; display source/time when they affect decisions.
Priority	P1 · operator review	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O09 with a valid payload/version, and verify that it is blocked when the guard for WF-27 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H09 — V2 visual composition

Context: command · **Role:** Dispatcher

Primary question: For priority evidence & decision, what do Exposure, Self-evacuation establish, and is “Assign proactive assistance” available to Dispatcher in this scope?

Primary visual: Exposure and self-evacuation evidence beside human priority decision

Primary action: Assign proactive assistance → O09

Progressive disclosure: Keep Exposure, Self-evacuation in the decision layer. Put the supporting details for SOS, Reason, Priority, full source/version history and FR-H09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a priority evidence & decision fixture for Dispatcher; change Exposure and make Priority unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Assign proactive assistance” reaches O09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H10 — Presence & accountability

EN: Presence & accountability · **JA:** 在場状況と安否確認
Requirement: FR-H10 / WF-23 · **Actor:** Operator · **Surface:** Care profiles · **Layout:** map

Purpose: Distinguish presence from registration.
Key rule: Presence and position have distinct uncertainty; registration is not presence.
Entry: Role-based menu / deep link after authorization.

Output: Open proactive queue → H08.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Observed present	21	Read-only within object scope; display source/time when they affect decisions.
Expected attendance	30	Read-only within object scope; display source/time when they affect decisions.
Presence unknown	9	Read-only within object scope; display source/time when they affect decisions.
Last-seen sources	QR + access control	Read-only within object scope; display source/time when they affect decisions.
Safe inference	Never automatic	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H08 with a valid payload/version, and verify that it is blocked when the guard for WF-23 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H10 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For presence & accountability, what do Observed present, Expected attendance establish, and is "Open proactive queue" available to Operator in this scope?

Primary visual: Presence evidence matrix separating observed, expected and unknown

Primary action: Open proactive queue → H08

Progressive disclosure: Keep Observed present, Expected attendance in the decision layer. Put the supporting details for Presence unknown, Last-seen sources, Safe inference, full source/version history and FR-H10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Expected attendance and last seen are not live presence; never infer safety from absence of a distress signal.

V2 acceptance scenario: Use a presence & accountability fixture for Operator; change Observed present and make Safe inference unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Open proactive queue" reaches H08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H11 — Restricted medical record

EN: Restricted medical record · **JA:** 制限付き医療情報
Requirement: FR-H11 / WF-19 · **Actor:** Authorized medical role · **Surface:** Care profiles · **Layout:** detail

Purpose: Emergency medical subset · audited access.
Key rule: Expose minimum authorized data and audit sensitive access.
Entry: Role-based menu / deep link after authorization.
Output: Return to assistance → H09.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Person	Alex Morgan	Read-only within object scope; display source/time when they affect decisions.
Record	Encrypted · restricted	Read-only within object scope; display source/time when they affect decisions.
Source	Self-reported · not clinically verified	Read-only within object scope; display source/time when they affect decisions.
Access reason	Assigned emergency task	Read-only within object scope; display source/time when they affect decisions.
Disclosure	Minimum necessary	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H11: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H09 with a valid payload/version, and verify that it is blocked when the guard for WF-19 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

H11 — V2 visual composition

Context: configuration · **Role:** Authorized medical role

Primary question: For restricted medical record, what do Person, Record establish, and is “Return to assistance” available to Authorized medical role in this scope?

Primary visual: Restricted medical record with access purpose and disclosure log

Primary action: Return to assistance → H09

Progressive disclosure: Keep Person, Record in the decision layer. Put the supporting details for Source, Access reason, Disclosure, full source/version history and FR-H11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a restricted medical record fixture for Authorized medical role; change Person and make Disclosure unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Return to assistance" reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

H12 — Welfare check outcome

EN: Welfare check outcome · **JA:** 安否確認の結果
Requirement: FR-H12 / WF-27 · **Actor:** Responder · **Surface:** Care profiles · **Layout:** form

Purpose: Record contact even without an SOS.
Key rule: Prioritize exposure and ability to self-evacuate, even without an SOS.
Entry: Role-based menu / deep link after authorization.
Output: Save welfare observation → R06.
Data: Sensitive: health/functional needs; field-level scope and access audit.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Case	PC-041	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Person	Alex Morgan	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Contact	Established	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Assistance	Required	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Location	Confirmed by responder	Validate the same tenant, coordinate system/floor, source and timestamp; do not infer floor from GPS.
Next	Assisted evacuation task	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-H12: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to R06 with a valid payload/version, and verify that it is blocked when the guard for WF-27 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back

navigation/deep links never pass the wrong tenant.

H12 — V2 visual composition

Context: field · **Role:** Responder

Primary question: For welfare check outcome, what do Case, Person establish, and is “Save welfare observation” available to Responder in this scope?

Primary visual: Welfare observation with contact outcome and next responsible actor

Primary action: Save welfare observation → R06

Progressive disclosure: Keep Case, Person in the decision layer. Put the supporting details for Contact, Assistance, Location, Next, full source/version history and FR-H12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Display functional assistance needs only within authorized scope; missing presence or a caregiver claim must not imply safety or legal authority.

V2 acceptance scenario: Use a welfare check outcome fixture for Responder; change Case and make Next unavailable. Verify the visual and warning update, all 6 source fields remain reachable, and “Save welfare observation” reaches R06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q01 — Positioning source health

EN: Positioning source health · **JA:** 測位ソースの状態
Requirement: FR-Q01 / WF-23 · **Actor:** Positioning admin · **Surface:** Intelligence · **Layout:** dashboard

Purpose: Know where location can be trusted.
Key rule: Presence and position have distinct uncertainty; registration is not presence.
Entry: Role-based menu / deep link after authorization.
Output: Inspect location quality → Q02.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
GNSS	Outdoor coverage	Read-only within object scope; display source/time when they affect decisions.
BLE	Floor 05 degraded	Read-only within object scope; display source/time when they affect decisions.
UWB	Calibrated zones only	Read-only within object scope; display source/time when they affect decisions.
QR anchors	Approved revision 12	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q01: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q02 with a valid payload/version, and verify that it is blocked when the guard for WF-23 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q01 — V2 visual composition

Context: command · **Role:** Positioning admin

Primary question: For positioning source health, what do GNSS, BLE establish, and is “Inspect location quality” available to Positioning admin in this scope?

Primary visual: Position-source coverage and health comparison by GNSS/BLE/UWB/QR

Primary action: Inspect location quality → Q02

Progressive disclosure: Keep GNSS, BLE in the decision layer. Put the supporting details for UWB, QR anchors, full source/version history and FR-Q01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a positioning source health fixture for Positioning admin; change GNSS and make QR anchors unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Inspect location quality” reaches Q02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q02 — Position quality & history

EN: Position quality & history · **JA:** 位置情報の品質と履歴
Requirement: FR-Q02 / WF-23 · **Actor:** Operator · **Surface:** Intelligence · **Layout:** detail

Purpose: Alex · position is evidence, not certainty.
Key rule: Presence and position have distinct uncertainty; registration is not presence.
Entry: Q01.
Output: Open assistance context → H09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Current source	QR anchor	Read-only within object scope; display source/time when they affect decisions.
Floor confidence	Confirmed by user scan	Read-only within object scope; display source/time when they affect decisions.
Observed	2 minutes ago	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Heading	Unknown	Read-only within object scope; display source/time when they affect decisions.
Conflicting source	BLE floor estimate rejected	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q02: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to H09 with a valid payload/version, and verify that it is blocked when the guard for WF-23 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q02 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For position quality & history, what do Current source, Floor confidence establish, and is “Open assistance context” available to Operator in this scope?

Primary visual: Location history with floor confidence and conflicting-source markers

Primary action: Open assistance context → H09

Progressive disclosure: Keep Current source, Floor confidence in the decision layer. Put the supporting details for Observed, Heading, Conflicting source, full source/version history and FR-Q02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a position quality & history fixture for Operator; change Current source and make Conflicting source unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open assistance context” reaches H09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q03 — Indoor calibration editor

EN: Indoor calibration editor · **JA:** 屋内測位の校正
Requirement: FR-Q03 / WF-23 · **Actor:** Positioning admin · **Surface:** Intelligence · **Layout:** floor

Purpose: Align anchors to a versioned floor plan.
Key rule: Presence and position have distinct uncertainty; registration is not presence.
Entry: Role-based menu / deep link after authorization.

Output: Review route graph → S15.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Coordinate frame	Building A local	Read-only within object scope; display source/time when they affect decisions.
Reference points	4 surveyed anchors	Read-only within object scope; display source/time when they affect decisions.
Floor	05	Read-only within object scope; display source/time when they affect decisions.
Calibration error	Requires field measurement	Read-only within object scope; display source/time when they affect decisions.
Publish	Safety review required	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q03: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S15 with a valid payload/version, and verify that it is blocked when the guard for WF-23 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q03 — V2 visual composition

Context: configuration · **Role:** Positioning admin

Primary question: For indoor calibration editor, what do Coordinate frame, Reference points establish, and is "Review route graph" available to Positioning admin in this scope?

Primary visual: Calibration reference geometry with residual-error preview

Primary action: Review route graph → S15

Progressive disclosure: Keep Coordinate frame, Reference points in the decision layer. Put the supporting details for Floor, Calibration error, Publish, full source/version history and FR-Q03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a indoor calibration editor fixture for Positioning admin; change Coordinate frame and make Publish unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review route graph" reaches S15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q04 — Evacuation optimization

EN: Evacuation optimization · **JA:** 避難戦略の最適化
Requirement: FR-Q04 / WF-24 · **Actor:** Incident commander · **Surface:** Intelligence · **Layout:** optimization

Purpose: Compare feasible evacuation strategies.
Key rule: Exclude unsafe or unsuitable routes before optimizing travel and congestion.
Entry: Role-based menu / deep link after authorization.
Output: Review route comparison → Q05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Primary	West stairs · reviewed graph	Read-only within object scope; display source/time when they affect decisions.
Alternative	North refuge + assisted transfer	Read-only within object scope; display source/time when they affect decisions.
Excluded	East corridor · hazard	Read-only within object scope; display source/time when they affect decisions.
Objective	Exposure constraints before travel time	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q04: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q05 with a valid payload/version, and verify that it is blocked when the guard for WF-24 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q04 — V2 visual composition

Context: command · **Role:** Incident commander

Primary question: For evacuation optimization, what do Primary, Alternative establish, and is “Review route comparison” available to Incident commander in this scope?

Primary visual: Candidate-route comparison with excluded options and constraints

Primary action: Review route comparison → Q05

Progressive disclosure: Keep Primary, Alternative in the decision layer. Put the supporting details for Excluded, Objective, full source/version history and FR-Q04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a evacuation optimization fixture for Incident commander; change Primary and make Objective unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and "Review route comparison" reaches Q05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q05 — Route comparison & decision

EN: Route comparison & decision · **JA:** 経路の比較と判断
Requirement: FR-Q05 / WF-24 · **Actor:** Safety reviewer · **Surface:** Intelligence · **Layout:** detail

Purpose: Understand the trade-offs before publishing.
Key rule: Exclude unsafe or unsuitable routes before optimizing travel and congestion.
Entry: Q04, Q07.
Output: Publish guidance version → O36.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Option A	West route · longer · feasible	Read-only within object scope; display source/time when they affect decisions.
Option B	East route · shorter · blocked	Read-only within object scope; display source/time when they affect decisions.
Option C	Refuge assistance · mobility profile	Read-only within object scope; display source/time when they affect decisions.
Uncertainty	Congestion estimate	Read-only within object scope; display source/time when they affect decisions.
Approval	Required for zone strategy	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q05: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to O36 with a valid payload/version, and verify that it is blocked when the guard for WF-24 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q05 — V2 visual composition

Context: command · **Role:** Safety reviewer

Primary question: For route comparison & decision, what do Option A, Option B establish, and is “Publish guidance version” available to Safety reviewer in this scope?

Primary visual: Route trade-off matrix showing uncertainty and human approval

Primary action: Publish guidance version → O36

Progressive disclosure: Keep Option A, Option B in the decision layer. Put the supporting details for Option C, Uncertainty, Approval, full source/version history and FR-Q05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a route comparison & decision fixture for Safety reviewer; change Option A and make Approval unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Publish guidance version” reaches O36 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q06 — AI incident copilot

EN: AI incident copilot · **JA:** インシデントAI支援
Requirement: FR-Q06 / WF-25 · **Actor:** Operator · **Surface:** Intelligence · **Layout:** copilot

Purpose: A grounded summary with visible uncertainty.
Key rule: AI cites evidence, exposes unknowns and cannot control equipment autonomously.
Entry: Role-based menu / deep link after authorization.
Output: Review AI recommendation → Q07.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Summary	Fire evidence and one proactive assistance need	Read-only within object scope; display source/time when they affect decisions.
Sources	EV-3101 + PC-041 + SOP v4	Read-only within object scope; display source/time when they affect decisions.
Data cutoff	10:44:08 local	Read-only within object scope; display source/time when they affect decisions.
Authority	Recommendation only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q06: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q07 with a valid payload/version, and verify that it is blocked when the guard for WF-25 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q06 — V2 visual composition

Context: command · **Role:** Operator

Primary question: For ai incident copilot, what do Summary, Sources establish, and is “Review AI recommendation” available to Operator in this scope?

Primary visual: Evidence-linked AI summary with cutoff and authority boundary

Primary action: Review AI recommendation → Q07

Progressive disclosure: Keep Summary, Sources in the decision layer. Put the supporting details for Data cutoff, Authority, full source/version history and FR-Q06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a ai incident copilot fixture for Operator; change Summary and make Authority unavailable. Verify the visual and warning update, all 4 source fields remain reachable, and “Review AI recommendation” reaches Q07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q07 — AI evidence & approval

EN: AI evidence & approval · **JA:** AI提案の根拠と承認
Requirement: FR-Q07 / WF-25 · **Actor:** Supervisor · **Surface:** Intelligence · **Layout:** detail

Purpose: Approve an action, not a fluent paragraph.
Key rule: AI cites evidence, exposes unknowns and cannot control equipment autonomously.
Entry: Q06.
Output: Review route decision → Q05.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Claim	West route remains feasible	Read-only within object scope; display source/time when they affect decisions.
Evidence	Route engine result R-18	Read-only within object scope; display source/time when they affect decisions.
Unknown	Current corridor congestion	Read-only within object scope; display source/time when they affect decisions.

Field / component	Sample value	Validation / editing permission
Guard	No unsupported exit nodes	Read-only within object scope; display source/time when they affect decisions.
Action	Draft guidance only	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q07: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q05 with a valid payload/version, and verify that it is blocked when the guard for WF-25 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q07 — V2 visual composition

Context: command · **Role:** Supervisor

Primary question: For ai evidence & approval, what do Claim, Evidence establish, and is “Review route decision” available to Supervisor in this scope?

Primary visual: Claim-evidence-unknown comparison before human decision

Primary action: Review route decision → Q05

Progressive disclosure: Keep Claim, Evidence in the decision layer. Put the supporting details for Unknown, Guard, Action, full source/version history and FR-Q07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a ai evidence & approval fixture for Supervisor; change Claim and make Action unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review route decision” reaches Q05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q08 — AI policies & model evaluation

EN: AI policies & model evaluation · **JA:** AIポリシーとモデル評価
Requirement: FR-Q08 / WF-25 · **Actor:** AI governance lead · **Surface:** Intelligence · **Layout:** detail

Purpose: Keep model behavior measurable.
Key rule: AI cites evidence, exposes unknowns and cannot control equipment autonomously.
Entry: Q09.

Output: Open model governance → Q09.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Model	Region-approved deployment	Read-only within object scope; display source/time when they affect decisions.
Grounding test	Release gate	Read-only within object scope; display source/time when they affect decisions.
Prompt injection	Adversarial fixtures	Read-only within object scope; display source/time when they affect decisions.
Health data	Minimized	Read-only within object scope; display source/time when they affect decisions.
Control tools	Not available to model	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q08: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q09 with a valid payload/version, and verify that it is blocked when the guard for WF-25 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q08 — V2 visual composition

Context: configuration · **Role:** AI governance lead

Primary question: For ai policies & model evaluation, what do Model, Grounding test establish, and is “Open model governance” available to AI governance lead in this scope?

Primary visual: Evaluation results for grounding, injection and data exposure

Primary action: Open model governance → Q09

Progressive disclosure: Keep Model, Grounding test in the decision layer. Put the supporting details for Prompt injection, Health data, Control tools, full source/version history and FR-Q08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a ai policies & model evaluation fixture for AI governance lead; change Model and make Control tools unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Open model governance” reaches Q09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q09 — AI deployment governance

EN: AI deployment governance · **JA:** AI導入のガバナンス
Requirement: FR-Q09 / WF-25 · **Actor:** AI governance lead · **Surface:** Intelligence · **Layout:** form

Purpose: Publish a reviewed model and prompt version.
Key rule: AI cites evidence, exposes unknowns and cannot control equipment autonomously.
Entry: Q08.
Output: Save deployment draft → Q08.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Region	EU	Read-only from the authoritative source/version; cannot be edited in this form.
Model version	Approved candidate	Read-only from the authoritative source/version; cannot be edited in this form.
Prompt version	7	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Data retention	Regional policy	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.
Fallback	Deterministic workflows remain active	Required when part of the business payload; enum/format follow the schema and policy version; contextual data is read-only.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q09: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to Q08 with a valid payload/version, and verify that it is blocked when the guard for WF-25 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Q09 — V2 visual composition

Context: configuration · **Role:** AI governance lead

Primary question: For ai deployment governance, what do Region, Model version establish, and is “Save deployment draft” available to AI governance lead in this scope?

Primary visual: Model/prompt version deployment preview with regional fallback

Primary action: Save deployment draft → Q08

Progressive disclosure: Keep Region, Model version in the decision layer. Put the supporting details for Prompt version, Data retention, Fallback, full source/version history and FR-Q09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a ai deployment governance fixture for AI governance lead; change Region and make Fallback unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Save deployment draft" reaches Q08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

Q10 — Simulation & crowd scenarios

EN: Simulation & crowd scenarios · **JA:** 群集とシナリオのシミュレーション
Requirement: FR-Q10 / WF-24 · **Actor:** Safety planner · **Surface:** Intelligence · **Layout:** optimization

Purpose: Test strategy without asserting real-world certainty.
Key rule: Exclude unsafe or unsuitable routes before optimizing travel and congestion.
Entry: Role-based menu / deep link after authorization.
Output: Review route validation → S16.
Data: Sensitive: location/incident; minimize by role.

Mockup VI · EN · JA

Field / component	Sample value	Validation / editing permission
Mode	Scenario simulation	Read-only within object scope; display source/time when they affect decisions.
Occupancy	Estimated cohort	Read-only within object scope; display source/time when they affect decisions.
Blocked exit	East corridor	Read-only within object scope; display source/time when they affect decisions.
Mobility support	4 people	Read-only within object scope; display source/time when they affect decisions.
Result	Assumptions require site validation	Read-only within object scope; display source/time when they affect decisions.

Actions and feedback: The primary CTA checks actor/scope/version/required inputs before creating a request; while pending, the button prevents repeated submissions. Receipts show the object ID and actual state, without treating a design interaction as an executed side effect. List screens support filters and empty results; evidence/media require permission before opening. Read-only panels navigate to the corresponding review step.

States: loading (do not fabricate fresh data); empty (clarify scope/filter); invalid (field-level error, retain draft); denied (do not reveal the record); stale (age + source); offline (queued ≠ received); conflict (compare versions, no overwrite); success (actual receipt). When a state does not apply, the product does not invent an action to display it. The prototype state selector is a shared review tool.

AC-Q10: An actor within scope can open the screen and see the fields above; an out-of-scope actor cannot see the data. Test the CTA to S16 with a valid payload/version, and verify that it is blocked when the guard for WF-24 fails. Verify that vi/en/ja neither obscure the CTA nor lose meaning; the screen reader reads labels and states; back navigation/deep links never pass the wrong tenant.

Physical firefighting and rescue — baseline 1.0

Q10 — V2 visual composition

Context: configuration · **Role:** Safety planner

Primary question: For simulation & crowd scenarios, what do Mode, Occupancy establish, and is “Review route validation” available to Safety planner in this scope?

Primary visual: Simulated crowd and blocked-exit scenario with mobility cohorts

Primary action: Review route validation → S16

Progressive disclosure: Keep Mode, Occupancy in the decision layer. Put the supporting details for Blocked exit, Mobility support, Result, full source/version history and FR-Q10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep observations, uncertainty and human decisions separate; stale or conflicting source data must withdraw unsupported recommendations.

V2 acceptance scenario: Use a simulation & crowd scenarios fixture for Safety planner; change Mode and make Result unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review route validation” reaches S16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F01 — Physical systems overview

EN: Physical systems overview · **JA:** 物理システム概要
FR: FR-F01 / WF-29 · **Actor:** Physical supervisor · **Surface:** PhysicalSystems · **Layout:** dashboard

Purpose / device guard: Never aggregate stale or impaired systems into a green all-clear; each zone retains its controller and observation time.

Entry: F13, F18. **Output:** Review F07 → F07.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Controller health	3 connected; 1 stale	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Zone status	A05 alarm; B02 impaired	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Evidence freshness	8 s; source controller C01	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Local authority	Site fire officer	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Field / component	Sample value	Validation / authority
Control permission	Read-only until approved capability	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F01: An authorized actor sees every field above and reaches F07; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Never aggregate stale or impaired systems into a green all-clear; each zone retains its controller and observation time. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F01 — V2 visual composition

Context: command · **Role:** Physical supervisor

Primary question: For physical systems overview, what do Controller health, Zone status establish, and is “Review F07” available to Physical supervisor in this scope?

Primary visual: Physical-system topology showing controller authority and evidence age

Primary action: Review F07 → F07

Progressive disclosure: Keep Controller health, Zone status in the decision layer. Put the supporting details for Evidence freshness, Local authority, Control permission, full source/version history and FR-F01 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a physical systems overview fixture for Physical supervisor; change Controller health and make Control permission unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F07” reaches F07 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F02 — Controller inventory

EN: Controller inventory · **JA:** 制御装置一覧
FR: FR-F02 / WF-28 · **Actor:** Integration engineer · **Surface:** PhysicalSystems · **Layout:** list

Purpose / device guard: Inventory discovery grants no write rights; bind exact model, firmware, site and interface evidence before enrollment.

Entry: F20. **Output:** Review F03 → F03.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Controller model	PRA-SCL; exact firmware pending verification	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Protocol binding	Vendor Open Interface; revision required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Operating mode	Supervision only	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Site assignment	Harbor / Building A	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Capability evidence	Supplier dossier pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F02: An authorized actor sees every field above and reaches F03; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Inventory discovery grants no write rights; bind exact model, firmware, site and interface evidence before enrollment. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F02 — V2 visual composition

Context: configuration · **Role:** Integration engineer

Primary question: For controller inventory, what do Controller model, Protocol binding establish, and is “Review F03” available to Integration engineer in this scope?

Primary visual: Controller inventory with model, protocol and verified capability

Primary action: Review F03 → F03

Progressive disclosure: Keep Controller model, Protocol binding in the decision layer. Put the supporting details for Operating mode, Site assignment, Capability evidence, full source/version history and FR-F02 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a controller inventory fixture for Integration engineer; change Controller model and make Capability evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review F03" reaches F03 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F03 — Controller capability detail

EN: Controller capability detail · **JA:** 制御装置の機能詳細
FR: FR-F03 / WF-28 · **Actor:** Integration engineer · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Unknown or undocumented capability is read-only; certification and API compatibility are separate evidence.

Entry: F02. **Output:** Review F04 → F04.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Controller model	C01 / vendor-approved configuration	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Allowed operations	Read status only	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Forbidden operations	Release; reset; silence until approved	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Capability evidence	Interface specification + signed FAT	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Local override	Hardwired local authority retained	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F03: An authorized actor sees every field above and reaches F04; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Unknown or undocumented capability is read-only; certification and API compatibility are separate evidence. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F03 — V2 visual composition

Context: configuration · **Role:** Integration engineer

Primary question: For controller capability detail, what do Controller model, Allowed operations establish, and is "Review F04" available to Integration engineer in this scope?

Primary visual: Allowed/forbidden operation matrix beside local override status

Primary action: Review F04 → F04

Progressive disclosure: Keep Controller model, Allowed operations in the decision layer. Put the supporting details for Forbidden operations, Capability evidence, Local override, full source/version history and FR-F03 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a controller capability detail fixture for Integration engineer; change Controller model and make Local override unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review F04" reaches F04 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F04 — Signal point mapping

EN: Signal point mapping · **JA:** 信号点マッピング
FR: FR-F04 / WF-28 · **Actor:** Integration engineer · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Verify point polarity and meaning against wiring and witnessed tests; a valve contact is not proof of water flow.

Entry: F03. **Output:** Review F05 → F05.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Signal address	C01 / point 017	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Signal meaning	Valve supervisory state; not flow	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Normal polarity	Normally closed; verify wiring	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Field / component	Sample value	Validation / authority
Quality mapping	Fault / stale / unknown remain distinct	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Mapping revision	Draft 3; dual review required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F04: An authorized actor sees every field above and reaches F05; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Verify point polarity and meaning against wiring and witnessed tests; a valve contact is not proof of water flow. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F04 — V2 visual composition

Context: configuration · **Role:** Integration engineer

Primary question: For signal point mapping, what do Signal address, Signal meaning establish, and is “Review F05” available to Integration engineer in this scope?

Primary visual: Signal-address-to-meaning mapping with polarity and quality states

Primary action: Review F05 → F05

Progressive disclosure: Keep Signal address, Signal meaning in the decision layer. Put the supporting details for Normal polarity, Quality mapping, Mapping revision, full source/version history and FR-F04 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a signal point mapping fixture for Integration engineer; change Signal address and make Mapping revision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F05” reaches F05 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F05 — Physical zone mapping

EN: Physical zone mapping · **JA:** 物理ゾーンマッピング
FR: FR-F05 / WF-28 · **Actor:** Site engineer · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Reject ambiguous floor or cross-tenant zone mappings; require signed plan revision and site walkdown.

Entry: F04. **Output:** Review F06 → F06.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Source zone	C01-Z05	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Target zone	Building A / Floor 05	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Coordinate frame	Approved plan revision 12	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Coverage boundary	No cross-floor inference	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Mapping revision	Draft 3; site walkdown pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F05: An authorized actor sees every field above and reaches F06; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Reject ambiguous floor or cross-tenant zone mappings; require signed plan revision and site walkdown. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F05 — V2 visual composition

Context: configuration · **Role:** Site engineer

Primary question: For physical zone mapping, what do Source zone, Target zone establish, and is “Review F06” available to Site engineer in this scope?

Primary visual: Physical/source zone overlay showing coordinate and coverage mismatch

Primary action: Review F06 → F06

Progressive disclosure: Keep Source zone, Target zone in the decision layer. Put the supporting details for Coordinate frame, Coverage boundary, Mapping revision, full source/version history and FR-F05 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a physical zone mapping fixture for Site engineer; change Source zone and make Mapping revision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F06” reaches F06 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F06 — Cause-and-effect matrix

EN: Cause-and-effect matrix · **JA:** 原因・動作マトリクス
FR: FR-F06 / WF-28 · **Actor:** Fire systems engineer · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Only the locally approved cause-and-effect matrix governs physical effects; cloud edits cannot silently rewrite controller logic.

Entry: F05. **Output:** Review F15 → F15.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Cause input	Zone A05 alarm evidence	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Approved effect	Notify operator; no automatic release	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Authority record	Site cause-and-effect revision CE-03	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Inhibit condition	Test / impaired / local lockout	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Change impact	Requires reviewed commissioning plan	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F06: An authorized actor sees every field above and reaches F15; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Only the locally approved cause-and-effect matrix governs physical effects; cloud edits cannot silently rewrite controller logic. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F06 — V2 visual composition

Context: configuration · **Role:** Fire systems engineer

Primary question: For cause-and-effect matrix, what do Cause input, Approved effect establish, and is “Review F15” available to Fire systems engineer in this scope?

Primary visual: Cause-to-effect graph with authority and inhibit gates

Primary action: Review F15 → F15

Progressive disclosure: Keep Cause input, Approved effect in the decision layer. Put the supporting details for Authority record, Inhibit condition, Change impact, full source/version history and FR-F06 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a cause-and-effect matrix fixture for Fire systems engineer; change Cause input and make Change impact unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F15” reaches F15 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F07 — Physical event detail

EN: Physical event detail · **JA:** 物理イベント詳細
FR: FR-F07 / WF-29 · **Actor:** Physical supervisor · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Deduplicate source sequence; retain raw evidence and distinguish alarm, supervisory, trouble, test and inferred incident.

Entry: F01. **Output:** Review F08 → F08.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Source event	C01 / sequence 28017	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Signal meaning	Flow switch active; source evidence	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Observed time	10:42:08 UTC+07	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Evidence quality	Current; source-bound	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Incident linkage	INC-041; operator verification pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F07: An authorized actor sees every field above and reaches F08; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Deduplicate source sequence; retain raw evidence and distinguish alarm, supervisory, trouble, test and inferred incident. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F07 — V2 visual composition

Context: command · **Role:** Physical supervisor

Primary question: For physical event detail, what do Source event, Signal meaning establish, and is “Review F08” available to Physical supervisor in this scope?

Primary visual: Physical event provenance card linking observed evidence and incident

Primary action: Review F08 → F08

Progressive disclosure: Keep Source event, Signal meaning in the decision layer. Put the supporting details for Observed time, Evidence quality, Incident linkage, full source/version history and FR-F07 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a physical event detail fixture for Physical supervisor; change Source event and make Incident linkage unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review F08" reaches F08 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F08 — Fire-water status

EN: Fire-water status · **JA:** 消防用水状態
FR: FR-F08 / WF-29 · **Actor:** Physical supervisor · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Pump-running, pressure, valve position and flow are independent signals; unknown water availability blocks readiness claims.

Entry: F07. **Output:** Review F09 → F09.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Pump feedback	Run contact active; discharge unverified	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Pressure observation	6.0 bar; PT-01; calibrated reference required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Valve feedback	Open limit active; flow not implied	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Water availability	Reservoir level unknown	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Equipment impairment	Line B02 isolated under permit	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F08: An authorized actor sees every field above and reaches F09; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Pump-running, pressure, valve position and flow are independent signals; unknown water availability blocks readiness claims. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F08 — V2 visual composition

Context: command · **Role:** Physical supervisor

Primary question: For fire-water status, what do Pump feedback, Pressure observation establish, and is “Review F09” available to Physical supervisor in this scope?

Primary visual: Water-system schematic separating pump, pressure and valve feedback

Primary action: Review F09 → F09

Progressive disclosure: Keep Pump feedback, Pressure observation in the decision layer. Put the supporting details for Valve feedback, Water availability, Equipment impairment, full source/version history and FR-F08 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a fire-water status fixture for Physical supervisor; change Pump feedback and make Equipment impairment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F09” reaches F09 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F09 — Controlled release review

EN: Controlled release review · **JA:** 制御放出の審査
FR: FR-F09 / WF-30 · **Actor:** Authorized fire officer · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Release is blocked without documented allowed operation, installed certification, fresh prerequisites, local permit and controller interlocks.

Entry: F08. **Output:** Review F10 → F10.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Requested operation	Release request; illustrative only	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Target equipment	Approved controller / zone A05	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Capability evidence	Unverified: execution blocked	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Release prerequisites	Local authority; occupancy; interlocks; site permit	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Command expiry	30 s example; approved policy determines TTL	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F09: An authorized actor sees every field above and reaches F10; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Release is blocked without documented allowed operation, installed certification, fresh prerequisites, local permit and controller interlocks. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F09 — V2 visual composition

Context: command · **Role:** Authorized fire officer

Primary question: For controlled release review, what do Requested operation, Target equipment establish, and is “Review F10” available to Authorized fire officer in this scope?

Primary visual: Release-prerequisite gate against exact equipment and expiring request

Primary action: Review F10 → F10

Progressive disclosure: Keep Requested operation, Target equipment in the decision layer. Put the supporting details for Capability evidence, Release prerequisites, Command expiry, full source/version history and FR-F09 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a controlled release review fixture for Authorized fire officer; change Requested operation and make Command expiry unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F10” reaches F10 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F10 — Command authorization

EN: Command authorization · **JA:** 装置コマンド承認
FR: FR-F10 / WF-30 · **Actor:** Independent command approver · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Bind distinct requester/approver to exact equipment, operation, parameters, zone, version and expiry; any change invalidates approval.

Entry: F09. **Output:** Review F11 → F11.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Exact command digest	CMD-041 / revision 3 / zone A05	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Requester identity	Fire officer A; verified scope	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Approver identity	Officer B; distinct identity required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Approval state	Blocked: capability dossier incomplete	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Execution boundary	No browser-to-controller connection	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F10: An authorized actor sees every field above and reaches F11; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Bind distinct requester/approver to exact equipment, operation, parameters, zone, version and expiry; any change invalidates approval. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F10 — V2 visual composition

Context: command · **Role:** Independent command approver

Primary question: For command authorization, what do Exact command digest, Requester identity establish, and is “Review F11” available to Independent command approver in this scope?

Primary visual: Immutable command digest with requester/approver and execution boundary

Primary action: Review F11 → F11

Progressive disclosure: Keep Exact command digest, Requester identity in the decision layer. Put the supporting details for Approver identity, Approval state, Execution boundary, full source/version history and FR-F10 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a command authorization fixture for Independent command approver; change Exact command digest and make Execution boundary unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F11” reaches F11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F11 — Execution feedback

EN: Execution feedback · **JA:** 実行フィードバック
FR: FR-F11 / WF-30 · **Actor:** Physical supervisor · **Surface:** PhysicalSystems · **Layout:** timeline

Purpose / device guard: Controller acceptance is not physical completion; missing independent feedback becomes unknown with no blind retry.

Entry: F10, F19. **Output:** Review F12 → F12.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Command receipt	Controller accepted; not completed	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Physical feedback	No independent feedback received	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Outcome state	Unknown; reconcile required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Execution correlation	CMD-041 / controller sequence pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Retry permission	No blind retry of non-idempotent operation	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F11: An authorized actor sees every field above and reaches F12; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Controller acceptance is not physical completion; missing independent feedback becomes unknown with no blind retry. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F11 — V2 visual composition

Context: command · **Role:** Physical supervisor

Primary question: For execution feedback, what do Command receipt, Physical feedback establish, and is “Review F12” available to Physical supervisor in this scope?

Primary visual: Receipt-versus-physical-feedback timeline with unknown outcome

Primary action: Review F12 → F12

Progressive disclosure: Keep Command receipt, Physical feedback in the decision layer. Put the supporting details for Outcome state, Execution correlation, Retry permission, full source/version history and FR-F11 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Accepted command and physical actuation are separate states; unknown outcome disables blind retry and opens reconciliation F12.

V2 acceptance scenario: Use a execution feedback fixture for Physical supervisor; change Command receipt and make Retry permission unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F12” reaches F12 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F12 — Unknown-result reconciliation

EN: Unknown-result reconciliation · **JA:** 結果不明の照合
FR: FR-F12 / WF-30 · **Actor:** Physical supervisor · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Reconcile through supported readback and qualified observation; never resend a non-idempotent release to discover the outcome.

Entry: F11. **Output:** Review F13 → F13.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Command reference	CMD-041; acceptance seen	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Readback evidence	Read-only query; latest source timestamp	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Field confirmation	Qualified responder observation pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Reconciliation decision	Remain unknown; no release retry	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Decision reason	Independent physical effect not verified	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F12: An authorized actor sees every field above and reaches F13; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Reconcile through supported readback and qualified observation; never resend a non-idempotent release to discover the outcome. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F12 — V2 visual composition

Context: command · **Role:** Physical supervisor

Primary question: For unknown-result reconciliation, what do Command reference, Readback evidence establish, and is “Review F13” available to Physical supervisor in this scope?

Primary visual: Unknown-result reconciliation comparing readback and field evidence

Primary action: Review F13 → F13

Progressive disclosure: Keep Command reference, Readback evidence in the decision layer. Put the supporting details for Field confirmation, Reconciliation decision, Decision reason, full source/version history and FR-F12 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Keep outcome unknown until correlated readback or authorized field evidence resolves it; retain contradictory evidence and reason.

V2 acceptance scenario: Use a unknown-result reconciliation fixture for Physical supervisor; change Command reference and make Decision reason unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F13” reaches F13 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F13 — Local override & handover

EN: Local override & handover · **JA:** 現地優先制御と引継ぎ
FR: FR-F13 / WF-30 · **Actor:** Site incident commander · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Local override revokes remote authority; reconnection cannot replay commands or restore a revoked lease.

Entry: F12. **Output:** Review F01 → F01.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Local controller state	Local control active	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Remote lease	Revoked; stale intents invalidated	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Receiving operator	Site fire officer; verified acceptance required	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Handover scope	Zone A05; command CMD-041	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Return-to-remote gate	Explicit local release + fresh authorization	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F13: An authorized actor sees every field above and reaches F01; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Local override revokes remote authority; reconnection cannot replay commands or restore a revoked lease. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F13 — V2 visual composition

Context: command · **Role:** Site incident commander

Primary question: For local override & handover, what do Local controller state, Remote lease establish, and is “Review F01” available to Site incident commander in this scope?

Primary visual: Local-control handover showing remote lease and receiving operator

Primary action: Review F01 → F01

Progressive disclosure: Keep Local controller state, Remote lease in the decision layer. Put the supporting details for Receiving operator, Handover scope, Return-to-remote gate, full source/version history and FR-F13 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Loss of remote lease cannot take control back from the local controller; returning remote requires the documented handover gate.

V2 acceptance scenario: Use a local override & handover fixture for Site incident commander; change Local controller state and make Return-to-remote gate unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F01” reaches F01 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F14 — Impairment & test mode

EN: Impairment & test mode · **JA:** 機能障害と試験モード
FR: FR-F14 / WF-31 · **Actor:** Maintenance coordinator · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Test mode requires bounded isolation and compensating measures; permit expiry escalates but does not assert restoration.

Entry: Authorized physical-systems menu. **Output:** Review F18 → F18.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Impairment scope	Line B02; no adjacent zone changes	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Permit reference	IMP-014; site approval pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Compensating measure	Fire watch; responsible team assigned	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Test isolation	Exercise partition; physical output inhibited	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Expiry and escalation	16:00 local; overdue remains impaired	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F14: An authorized actor sees every field above and reaches F18; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Test mode requires bounded isolation and compensating measures; permit expiry escalates but does not assert restoration. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F14 — V2 visual composition

Context: command · **Role:** Maintenance coordinator

Primary question: For impairment & test mode, what do Impairment scope, Permit reference establish, and is "Review F18" available to Maintenance coordinator in this scope?

Primary visual: Impairment map with compensating measures, permit and expiry

Primary action: Review F18 → F18

Progressive disclosure: Keep Impairment scope, Permit reference in the decision layer. Put the supporting details for Compensating measure, Test isolation, Expiry and escalation, full source/version history and FR-F14 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a impairment & test mode fixture for Maintenance coordinator; change Impairment scope and make Expiry and escalation unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and "Review F18" reaches F18 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F15 — Commissioning plan

EN: Commissioning plan · **JA:** 試運転計画
FR: FR-F15 / WF-32 · **Actor:** Commissioning lead · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Freeze BOM, firmware, mappings and matrix; physical tests require site permit, witnesses, isolation and local abort.

Entry: F06. **Output:** Review F16 → F16.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Test configuration	Controller C01 / firmware / mappings pinned	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Test scope	FAT then SAT; signed local procedure	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Test prerequisites	Isolation; fire watch; emergency abort	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Witnesses	Integrator + site authority	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Field / component	Sample value	Validation / authority
Evidence requirement	Point-by-point result; timestamp; signed attachments	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F15: An authorized actor sees every field above and reaches F16; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Freeze BOM, firmware, mappings and matrix; physical tests require site permit, witnesses, isolation and local abort. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F15 — V2 visual composition

Context: configuration · **Role:** Commissioning lead

Primary question: For commissioning plan, what do Test configuration, Test scope establish, and is “Review F16” available to Commissioning lead in this scope?

Primary visual: Commissioning sequence with test isolation and witness checkpoints

Primary action: Review F16 → F16

Progressive disclosure: Keep Test configuration, Test scope in the decision layer. Put the supporting details for Test prerequisites, Witnesses, Evidence requirement, full source/version history and FR-F15 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a commissioning plan fixture for Commissioning lead; change Test configuration and make Evidence requirement unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F16” reaches F16 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F16 — Test-run evidence

EN: Test-run evidence · **JA:** 試験実施証跡
FR: FR-F16 / WF-32 · **Actor:** Commissioning witness · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Expected results cannot populate observed results; failed or unexecuted tests block release and preserve attachments.

Entry: F15. **Output:** Review F17 → F17.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Test run	SAT-041 / approved plan revision 2	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Expected effect	Selected zone only; local priority retained	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Observed effect	Pending witnessed measurement	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Fault injection	Power loss / link loss / stuck feedback	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Result status	Not executed; not a passed certificate	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F16: An authorized actor sees every field above and reaches F17; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Expected results cannot populate observed results; failed or unexecuted tests block release and preserve attachments. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F16 — V2 visual composition

Context: configuration · **Role:** Commissioning witness

Primary question: For test-run evidence, what do Test run, Expected effect establish, and is “Review F17” available to Commissioning witness in this scope?

Primary visual: Expected/observed physical-effect comparison with fault injections

Primary action: Review F17 → F17

Progressive disclosure: Keep Test run, Expected effect in the decision layer. Put the supporting details for Observed effect, Fault injection, Result status, full source/version history and FR-F16 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a test-run evidence fixture for Commissioning witness; change Test run and make Result status unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F17” reaches F17 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F17 — Compatibility & certificate dossier

EN: Compatibility & certificate dossier · **JA:** 互換性・証明書台帳
FR: FR-F17 / WF-32 · **Actor:** Compliance reviewer · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: A product certificate does not certify the installation or cloud adapter; exact scope, revisions and acceptance evidence must match.

Entry: F16. **Output:** Review F20 → F20.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Installed bill of materials	Controller + modules + PSU + firmware	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Certificate scope	Exact assembly and jurisdiction to verify	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Interface limitation	Unknown capabilities remain read-only	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Evidence validity	Supplier document revision; expiry tracked	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Release decision	Blocked until FAT / SAT / authority acceptance	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote

authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F17: An authorized actor sees every field above and reaches F20; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: A product certificate does not certify the installation or cloud adapter; exact scope, revisions and acceptance evidence must match. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F17 — V2 visual composition

Context: configuration · **Role:** Compliance reviewer

Primary question: For compatibility & certificate dossier, what do Installed bill of materials, Certificate scope establish, and is “Review F20” available to Compliance reviewer in this scope?

Primary visual: Certificate scope mapped to installed bill of materials and limitations

Primary action: Review F20 → F20

Progressive disclosure: Keep Installed bill of materials, Certificate scope in the decision layer. Put the supporting details for Interface limitation, Evidence validity, Release decision, full source/version history and FR-F17 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a compatibility & certificate dossier fixture for Compliance reviewer; change Installed bill of materials and make Release decision unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F20” reaches F20 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F18 — Maintenance & restoration

EN: Maintenance & restoration · **JA:** 保守と復旧
FR: FR-F18 / WF-31 · **Actor:** Maintenance coordinator · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Only witnessed restoration and site approval clear impairment; time elapsed or technician closure alone cannot restore availability.

Entry: F14. **Output:** Review F01 → F01.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Maintenance scope	Line B02 isolation IMP-014	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Field / component	Sample value	Validation / authority
Repair evidence	Technician report; component serial	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Restoration test	Independent feedback + local panel check	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Closure approval	Site authority; not technician alone	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Residual impairment	Unknown until witnessed restoration	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F18: An authorized actor sees every field above and reaches F01; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Only witnessed restoration and site approval clear impairment; time elapsed or technician closure alone cannot restore availability. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F18 — V2 visual composition

Context: configuration · **Role:** Maintenance coordinator

Primary question: For maintenance & restoration, what do Maintenance scope, Repair evidence establish, and is “Review F01” available to Maintenance coordinator in this scope?

Primary visual: Restoration checklist with residual impairment and approval evidence

Primary action: Review F01 → F01

Progressive disclosure: Keep Maintenance scope, Repair evidence in the decision layer. Put the supporting details for Restoration test, Closure approval, Residual impairment, full source/version history and FR-F18 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a maintenance & restoration fixture for Maintenance coordinator; change Maintenance scope and make Residual impairment unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F01” reaches F01 only under its original guards. Repeat in VI/EN, keyboard and

320 CSS-pixel reflow; record observed results separately from this specification.

F19 — Rescue equipment mission

EN: Rescue equipment mission · **JA:** 救助機材ミッション
FR: FR-F19 / WF-33 · **Actor:** Rescue mission commander · **Surface:** PhysicalSystems · **Layout:** detail

Purpose / device guard: Colossus control API is unverified: dispatch to trained vendor-console operator; no cloud movement, nozzle or casualty actuation.

Entry: Authorized physical-systems menu. **Output:** Review F11 → F11.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Equipment configuration	Colossus water monitor / stretcher as approved	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Control interface	Vendor console; public API unverified	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Operator qualification	Trained operator; task acceptance pending	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Mission constraints	Water supply; access; load; local stop; casualty procedure	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Mission evidence	Operator report + supported telemetry; not automatic success	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F19: An authorized actor sees every field above and reaches F11; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Colossus control API is unverified: dispatch to trained vendor-console operator; no cloud movement, nozzle or casualty actuation. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F19 — V2 visual composition

Context: command · **Role:** Rescue mission commander

Primary question: For rescue equipment mission, what do Equipment configuration, Control interface establish, and is “Review F11” available to Rescue mission commander in this scope?

Primary visual: Rescue-equipment mission envelope with qualified operator and limits

Primary action: Review F11 → F11

Progressive disclosure: Keep Equipment configuration, Control interface in the decision layer. Put the supporting details for Operator qualification, Mission constraints, Mission evidence, full source/version history and FR-F19 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a rescue equipment mission fixture for Rescue mission commander; change Equipment configuration and make Mission evidence unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F11” reaches F11 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

F20 — Physical connector onboarding

EN: Physical connector onboarding · **JA:** 物理コネクタ登録
FR: FR-F20 / WF-28 · **Actor:** Integration engineer · **Surface:** PhysicalSystems · **Layout:** form

Purpose / device guard: Connector activation starts read-only; credentials stay in vault and only signed capability policy can enable an approved write operation.

Entry: F17. **Output:** Review F02 → F02.

Mockup VI · EN · JA

Field / component	Sample value	Validation / authority
Vendor interface	Exact published model / firmware / protocol	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Transport security	Certificate and OT allowlist verified before connection	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Credential custody	Secret reference only; never display secret	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.
Import scope	Read-only points; no discovered write grant	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Field / component	Sample value	Validation / authority
Activation gate	Sandbox test + mapping review + signed capability dossier	Source-bound read-only evidence; retain observation time and quality. Edits to plans/decisions require actor scope, expected version, reason and independent review; never edit physical feedback.

Actions and feedback: The CTA opens the linked review screen; prototype navigation never sends device commands. Production mutations use scoped idempotency and expected version; control additionally requires the exact approved capability, site authority and unexpired authorization. Fields from telemetry and dossiers remain read-only. Display request ID, controller receipt and independent physical evidence separately.

States: loading: no fabricated telemetry; empty: no assigned equipment; denied: no object leakage; stale: show age/source and block dependent control; offline: read-only cached evidence, no queued actuation; conflict: compare revisions, invalidate prior approval; impaired/test: explicit scope and compensating measure; local override: remote authority revoked; unknown outcome: reconciliation only; success: appropriate durable record or verified physical evidence, never inference from ACK.

AC-F20: An authorized actor sees every field above and reaches F02; an out-of-scope actor cannot read or mutate the object. Exercise the specific guard: Connector activation starts read-only; credentials stay in vault and only signed capability policy can enable an approved write operation. Test stale evidence, cross-tenant point IDs, changed mapping/version, expired authorization and loss of local authority. Verify no prohibited write reaches the adapter; preserve audit and source event correlation. Check vi/en/ja labels, keyboard focus, screen-reader state, retained draft and explicit unknown/error recovery.

F20 — V2 visual composition

Context: configuration · **Role:** Integration engineer

Primary question: For physical connector onboarding, what do Vendor interface, Transport security establish, and is “Review F02” available to Integration engineer in this scope?

Primary visual: Vendor connector onboarding gates for credentials, import and activation

Primary action: Review F02 → F02

Progressive disclosure: Keep Vendor interface, Transport security in the decision layer. Put the supporting details for Credential custody, Import scope, Activation gate, full source/version history and FR-F20 rules in labeled expandable sections; never hide a safety-critical constraint.

Critical failure behavior: Retain local controller authority; absent physical feedback means unknown outcome, never success, and forbids blind repeat execution.

V2 acceptance scenario: Use a physical connector onboarding fixture for Integration engineer; change Vendor interface and make Activation gate unavailable. Verify the visual and warning update, all 5 source fields remain reachable, and “Review F02” reaches F02 only under its original guards. Repeat in VI/EN, keyboard and 320 CSS-pixel reflow; record observed results separately from this specification.

BMS implementation screen contracts

This extension is integrated with the original233screen contracts. Total241screens;37workflows. Screen IDs FR-BMS01..08 are distinct from domain IDs FR-BMS-01..40. API paths below are SmartAlarm internal contracts, not vendor endpoints. Authoritative API contract.

BMS01 — BMS identity and capabilities

Screen contract: FR-BMS01 · **Workflow:** WF-34 · **Actor:** Integration engineer.

Domain requirements: FR-BMS-01, FR-BMS-02, FR-BMS-03, FR-BMS-04, FR-BMS-32, FR-BMS-36, FR-BMS-39, FR-BMS-40.

Mockup VI · EN · JA

Visual and hierarchy: BMS → edge → SmartAlarm chain; capability matrix beside the identity dossier.

Primary question: Is the exact product/build/site identified, and which operations have sufficient evidence?

End-to-end flow: Select one of11profiles → enter build/license references → review dossier → scoped import → inspect validation/qualification → request read-only activation. Changing the profile clears its draft evidence and preview to avoid cross-product reuse.

Permission: bms.configure, acceptance approver. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Product profile	bms.siemens.desigo-wsi	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Verified product build	Unverified	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
License evidence reference	EVID-LIC-PENDING	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Admitted capabilities	None admitted	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Credential custody	Vault reference only	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Activation gate	Exact build + license + signed acceptance	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding:
`BMSConnectionCreate.profile_id/profile_version/vendor_product/vendor_build/endpoint_ref/credential_ref/network_policy_ref; BMSConnectionView.state/capabilities/revision; BMSEvidenceRef.evidence_id/sha256/version; BMSJob.status/problem`.

Validation and guards: Never infer product/build/interface/license from vendor name. endpoint_ref and credential_ref are server-side references; secrets are never displayed. Owner, site scope, network_policy_ref and point/request budgets are required before create. Invalid qualification or a changed adapter digest blocks activation with a reason.

Method / SmartAlarm path	operationId
POST /bms/connections	bms_post_bms_connections
POST /bms/connections/{connection_id}/discoveries	bms_post_bms_connections_connection_id_discoveries
POST /bms/connections/{connection_id}/validations	bms_post_bms_connections_connection_id_validations
POST /bms/connections/{connection_id}/qualifications	bms_post_bms_connections_connection_id_qualifications
POST /bms/connections/{connection_id}/activations	bms_post_bms_connections_connection_id_activations

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Unknown license keeps a draft; expired certificates lead to degraded/suspended status per policy without bypassing authentication; mismatched builds fail validation; partial admission explicitly lists excluded points.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Select Metasys and enter build/license; preview preserves a draft without activating capabilities. Switching to EBO clears that evidence draft. A bms.read-only user cannot create or activate.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS02 — Semantic topology

Screen contract: FR-BMS02 · **Workflow:** WF-34 · **Actor:** Mapping engineer.

Domain requirements: FR-BMS-05, FR-BMS-06, FR-BMS-07, FR-BMS-08, FR-BMS-39.

Mockup VI · EN · JA

Visual and hierarchy: Native point → equipment → served zone → route edge topology with a separate mapping/evidence table.

Primary question: Is this a service, room-containment or fire-compartment relation, and what evidence establishes it?

End-to-end flow: Import a cursor-based inventory → select point → map asset/space/zone → validate relation type and cycles → inspect dependencies → create a candidate revision → independent authorized approval. Drafts do not modify active routes.

Permission: bms.configure, bms.mapping.approve. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Native point identity	BI:102 / AHU-05	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Served zone	A05	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Fire compartment	A05-FC1	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Relationship evidence	MEP-A05-REV4	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Mapping revision	map-r18	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Dependent route edge	ROUTE-E17	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding:
`BMSPointMapping.point_id/asset_id/space_id/zone_id/source_address/semantic_role/source_unit/canonical_unit/scale/offset/stale_after_seconds; BMSMapRevision.map_revision/mapping_digest/state/validation_ref`.

Validation and guards: Native ID alone is not a global key. asset_id, space_id and zone_id must share the tenant/site or belong to an authorized projection. Validate value_type, units, scale, offset, stale_after_seconds and semantic_role. Name prefixes are not relationship evidence.

Method / SmartAlarm path	operationId
`GET /bms/connections/{connection_id}/points`	`bms_get_bms_connections_connection_id_points`
`POST /bms/connections/{connection_id}/map-revisions`	`bms_post_bms_connections_connection_id_map_revisions`
`POST /bms/connections/{connection_id}/validations`	`bms_post_bms_connections_connection_id_validations`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Orphan points or conflicting zones quarantine mapping; containment cycles block revisions; stale sources cannot become route evidence. Changing A05 to undocumented A06 blocks a valid preview.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Change served zone A05→A06; preview reports missing evidence without changing routes. Restoring A05 permits an unpublished candidate. Mapping a read-only point must not grant write authority.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS03 — Cross-system alarm reconciliation

Screen contract: FR-BMS03 · **Workflow:** WF-35 · **Actor:** Incident operator.

Domain requirements: FR-BMS-09, FR-BMS-10, FR-BMS-11, FR-BMS-12, FR-BMS-13, FR-BMS-14, FR-BMS-15.

Mockup VI · EN · JA

Visual and hierarchy: Native condition and incident comparison with separate receive/link/native-ACK/response-closure timeline states.

Primary question: Which occurrence is this, who acknowledged it in which system, and what remains unknown?

End-to-end flow: Select condition plus occurrence/branch → verify revision and transition token → inspect actor/receipt → prepare native ACK request with approval → await mutation receipt → reconcile BMS. Internal review and incident closure never synchronize implicitly.

Permission: bms.read, bms.ack. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Native condition identity	COND-77	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Occurrence generation	42 / 43	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Source cursor	9018	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Snapshot completeness	Gap9019–9022	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Linked incident	INC-2041	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Native acknowledgment state	UNKNOWN	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding:
`BMSAlarmView.alarm_id/native_alarm_id/occurrence_id/branch_id/active_state/native_ack_state/smartalarm_reviewed/snapshot_generation/revision; BMSNativeAckRequest; BMSMutationReceipt.state/effect_confirmed`.

Validation and guards: ACK targets the current occurrence_id, branch_id and native_transition_token; expected_revision, expiry and qualification must remain valid. A new generation invalidates the previous preview. Block the action for unreconciled snapshot gaps or missing bms.ack authority.

Method / SmartAlarm path	operationId
`GET /bms/connections/{connection_id}/alarms`	`bms_get_bms_connections_connection_id_alarms`
`POST /bms/connections/{connection_id}/alarms/{alarm_id}/native-acknowledgements`	`bms_post_bms_connections_connection_id_alarms_alarm_id_native_acknowledgements`
`POST /bms/connections/{connection_id}/reconciliations`	`bms_post_bms_connections_connection_id_reconciliations`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: A timeout after dispatch is unknown and is not automatically retried. BMS normal does not close an incident. Stream gaps show missing scope; recurrence is not merged into an old generation.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Choose generation43 and preview ACK: target43 remains blocked without qualification and INC-2041 stays active. Changing42→43 clears preview42. Both panel and BMS raw provenance remain available.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS04 — Control authority and priority

Screen contract: FR-BMS04 · **Workflow:** WF-36 · **Actor:** Authorized command reviewer.

Domain requirements: FR-BMS-22, FR-BMS-23, FR-BMS-24, FR-BMS-25, FR-BMS-26, FR-BMS-27.

Mockup VI · EN · JA

Visual and hierarchy: Priority ladder and winning authority, with separate demand/controller-report/physical-feedback blocks and prominent local override.

Primary question: Who currently controls the equipment, and is this requested effect admissible?

End-to-end flow: Select target/capability → preview against map revision/cause-effect → display winner/interlocks/approvers → approve digest → submit before expiry → follow the PhysicalCommand aggregate through physical_command_id → reconcile feedback.

Permission: bms.command.request, bms.command.approve. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Control target	AHU-05 / BO:18	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Winning authority	Local operator	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Remote slot ownership	Not admitted	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Command lease	NOT ISSUED	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

Field	Example	UI behavior
Physical feedback	UNKNOWN	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Relinquish mechanism	Unverified	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding:
`BMSCommandPreview.point_id/capability_id/value/unit/priority/map_revision/cause_effect_revision/expires_at; BMSCommandPreviewResult.effective_owner/quality/interlock_result/required_approvers/approval_digest; BMSMutationReceipt.physical_command_id`.

Validation and guards: Value, unit and priority must be on the site allowlist; a priority slot does not prove ownership. Approval binds the exact preview digest, adapter/qualification and revision; changes require a new preview. Requesters cannot replace approval.

Method / SmartAlarm path	operationId
`POST /bms/connections/{connection_id}/command-previews`	`bms_post_bms_connections_connection_id_command_previews`
`POST /bms/connections/{connection_id}/commands`	`bms_post_bms_connections_connection_id_commands`
`GET /bms/connections/{connection_id}/commands/{command_id}`	`bms_get_bms_connections_connection_id_commands_command_id`
`POST /bms/connections/{connection_id}/commands/{command_id}/reconciliations`	`bms_post_bms_connections_connection_id_commands_command_id_reconciliations`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Winning local override blocks remote control. Lease expiry does not establish relinquishment. Stale/unknown feedback is never green. Foreign slots cannot be cleared. Edge power loss requires a proven local relinquish mechanism.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Toggling local override changes the winner explanation, but execute remains locked without policy/lease. Preview sends no command. Replaying an idempotency key returns its original aggregate; different input under the same key conflicts.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS05 — Point drift and impact

Screen contract: FR-BMS05 · **Workflow:** WF-37 · **Actor:** Mapping approver.

Domain requirements: FR-BMS-07, FR-BMS-08, FR-BMS-31.

Mockup VI · EN · JA

Visual and hierarchy: Side-by-side before/after diff highlights unit/type changes, with dependent rules and routes below.

Primary question: How did this same-ID point change meaning, and which rules/routes are affected?

End-to-end flow: Receive schema drift → preserve immutable active mapping → compare base and candidate → validate transform/identity → quarantine point → validation job → independent approval → staged activation or rollback.

Permission: bms.mapping.approve. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Previous native type	analog-input / °C	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Candidate native type	analog-input / °F	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Previous mapping hash	85d1…	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Candidate mapping hash	b93a…	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Affected dependency	RULE-TEMP-05 / ROUTE-E17	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Migration disposition	Approval pending	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding: `BMSMapDraftCreate.base_map_revision/import_digest/change_reason/points; BMSMapRevision.state/mapping_digest/validation_ref; BMSValidationRequest.checks/allow_write_probe`.

Validation and guards: Same ID is insufficient to establish the same device. Validate type, units, scale/offset, native address and digest. °F→°C vectors cover range, NaN and missing values;75.2→24.0 is only one example vector.

Method / SmartAlarm path	operationId
`POST /bms/connections/{connection_id}/map-revisions`	`bms_post_bms_connections_connection_id_map_revisions`
`POST /bms/connections/{connection_id}/validations`	`bms_post_bms_connections_connection_id_validations`
`POST /bms/connections/{connection_id}/activations`	`bms_post_bms_connections_connection_id_activations`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires

refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Unit changes without a transform are blocked. Digest mismatch or stale base revision conflicts rather than overwriting. Mapping rollback re-evaluates the drifted source and does not assert old evidence is valid.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: A °C→°F change without a transform is blocked. Inspecting the transform shows24.0°C, but publish remains locked pending approval. Changing transform/type/unit invalidates the old review.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS06 — Budgets and data freshness

Screen contract: FR-BMS06 · **Workflow:** WF-37 · **Actor:** Integration operations.

Domain requirements: FR-BMS-28, FR-BMS-33, FR-BMS-34, FR-BMS-35.

Mockup VI · EN · JA

Visual and hierarchy: Zone-quality heatmap with stale hatching, alarm/telemetry/history queues and request-budget calculation.

Primary question: Which sources are stale, and which workload can be reduced to protect alarms?

End-to-end flow: Read health with upstream timestamp → inspect stale scope/backlog → estimate budgets with batching/source limits → preview policy → authorized policy revision → monitor queues/freshness. Pausing backfill does not stop alarm ingestion.

Permission: bms.configure, operations. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Polling point count	2400	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Read batch size	40	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Polling interval	10 s	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Alarm queue policy	Priority over history	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
History budget	2 requests/s	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

Field	Example	UI behavior
Stale evidence zones	A05 / B02	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding:
`BMSHealth.state/heartbeat_at/last_upstream_success_at/backlog_events/oldest_backlog_seconds/stale_point_count/native_subscription_state;`
`BMSConnectionCreate.point_budget/request_rate_budget/poll_interval_seconds;`
`BMSPointView.quality/source_time/received_at`.

Validation and guards: N,B,T are positive finite values; estimate=ceil(N/B)/T, separate from COV/history/retry/reconnect. Online heartbeat does not replace data quality. New policies preserve source-rate limits and reserved alarm capacity.

Method / SmartAlarm path	operationId
`GET /bms/connections/{connection_id}/health`	`bms_get_bms_connections_connection_id_health`
`GET /bms/connections/{connection_id}/points`	`bms_get_bms_connections_connection_id_points`
`GET /bms/connections/{connection_id}/policies/request-budget`	`bms_request_budget_get`
`PATCH /bms/connections/{connection_id}/policies/request-budget`	`bms_request_budget_patch`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Rate limiting applies bounded jittered backoff and quotas, not a request storm. History cannot starve alarms. Healthy heartbeat with old data remains stale. Missing source_time displays received_at under its own label.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: 2401points/40batch/10s yields6.10requests/s. Pausing history sets its sample budget to0 without stopping alarms. N=0 or interval≤0 is invalid. Hatching and labels remain usable without color discrimination.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS07 — Edge recovery and handover

Screen contract: FR-BMS07 · **Workflow:** WF-37 · **Actor:** Site OT approver.

Domain requirements: FR-BMS-26, FR-BMS-27, FR-BMS-28, FR-BMS-29, FR-BMS-30, FR-BMS-31, FR-BMS-40.

Mockup VI · EN · JA

Visual and hierarchy: Old edge → journal/cursor → standby edge recovery lane, with separate evidence checklist and writer lock.

Primary question: Is the old writer actually fenced, and which alarms/commands remain unreconciled?

End-to-end flow: Revoke/fence old writer → read checkpoint/gaps → snapshot active conditions → reconcile unknown commands/leases → record OT handover evidence → issue a new epoch and capabilities only after gates. Retirement hands back control before credential revocation.

Permission: bms.reconcile, site OT approver. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Writer ownership epoch	17 → 18	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
BMS source epoch	9 → 10	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Durable journal checkpoint	9018	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Outstanding command	CMD-88 / UNKNOWN	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Recovery snapshot status	Unverified	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
OT handover approval	PENDING	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding: `BMSHealth.epoch/write_authority; BMSConnectionView.reconciliation_required; BMSReconciliationRequest.scope/cursor_ref/allow_side_effect_replay; BMSCommandReconcile.observation_refs/action/physical_effect; BMSLifecycleRequest.local_handback_ref/approval_id/target_adapter_digest`.

Validation and guards: Require egress fencing proof, monotonic epoch, durable journal, complete snapshot and local handback evidence. UI checkboxes do not grant authority. Telemetry/history replay is tagged and side-effect replay is blocked.

Method / SmartAlarm path	operationId
`POST /bms/connections/{connection_id}/reconciliations`	`bms_post_bms_connections_connection_id_reconciliations`
`POST /bms/connections/{connection_id}/commands/{command_id}/reconciliations`	`bms_post_bms_connections_connection_id_commands_command_id_reconciliations`
`POST /bms/connections/{connection_id}/lifecycle-transitions`	`bms_post_bms_connections_connection_id_lifecycle_transitions`
`GET /bms/connections/{connection_id}/health`	`bms_get_bms_connections_connection_id_health`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires

refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Split brain/old epochs block command egress. Missing journal or incomplete snapshot retains read-only operation and escalates reconciliation. Missing remote lease does not prove local slot release; unknown commands are not replayed.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Any missing one of4evidence items keeps the writer locked. Completing4items prepares an OT-approval dossier without enabling writes. Epoch17 after epoch18 must be blocked at egress, not only at the API.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.

BMS08 — Exchange tracing and loop control

Screen contract: FR-BMS08 · **Workflow:** WF-35 · **Actor:** Integration incident reviewer.

Domain requirements: FR-BMS-10, FR-BMS-16, FR-BMS-37.

Mockup VI · EN · JA

Visual and hierarchy: Panel → BMS → SmartAlarm → writeback causation path; the echo leg has its own shape/label while raw receipts are retained.

Primary question: Is this event new evidence, a duplicate, or our own echo?

End-to-end flow: Trace origin/generation/causation → verify mapping revision/namespace → retain raw receipt → link canonical event only with evidence → block echo workflow triggering → quarantine mapping with a reason → independently approve restoration.

Permission: bms.configure, incident reviewer. Tenant,region,site and object scope are checked server-side; UI visibility is not an authorization boundary.

Field	Example	UI behavior
Origin event identity	PANEL-A / P-188	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Causation reference	EV-991	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Outbound namespace	smartalarm.outbound	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Native outbound object	SA-STATUS-41	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.
Echo receipt	EV-993	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

Field	Example	UI behavior
Quarantine reason	Echo re-entry	Preserve raw ID/unit; display source/time/revision when provided. Edits are drafts until server validation and approval.

DTO binding: `BMSAlarmView.native_alarm_id/occurrence_id/incident_id/map_revision; BMSPointView.provenance_ref; canonical event origin/causation IDs and outbound namespace are evidence projection fields, not writable alarm state.`

Validation and guards: Do not deduplicate on timestamp/name/value alone. Outbound namespace uses a dedicated BMS object. Native alarms are never overwritten. Exports redact secrets and out-of-scope health details.

Method / SmartAlarm path	operationId
`GET /bms/connections/{connection_id}/alarms`	`bms_get_bms_connections_connection_id_alarms`
`GET /bms/connections/{connection_id}/points`	`bms_get_bms_connections_connection_id_points`
`POST /bms/connections/{connection_id}/reconciliations`	`bms_post_bms_connections_connection_id_reconciliations`
`GET /bms/connections/{connection_id}/policies/incident-export`	`bms_incident_export_get`
`PATCH /bms/connections/{connection_id}/policies/incident-export`	`bms_incident_export_patch`

Use typed request/response schemas from API-Contract.json. Persist Idempotency-Key for the logical mutation; carry expected_revision and applicable qualification/approval/expiry fields. A202job/receipt means accepted processing, not completed effect. Follow job/result resources until terminal or unknown state. A409revision conflict requires refresh/review; do not silently overwrite. Authorization/validation failures remain actionable and do not trigger blind retries.

Screen-specific failure behavior: Echo blocks/quarantines re-entry while preserving raw events. Conflicting identity prevents automatic merge. Unsupported writeback stays read-only with a capability reason. Genuine recurrence keeps a distinct generation.

Shared states: loading hides currentness claims; empty states name their filter/scope rather than asserting safety; partial data identifies missing source/point coverage; stale/offline retain explicit age/quality and block unsafe actions; denied suppresses scoped data; unknown command/ACK routes to reconciliation. Never convert a transport success into a safe-zone indicator.

Acceptance scenario: Quarantining SA-STATUS-41 retains EV-993 in lineage and does not create a new incident from echo. A genuinely new generation is not suppressed merely for sharing a point. Native conditions are not overwritten.

Prototype evidence: local interactions,24locale views and responsive checks are recorded separately in BMS-UI-QA.json. Backend/authorization/device behavior above is a production acceptance contract, not a passed runtime test.